Notes on the Classical Theory of Linear Models

1 The Multiple Linear Regression Model

1.1 Model Specification and Matrix Notation

Definition 1.1 (Multiple Linear Regression Model)**.** Given n observations, each consisting of a dependent variable y_i and k independent variables $x_{i1}, \dots, x_{ik}$, the **multiple linear regression model** assumes

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i, \quad i = 1, 2, \dots, n, \tag{1}$$

where β_0 is the intercept, $\beta_1, \dots, \beta_k$ are slope coefficients, and ε_i is the error term. Equivalently, in matrix form,

$$\boxed{\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}} \tag{2}$$

where

$$\mathbf{y} = \underset{n\times 1}{\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}}, \quad \mathbf{X} = \underset{n\times(k+1)}{\begin{pmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{pmatrix}}, \quad \boldsymbol{\beta} = \underset{(k+1)\times 1}{\begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}}, \quad \boldsymbol{\varepsilon} = \underset{n\times 1}{\begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}}.$$

The matrix $\mathbf{X}$ is called the **design matrix**. Its first column is the all-ones vector $\mathbf{1} = (1, 1, \dots, 1)'$, corresponding to the intercept. We write $\mathbf{x}_i'$ for the i-th row of $\mathbf{X}$, so that $y_i = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i$.

1.2 Classical Assumptions

Definition 1.2 (Classical Assumptions)**.** The following assumptions are imposed on model (2):

(A1) **Linearity**: $\mathsf{E}[\mathbf{y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta}$.

(A2) **Zero conditional mean**: $\mathsf{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$.

(A3) **Spherical errors**: $\mathrm{var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$ for a constant $\sigma^2 > 0$. This comprises two sub-conditions:

- *Homoscedasticity*: $\mathrm{var}(\varepsilon_i) = \sigma^2$ for all i (a constant).
- *No serial correlation*: $\mathrm{cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$.

(A4) **Full rank**: $\mathrm{rank}(\mathbf{X}) = k + 1$ (the columns of $\mathbf{X}$ are linearly independent). This requires $n \geqslant k + 1$. For variance estimation and inference (Sections 3–7 and 9, and wherever s appears in later sections), we additionally require $n > k + 1$ so that $\mathrm{MSE} = \mathrm{SSE}/(n - k - 1)$ (see Definition 3.1) is well-defined.

(A5) **Normality**: $\boldsymbol{\varepsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ (the multivariate normal distribution $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is defined in Definition 4.1, Section 4).

Assumptions (A1)–(A4) are called the **Gauss–Markov assumptions**. Assumption (A5) is additionally needed for exact distributional results in finite samples.

Remark 1.3. Throughout this article, we treat $\mathbf{X}$ as fixed (non-stochastic) and condition on it. All expectations, variances, and distributions are conditional on $\mathbf{X}$, but we suppress the conditioning notation for brevity.

Remark 1.4 (Relationship between (A1) and (A2))**.** Given the model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, assumptions (A1) and (A2) are logically equivalent: taking conditional expectations of both sides gives $\mathsf{E}[\mathbf{y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta} + \mathsf{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}]$, so $\mathsf{E}[\mathbf{y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta}$ holds if and only if $\mathsf{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$. We state both because each form is convenient in different proofs: (A1) is closer to how the model is written, and (A2) is closer to how residuals are handled. Conditioning on $\mathbf{X}$ (previous remark) also means that inside any proof, $\mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\mathrm{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ can be used without further notation—these are the conditional expectation and variance with the conditioning suppressed.

1.3 Prerequisites from Linear Algebra (assumed without proof)

The following standard facts are used below without proof; they are the only linear-algebra results imported from outside these notes. (The probabilistic prerequisites are listed in Section 4.1.) Notation: for $\mathbf{A} \in \mathbb{R}^{m\times p}$, $\mathcal{C}(\mathbf{A}) = \{\mathbf{A}\mathbf{v} : \mathbf{v} \in \mathbb{R}^p\} \subseteq \mathbb{R}^m$ is the **column space** of $\mathbf{A}$ and $\mathcal{N}(\mathbf{A}) = \{\mathbf{v} \in \mathbb{R}^p : \mathbf{A}\mathbf{v} = \mathbf{0}\}$ its **null space**; a symmetric matrix $\mathbf{A}$ is **positive semi-definite**, written $\mathbf{A} \succeq \mathbf{0}$, if $\mathbf{v}'\mathbf{A}\mathbf{v} \geqslant 0$ for all $\mathbf{v}$, and **positive definite** if $\mathbf{v}'\mathbf{A}\mathbf{v} > 0$ for all $\mathbf{v} \neq \mathbf{0}$. Vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^m$ are **orthogonal**, written $\mathbf{v} \perp \mathbf{w}$, if $\mathbf{v}'\mathbf{w} = 0$; for a subspace $\mathcal{S} \subseteq \mathbb{R}^m$, $\mathbf{v} \perp \mathcal{S}$ means $\mathbf{v} \perp \mathbf{w}$ for every $\mathbf{w} \in \mathcal{S}$, and $\mathcal{S}^\perp = \{\mathbf{v} \in \mathbb{R}^m : \mathbf{v} \perp \mathcal{S}\}$ is the **orthogonal complement** of $\mathcal{S}$; for a single vector $\mathbf{a}$ we write $\mathbf{a}^\perp = \mathcal{C}(\mathbf{a})^\perp = \{\mathbf{v} \in \mathbb{R}^m : \mathbf{a}'\mathbf{v} = 0\}$.

(L1) *Rank–nullity.* For $\mathbf{A} \in \mathbb{R}^{m\times p}$, $\dim \mathcal{N}(\mathbf{A}) + \operatorname{rank}(\mathbf{A}) = p$. Hence two matrices with the same number of columns and the same null space have the same rank. (Used in Theorem 2.4, Step 3, and in the remark following Corollary 7.6.)

(L2) *Trace.* $\operatorname{tr}(\mathbf{A}\mathbf{B}) = \operatorname{tr}(\mathbf{B}\mathbf{A})$ whenever both products are defined; in particular $\operatorname{tr}(\mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}) = \operatorname{tr}(\mathbf{\Lambda})$. (Used in Lemma 2.8, in Theorem 4.5 and in the solution to Exercise A2.)

(L3) *Spectral theorem.* Every symmetric $\mathbf{A} \in \mathbb{R}^{n\times n}$ can be written $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ with $\mathbf{P}$ orthogonal ($\mathbf{P}'\mathbf{P} = \mathbf{P}\mathbf{P}' = \mathbf{I}$) and $\mathbf{\Lambda}$ diagonal, its entries being the eigenvalues of $\mathbf{A}$. (Used in Theorem 4.5, in (L4), and in the solutions to Exercises A2 and A8.)

(L4) *Square roots of positive definite matrices.* If $\mathbf{A}$ is symmetric positive definite, then $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ as in (L3) with all eigenvalues positive, and $\mathbf{A}^{1/2} := \mathbf{P}\mathbf{\Lambda}^{1/2}\mathbf{P}'$ is symmetric positive definite with $\mathbf{A}^{1/2}\mathbf{A}^{1/2} = \mathbf{A}$; its inverse $\mathbf{A}^{-1/2} := (\mathbf{A}^{1/2})^{-1}$ satisfies $\mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2} = \mathbf{I}$. In particular $\mathbf{A} = \mathbf{L}\mathbf{L}'$ for some invertible $\mathbf{L}$ (for instance $\mathbf{L} = \mathbf{A}^{1/2}$, or the Cholesky factor). (Used in Theorems 7.4 and 7.10 and in the solutions to Exercises B4 and B5.)

2 Ordinary Least Squares

Standing assumption: from here to the end of these notes, assumption (A4) (full column rank of $\mathbf{X}$) is in force. It is what makes $(\mathbf{X}'\mathbf{X})^{-1}$—and hence $\hat{\boldsymbol{\beta}}$, $\mathbf{H}$ and $\mathbf{M}$—exist, and it is restated in a theorem only when the proof uses it in an additional way. Assumptions (A1)–(A3) and (A5) are invoked explicitly where needed; the results of this section use none of them.

2.1 Definition and the Normal Equations

Definition 2.1 (Ordinary Least Squares)**.** The **OLS estimator** $\hat{\boldsymbol{\beta}}$ is the value of $\boldsymbol{\beta}$ that minimizes the sum of squared residuals:

$$Q(\boldsymbol{\beta}) = \sum_{i=1}^{n} (y_i - \mathbf{x}_i'\boldsymbol{\beta})^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2. \tag{3}$$

Lemma 2.2 (Matrix Differentiation Formulas)**.** For $\boldsymbol{\beta} \in \mathbb{R}^p$, constant $\mathbf{a} \in \mathbb{R}^p$ and constant $\mathbf{A} \in \mathbb{R}^{p\times p}$, write $Df(\boldsymbol{\beta}) := \partial f/\partial \boldsymbol{\beta}'$ (numerator layout, row vector). Then

(i) $D(\mathbf{a}'\boldsymbol{\beta}) = D(\boldsymbol{\beta}'\mathbf{a}) = \mathbf{a}'$;

(ii) $D(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = \boldsymbol{\beta}'(\mathbf{A} + \mathbf{A}')$, which equals $2\boldsymbol{\beta}'\mathbf{A}$ when $\mathbf{A}' = \mathbf{A}$.

Proof. Both functions are polynomials in the coordinates $\beta_1, \ldots, \beta_p$ of $\boldsymbol{\beta}$, so we compute the partial derivatives coordinate by coordinate; $Df(\boldsymbol{\beta})$ is the row vector $(\partial f/\partial\beta_1, \ldots, \partial f/\partial\beta_p)$.

(i). $\mathbf{a}'\boldsymbol{\beta} = \sum_{i=1}^p a_i\beta_i$, so $\partial(\mathbf{a}'\boldsymbol{\beta})/\partial\beta_j = a_j$ for each j; assembling the row vector gives $D(\mathbf{a}'\boldsymbol{\beta}) = (a_1, \ldots, a_p) = \mathbf{a}'$. The case $D(\boldsymbol{\beta}'\mathbf{a})$ follows from $\boldsymbol{\beta}'\mathbf{a} = \mathbf{a}'\boldsymbol{\beta}$.

(ii). $\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta} = \sum_{i=1}^p \sum_{l=1}^p \beta_i A_{il}\beta_l$. The coordinate β_j appears in the terms with $i = j$ and in the terms with $l = j$ (the term $i = l = j$ is $A_{jj}\beta_j^2$, whose derivative $2A_{jj}\beta_j$ is counted once in each group), so

$$\frac{\partial(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta})}{\partial\beta_j} = \sum_{l=1}^{p} A_{jl}\beta_l + \sum_{i=1}^{p} \beta_i A_{ij} = (\mathbf{A}\boldsymbol{\beta})_j + (\mathbf{A}'\boldsymbol{\beta})_j.$$

Hence $D(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = (\mathbf{A}\boldsymbol{\beta} + \mathbf{A}'\boldsymbol{\beta})' = \boldsymbol{\beta}'(\mathbf{A} + \mathbf{A}')$, which equals $2\boldsymbol{\beta}'\mathbf{A}$ when $\mathbf{A}' = \mathbf{A}$. □

Remark 2.3 (The same formulas via differentials)**.** Readers who know the differential calculus of Magnus and Neudecker (developed in the Matrix Calculus notes, and *not* assumed here) can obtain the same formulas from the linearity and product rule of d together with the *first identification theorem* $\mathrm{d}f(\boldsymbol{\beta}) = \mathbf{c}(\boldsymbol{\beta})'\mathrm{d}\boldsymbol{\beta} \iff Df(\boldsymbol{\beta}) = \mathbf{c}(\boldsymbol{\beta})'$: since $\mathrm{d}\mathbf{a} = 0$ and $\mathrm{d}\mathbf{A} = 0$, $\mathrm{d}(\mathbf{a}'\boldsymbol{\beta}) = \mathbf{a}'\mathrm{d}\boldsymbol{\beta}$, and, using the scalar identity $(\mathrm{d}\boldsymbol{\beta})'\mathbf{A}\boldsymbol{\beta} = \boldsymbol{\beta}'\mathbf{A}'\mathrm{d}\boldsymbol{\beta}$,

$$\mathrm{d}(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = (\mathrm{d}\boldsymbol{\beta})'\mathbf{A}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{A}\,\mathrm{d}\boldsymbol{\beta} = \boldsymbol{\beta}'\mathbf{A}'\mathrm{d}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{A}\,\mathrm{d}\boldsymbol{\beta} = \boldsymbol{\beta}'(\mathbf{A} + \mathbf{A}')\,\mathrm{d}\boldsymbol{\beta}.$$

Nothing in these notes depends on this remark.

2.2 Derivation via Differentiation

Theorem 2.4 (OLS Estimator)**.** Under assumption (A4), the OLS estimator is

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.} \tag{4}$$

Proof. Step 1: Expand $Q(\boldsymbol{\beta})$.

$$\begin{aligned} Q(\boldsymbol{\beta}) &= (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \\ &= \mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}'\mathbf{X}'\mathbf{y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} \\ &= \mathbf{y}'\mathbf{y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}. \end{aligned} \tag{5}$$

In the last line, $\mathbf{y}'\mathbf{X}\boldsymbol{\beta}$ is a scalar, hence equals its transpose $\boldsymbol{\beta}'\mathbf{X}'\mathbf{y}$.

Step 2: First-order condition. Apply Lemma 2.2 term by term, with $\mathbf{a} = \mathbf{X}'\mathbf{y}$ and $\mathbf{A} = \mathbf{X}'\mathbf{X}$ (symmetric):

$$DQ(\boldsymbol{\beta}) = -2(\mathbf{X}'\mathbf{y})' + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X} = -2\mathbf{y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}.$$

Setting $DQ = \mathbf{0}'$ and transposing gives the **normal equation**

$$\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}. \tag{6}$$

Step 3: Solve. By (A4), $\operatorname{rank}(\mathbf{X}) = k + 1$. We show $\operatorname{rank}(\mathbf{X}'\mathbf{X}) = \operatorname{rank}(\mathbf{X})$: $\mathbf{X}\mathbf{v} = \mathbf{0}$ trivially implies $\mathbf{X}'\mathbf{X}\mathbf{v} = \mathbf{0}$; conversely, $\mathbf{X}'\mathbf{X}\mathbf{v} = \mathbf{0}$ gives $\mathbf{v}'\mathbf{X}'\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 = 0$, hence $\mathbf{X}\mathbf{v} = \mathbf{0}$. So $\mathbf{X}$ and $\mathbf{X}'\mathbf{X}$ have the same null space, hence the same rank ((L1)). Thus $\mathbf{X}'\mathbf{X}$ is $(k+1) \times (k+1)$ of full rank, hence invertible. Multiplying (6) by $(\mathbf{X}'\mathbf{X})^{-1}$:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Step 4: Verify the second-order condition. The Hessian matrix is

$$\frac{\partial^2 Q}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}'} = 2\mathbf{X}'\mathbf{X}.$$

Under assumption (A4), for any nonzero $\mathbf{v} \in \mathbb{R}^{k+1}$, $\mathbf{X}\mathbf{v} \neq \mathbf{0}$, so $\mathbf{v}'\mathbf{X}'\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 > 0$. Hence $\mathbf{X}'\mathbf{X}$ is positive definite, so $\hat{\boldsymbol{\beta}}$ is a local minimum by the second-order sufficient condition of calculus, which is not among the imported results; nothing below depends on it, since (7) below proves directly that $\hat{\boldsymbol{\beta}}$ is the unique global minimizer. (Its inverse $(\mathbf{X}'\mathbf{X})^{-1}$ is positive definite as well, a fact needed in Theorems 7.4 and 10.3: it is symmetric, being the inverse of a symmetric matrix, and for $\mathbf{v} \neq \mathbf{0}$ the vector $\mathbf{w} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{v}$ is nonzero (as $\mathbf{v} = \mathbf{X}'\mathbf{X}\mathbf{w}$), so $\mathbf{v}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{v} = \mathbf{w}'\mathbf{X}'\mathbf{X}\mathbf{w} = \|\mathbf{X}\mathbf{w}\|^2 > 0$.) Positive definiteness of the Hessian by itself certifies only a *local* minimum; that the minimum is global and unique follows from an exact expansion. For any $\boldsymbol{\beta} \in \mathbb{R}^{k+1}$, write $\mathbf{y} - \mathbf{X}\boldsymbol{\beta} = (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) - \mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})$; then

$$Q(\boldsymbol{\beta}) = Q(\hat{\boldsymbol{\beta}}) - 2(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})'\mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) + \|\mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})\|^2 = Q(\hat{\boldsymbol{\beta}}) + \|\mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})\|^2 \geqslant Q(\hat{\boldsymbol{\beta}}), \tag{7}$$

because $\mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$ is the normal equation (6). Equality holds iff $\mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) = \mathbf{0}$, i.e. (by (A4)) iff $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$. Thus $\hat{\boldsymbol{\beta}}$ is the unique global minimizer of Q. □

2.3 Derivation via Orthogonal Projection

Theorem 2.5 (OLS as Orthogonal Projection)**.** The fitted vector $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ is the orthogonal projection of $\mathbf{y}$ onto $\mathcal{C}(\mathbf{X})$, that is, $\hat{\mathbf{y}} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{y} - \hat{\mathbf{y}} \perp \mathcal{C}(\mathbf{X})$ (the term is defined for a general subspace, in matrix form, in Remark 2.10), and the OLS estimator is $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.

Proof. The OLS problem is to minimize $Q(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ over $\boldsymbol{\beta} \in \mathbb{R}^{k+1}$, equivalently, to find the point in $\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^{k+1}\}$ closest to $\mathbf{y}$ in Euclidean norm.

Step 1: Orthogonality characterization of the minimum. We claim $\hat{\mathbf{y}} \in \mathcal{C}(\mathbf{X})$ minimizes $\|\mathbf{y} - \tilde{\mathbf{y}}\|^2$ over $\tilde{\mathbf{y}} \in \mathcal{C}(\mathbf{X})$ iff $\mathbf{y} - \hat{\mathbf{y}} \perp \mathcal{C}(\mathbf{X})$.

($\Leftarrow$) Assume the orthogonality condition and let $\tilde{\mathbf{y}} \in \mathcal{C}(\mathbf{X})$. Since $\mathcal{C}(\mathbf{X})$ is a subspace, $\hat{\mathbf{y}} - \tilde{\mathbf{y}} \in \mathcal{C}(\mathbf{X})$, so $(\mathbf{y} - \hat{\mathbf{y}}) \perp (\hat{\mathbf{y}} - \tilde{\mathbf{y}})$. Pythagoras (for $\mathbf{a}'\mathbf{b} = 0$, expanding gives $\|\mathbf{a} + \mathbf{b}\|^2 = \|\mathbf{a}\|^2 + 2\mathbf{a}'\mathbf{b} + \|\mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2$; this identity is what "Pythagoras" means throughout) applied to $\mathbf{y} - \tilde{\mathbf{y}} = (\mathbf{y} - \hat{\mathbf{y}}) + (\hat{\mathbf{y}} - \tilde{\mathbf{y}})$ gives

$$\|\mathbf{y} - \tilde{\mathbf{y}}\|^2 = \|\mathbf{y} - \hat{\mathbf{y}}\|^2 + \|\hat{\mathbf{y}} - \tilde{\mathbf{y}}\|^2 \;\geqslant\; \|\mathbf{y} - \hat{\mathbf{y}}\|^2,$$

with equality iff $\tilde{\mathbf{y}} = \hat{\mathbf{y}}$. So $\hat{\mathbf{y}}$ is the unique minimizer.

($\Rightarrow$) Let $\hat{\mathbf{y}} \in \mathcal{C}(\mathbf{X})$ minimize. Suppose, for contradiction, some $\mathbf{v} \in \mathcal{C}(\mathbf{X})$ has $\mathbf{v}'(\mathbf{y} - \hat{\mathbf{y}}) \neq 0$; normalize to $\|\mathbf{v}\|^2 = 1$. For $\tilde{\mathbf{y}} = \hat{\mathbf{y}} + t\mathbf{v} \in \mathcal{C}(\mathbf{X})$,

$$\|\mathbf{y} - \tilde{\mathbf{y}}\|^2 = \|\mathbf{y} - \hat{\mathbf{y}}\|^2 - 2t\,\mathbf{v}'(\mathbf{y} - \hat{\mathbf{y}}) + t^2,$$

minimized at $t^* = \mathbf{v}'(\mathbf{y} - \hat{\mathbf{y}}) \neq 0$ with value $\|\mathbf{y} - \hat{\mathbf{y}}\|^2 - (t^*)^2 < \|\mathbf{y} - \hat{\mathbf{y}}\|^2$, contradicting optimality. Hence $\mathbf{y} - \hat{\mathbf{y}} \perp \mathcal{C}(\mathbf{X})$.

Step 2: Translate the orthogonality condition to a normal equation. The residual is orthogonal to every vector in $\mathcal{C}(\mathbf{X})$ if and only if it is orthogonal to every column of $\mathbf{X}$ (since those columns span $\mathcal{C}(\mathbf{X})$):

$$\mathbf{y} - \hat{\mathbf{y}} \perp \mathcal{C}(\mathbf{X}) \iff \mathbf{X}'(\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{0}.$$

Writing $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ for some $\hat{\boldsymbol{\beta}}$, this becomes $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$, the same normal equation (6) obtained by differentiation.

Step 3: Solve. Under assumption (A4), $\mathbf{X}'\mathbf{X}$ is invertible, so $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$, and the minimizer $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ exists and is unique. □

2.4 Hat Matrix, Residual Maker, and Idempotent Matrices

Definition 2.6 (Hat Matrix and Residual Maker Matrix)**.** The **hat matrix** is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'. \tag{8}$$

The **residual maker matrix** (or **annihilator matrix**) is

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}. \tag{9}$$

Then $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{H}\mathbf{y}$, and the **residual vector** $\mathbf{e} := \mathbf{y} - \hat{\mathbf{y}}$, with components $e_i = y_i - \hat{y}_i$, satisfies $\mathbf{e} = \mathbf{y} - \mathbf{H}\mathbf{y} = \mathbf{M}\mathbf{y}$.

Definition 2.7 (Idempotent Matrix)**.** A square matrix $\mathbf{A}$ is **idempotent** if $\mathbf{A}^2 = \mathbf{A}$.

Lemma 2.8 (Trace Equals Rank for Idempotent Matrices)**.** If $\mathbf{A}$ is an $n \times n$ idempotent matrix, then $\operatorname{tr}(\mathbf{A}) = \operatorname{rank}(\mathbf{A})$.

Proof. We show every idempotent matrix is diagonalizable with eigenvalues in $\{0, 1\}$.

Step 1: Eigenvalues are in $\{0, 1\}$. If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $\lambda\mathbf{v} = \mathbf{A}\mathbf{v} = \mathbf{A}^2\mathbf{v} = \lambda^2\mathbf{v}$, forcing $\lambda \in \{0, 1\}$.

Step 2: $\mathbb{R}^n = \mathcal{N}(\mathbf{A}) \oplus \mathcal{C}(\mathbf{A})$**.** Any $\mathbf{v} \in \mathbb{R}^n$ decomposes as $\mathbf{v} = (\mathbf{v} - \mathbf{A}\mathbf{v}) + \mathbf{A}\mathbf{v}$, with $\mathbf{A}\mathbf{v} \in \mathcal{C}(\mathbf{A})$ and $\mathbf{v} - \mathbf{A}\mathbf{v} \in \mathcal{N}(\mathbf{A})$ (since $\mathbf{A}(\mathbf{v} - \mathbf{A}\mathbf{v}) = \mathbf{A}\mathbf{v} - \mathbf{A}^2\mathbf{v} = \mathbf{0}$). The sum is direct: if $\mathbf{v} = \mathbf{A}\mathbf{w} \in \mathcal{C}(\mathbf{A}) \cap \mathcal{N}(\mathbf{A})$, then $\mathbf{v} = \mathbf{A}\mathbf{w} = \mathbf{A}^2\mathbf{w} = \mathbf{A}\mathbf{v} = \mathbf{0}$.

Step 3: Diagonalizability. Any $\mathbf{u} \in \mathcal{C}(\mathbf{A})$ satisfies $\mathbf{A}\mathbf{u} = \mathbf{u}$ ($\mathbf{u} = \mathbf{A}\mathbf{w}$, so $\mathbf{A}\mathbf{u} = \mathbf{A}^2\mathbf{w} = \mathbf{A}\mathbf{w} = \mathbf{u}$). Combining a basis of $\mathcal{N}(\mathbf{A})$ (eigenvalue 0) and a basis of $\mathcal{C}(\mathbf{A})$ (eigenvalue 1) gives an eigenvector basis of $\mathbb{R}^n$ by Step 2. Thus $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$ with $\mathbf{\Lambda}$ diagonal and entries in $\{0, 1\}$.

Step 4: Conclude. Let $r = \operatorname{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$. Then $\mathbf{\Lambda}$ has r ones and $n - r$ zeros, so by the cyclic property of trace ((L2)),

$$\operatorname{tr}(\mathbf{A}) = \operatorname{tr}(\mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}) = \operatorname{tr}(\mathbf{\Lambda}) = r = \operatorname{rank}(\mathbf{A}). \qquad \square$$

Theorem 2.9 (Properties of $\mathbf{H}$ and $\mathbf{M}$)**.** The matrices $\mathbf{H}$ and $\mathbf{M}$ satisfy:

(i) $\mathbf{H}$ and $\mathbf{M}$ are both symmetric and idempotent.

(ii) $\mathbf{H}\mathbf{X} = \mathbf{X}$ and $\mathbf{M}\mathbf{X} = \mathbf{0}$.

(iii) $\mathbf{H}\mathbf{M} = \mathbf{M}\mathbf{H} = \mathbf{0}$.

(iv) $\operatorname{tr}(\mathbf{H}) = k + 1$ and $\operatorname{tr}(\mathbf{M}) = n - k - 1$.

Proof. (i) Symmetry of H: Since $\mathbf{X}'\mathbf{X}$ is symmetric, so is $(\mathbf{X}'\mathbf{X})^{-1}$:

$$\mathbf{H}' = [\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']' = \mathbf{X}[(\mathbf{X}'\mathbf{X})^{-1}]'\mathbf{X}' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{H}.$$

Idempotency of H:

$$\mathbf{H}^2 = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \underbrace{\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}}_{\mathbf{I}_{k+1}} \mathbf{X}' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{H}.$$

Symmetry and idempotency of M: $\mathbf{M}' = (\mathbf{I} - \mathbf{H})' = \mathbf{I} - \mathbf{H}' = \mathbf{I} - \mathbf{H} = \mathbf{M}$. Also, $\mathbf{M}^2 = (\mathbf{I} - \mathbf{H})^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H}^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H} = \mathbf{I} - \mathbf{H} = \mathbf{M}$.

(ii) $\mathbf{H}\mathbf{X} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{X}\mathbf{I}_{k+1} = \mathbf{X}$, and $\mathbf{M}\mathbf{X} = (\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{X} - \mathbf{X} = \mathbf{0}$.

(iii) $\mathbf{H}\mathbf{M} = \mathbf{H}(\mathbf{I} - \mathbf{H}) = \mathbf{H} - \mathbf{H}^2 = \mathbf{H} - \mathbf{H} = \mathbf{0}$, and $\mathbf{M}\mathbf{H} = (\mathbf{I} - \mathbf{H})\mathbf{H} = \mathbf{H} - \mathbf{H}^2 = \mathbf{0}$.

(iv) By Lemma 2.8, $\operatorname{tr}(\mathbf{H}) = \operatorname{rank}(\mathbf{H})$. We show $\operatorname{rank}(\mathbf{H}) = k + 1$ in two steps.

Step 1: $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$. The inclusion $\mathcal{C}(\mathbf{H}) \subseteq \mathcal{C}(\mathbf{X})$ is immediate: every column of $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is of the form $\mathbf{X}\mathbf{v}$ for some $\mathbf{v} \in \mathbb{R}^{k+1}$. Conversely, $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{H})$ because $\mathbf{H}\mathbf{X} = \mathbf{X}$ by (ii), so every column of $\mathbf{X}$ equals $\mathbf{H}$ times a column of $\mathbf{X}$, hence lies in $\mathcal{C}(\mathbf{H})$.

Step 2: $\dim \mathcal{C}(\mathbf{X}) = k + 1$ by assumption (A4).

Combining, $\operatorname{rank}(\mathbf{H}) = \dim \mathcal{C}(\mathbf{H}) = \dim \mathcal{C}(\mathbf{X}) = k + 1$, so $\operatorname{tr}(\mathbf{H}) = k + 1$. Finally, $\operatorname{tr}(\mathbf{M}) = \operatorname{tr}(\mathbf{I}) - \operatorname{tr}(\mathbf{H}) = n - (k + 1) = n - k - 1$. $\square$

Remark 2.10 ($\mathbf{H}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$)**.** For a subspace $\mathcal{S} \subseteq \mathbb{R}^n$, a matrix $\mathbf{P}$ is called the **orthogonal projection onto** $\mathcal{S}$ if $\mathbf{P}\mathbf{v} \in \mathcal{S}$ and $\mathbf{v} - \mathbf{P}\mathbf{v} \perp \mathcal{S}$ for every $\mathbf{v} \in \mathbb{R}^n$. By Step 1 of the proof of Theorem 2.5 (which used only that $\mathcal{C}(\mathbf{X})$ is a subspace), $\mathbf{P}\mathbf{v}$ is then the unique point of $\mathcal{S}$ closest to $\mathbf{v}$; hence there is at most one such matrix. One always exists: if $\mathcal{S} = \{\mathbf{0}\}$, the zero matrix is the orthogonal projection onto $\mathcal{S}$ (it maps every $\mathbf{v}$ to $\mathbf{0} \in \mathcal{S}$, and $\mathbf{v} - \mathbf{0} = \mathbf{v} \perp \{\mathbf{0}\}$ since $\mathbf{v}'\mathbf{0} = 0$), and it is symmetric and idempotent with column space $\{\mathbf{0}\} = \mathcal{S}$; otherwise, writing $\mathcal{S} = \mathcal{C}(\mathbf{B})$ with the columns of $\mathbf{B}$ a basis of $\mathcal{S}$ (so that $\mathbf{B}$ has full column rank), Definition 2.6 and Theorem 2.9 apply with $\mathbf{B}$ in place of $\mathbf{X}$, and $\mathbf{B}(\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}'$ is the orthogonal projection onto $\mathcal{S}$ (it maps $\mathbf{v}$ into $\mathcal{C}(\mathbf{B})$, and $\mathbf{B}'[\mathbf{v} - \mathbf{B}(\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}'\mathbf{v}] = \mathbf{0}$); it is symmetric and idempotent with column space $\mathcal{S}$. In particular $\mathbf{H}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$: $\mathbf{H}\mathbf{v} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{X}'(\mathbf{v} - \mathbf{H}\mathbf{v}) = \mathbf{X}'\mathbf{M}\mathbf{v} = \mathbf{0}$ by (ii).

Combining (i), (ii), and the identity $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$ proved above: $\mathbf{H}$ is symmetric, idempotent, and has column space $\mathcal{C}(\mathbf{X})$. These three properties characterize the orthogonal projection onto $\mathcal{C}(\mathbf{X})$, in the sense that any symmetric idempotent $\mathbf{P}$ with $\mathcal{C}(\mathbf{P}) = \mathcal{C}(\mathbf{X})$ equals $\mathbf{H}$. Indeed, for $\mathbf{v} \in \mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{P})$, say $\mathbf{v} = \mathbf{P}\mathbf{w}$, we get $\mathbf{P}\mathbf{v} = \mathbf{P}^2\mathbf{w} = \mathbf{P}\mathbf{w} = \mathbf{v}$; for $\mathbf{v} \perp \mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{P})$, we get $\|\mathbf{P}\mathbf{v}\|^2 = \mathbf{v}'\mathbf{P}'\mathbf{P}\mathbf{v} = \mathbf{v}'(\mathbf{P}\mathbf{v}) = 0$ because $\mathbf{P}\mathbf{v} \in \mathcal{C}(\mathbf{P})$; and every $\mathbf{v} \in \mathbb{R}^n$ decomposes as $\mathbf{v} = \mathbf{H}\mathbf{v} + \mathbf{M}\mathbf{v}$ with $\mathbf{H}\mathbf{v} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{M}\mathbf{v} \perp \mathcal{C}(\mathbf{X})$, so $\mathbf{P}\mathbf{v} = \mathbf{P}(\mathbf{H}\mathbf{v}) + \mathbf{P}(\mathbf{M}\mathbf{v}) = \mathbf{H}\mathbf{v} + \mathbf{0} = \mathbf{H}\mathbf{v}$. Geometrically, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ is the closest point in $\mathcal{C}(\mathbf{X})$ to $\mathbf{y}$ in Euclidean distance (this is the content of §2, specifically the orthogonal projection derivation). Analogously, $\mathbf{M} = \mathbf{I} - \mathbf{H}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})^\perp$: $\mathbf{M}\mathbf{v} \perp \mathcal{C}(\mathbf{X})$ as just noted, and $\mathbf{v} - \mathbf{M}\mathbf{v} = \mathbf{H}\mathbf{v} \in \mathcal{C}(\mathbf{X})$ is orthogonal to $\mathcal{C}(\mathbf{X})^\perp$.

2.5 Errors Versus Residuals

Remark 2.11 (Errors versus residuals)**.** The **error** $\varepsilon_i = y_i - \mathbf{x}_i'\boldsymbol{\beta}$ is the deviation from the true regression function; computing it requires the unknown $\boldsymbol{\beta}$, so it is *unobservable*. The **residual** $e_i = y_i - \mathbf{x}_i'\hat{\boldsymbol{\beta}} = y_i - \hat{y}_i$ is the deviation from the fitted regression function and is a function of the data $(\mathbf{y}, \mathbf{X})$, hence *observable*. In matrix form, $\mathbf{e} = \mathbf{M}\mathbf{y} = \mathbf{M}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \mathbf{M}\boldsymbol{\varepsilon}$ (since $\mathbf{M}\mathbf{X} = \mathbf{0}$ by Theorem 2.9(ii)). This identity does not make $\boldsymbol{\varepsilon}$ observable: $\mathbf{M}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})^\perp$, of rank $n - k - 1$, so $\mathbf{e}$ retains only the component of $\boldsymbol{\varepsilon}$ orthogonal to the column space, and the discarded component $\mathbf{H}\boldsymbol{\varepsilon} = \mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ is precisely the estimation error. The residual is thus the observable part of the error, and the two behave differently: under assumption (A3) the ε_i are uncorrelated with equal variance, whereas $\operatorname{var}(\mathbf{e} \mid \mathbf{X}) = \sigma^2\mathbf{M}$ (Theorem 3.8) makes the e_i correlated in general (with an intercept and $n > k + 1$, $\mathbf{M}\mathbf{1} = \mathbf{0}$ and $\operatorname{tr}\mathbf{M} = n - k - 1 > 0$ force a nonzero off-diagonal entry of $\mathbf{M}$) and, unless the diagonal entries of $\mathbf{M}$ are all equal, heteroskedastic.

2.6 Algebraic Properties of OLS Residuals

Theorem 2.12 (Implications of the Normal Equations)**.** The OLS residuals satisfy:

(i) $\mathbf{X}'\mathbf{e} = \mathbf{0}$ (residuals are orthogonal to every column of $\mathbf{X}$).

(ii) $\sum_{i=1}^n e_i = 0$ (residuals sum to zero), provided the model includes an intercept.

(iii) $\hat{\mathbf{y}}'\mathbf{e} = 0$ (residuals are orthogonal to fitted values).

Proof. **(i)** From the normal equation $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$, rearranging:

$$\mathbf{X}'\mathbf{y} - \mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0} \iff \mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0} \iff \mathbf{X}'\mathbf{e} = \mathbf{0}.$$

(ii) The first column of $\mathbf{X}$ is $\mathbf{1}$. The first component of the vector equation $\mathbf{X}'\mathbf{e} = \mathbf{0}$ is $\mathbf{1}'\mathbf{e} = \sum_{i=1}^n e_i = 0$. This relies on the presence of the intercept column.

(iii) Since $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$:

$$\hat{\mathbf{y}}'\mathbf{e} = (\mathbf{X}\hat{\boldsymbol{\beta}})'\mathbf{e} = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{e} = \hat{\boldsymbol{\beta}}' \cdot \mathbf{0} = 0.$$

Geometrically, $\hat{\mathbf{y}} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{e} \perp \mathcal{C}(\mathbf{X})$. □

2.7 Leverages

Definition 2.13 (Leverage)**.** The **leverage** of the i-th observation is h_{ii}, the i-th diagonal entry of $\mathbf{H}$.

Theorem 2.14 (Properties of Leverages)**.**

(i) $0 \leqslant h_{ii} \leqslant 1$ for all i.

(ii) $\sum_{i=1}^n h_{ii} = k + 1$.

Proof. **(i)** Since $\mathbf{H}$ is symmetric and idempotent, $\mathbf{H} = \mathbf{H}^2 = \mathbf{H}'\mathbf{H}$. Comparing the (i,i) entry on both sides using $h_{ji} = h_{ij}$:

$$h_{ii} = \sum_{j=1}^{n} h_{ij}h_{ji} = \sum_{j=1}^{n} h_{ij}^2 = h_{ii}^2 + \sum_{j\neq i} h_{ij}^2.$$

The rightmost sum is $\geqslant 0$, so $h_{ii} \geqslant h_{ii}^2$, which rearranges to $h_{ii}(1-h_{ii}) \geqslant 0$. The leftmost expression $h_{ii} = \sum_j h_{ij}^2 \geqslant 0$ is also a sum of squares. Combining $h_{ii} \geqslant 0$ with $h_{ii}(1-h_{ii}) \geqslant 0$ forces $h_{ii} \leqslant 1$.

(ii) $\sum_{i=1}^{n} h_{ii} = \operatorname{tr}(\mathbf{H}) = k+1$ by Theorem 2.9(iv). □

2.8 The Case $k = 1$

When $k = 1$, the model of Definition 1.1 is the **simple linear regression model** $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, where we write x_i for x_{i1} (a scalar, not to be confused with the row $\mathbf{x}_i' = (1, x_i)$ of $\mathbf{X}$) and $\mathbf{x} = (x_1, \ldots, x_n)'$, so that $\mathbf{X} = [\mathbf{1} \;\; \mathbf{x}]$. This subsection specializes the formulas above to this case. They are expressed through the following sample quantities.

Definition 2.15 (Sample Moments and Sample Correlation)**.** For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ with means $\bar{u} = n^{-1}\sum_i u_i$ and $\bar{v} = n^{-1}\sum_i v_i$, write

$$S_{uv} = \sum_{i=1}^{n}(u_i - \bar{u})(v_i - \bar{v}) = \sum_{i=1}^{n} u_i v_i - n\bar{u}\bar{v}, \qquad S_{uu} = \sum_{i=1}^{n}(u_i - \bar{u})^2 = \sum_{i=1}^{n} u_i^2 - n\bar{u}^2$$

(the second forms follow on expanding the products and using $\sum_i u_i = n\bar{u}$ and $\sum_i v_i = n\bar{v}$), and, for $n \geqslant 2$, $s_u^2 = S_{uu}/(n-1)$ for the **sample variance** of $\mathbf{u}$. Since S_{uu} is a sum of squares, $S_{uu} = 0$ iff $\mathbf{u}$ is **constant**, i.e. $\mathbf{u} = \bar{u}\mathbf{1}$. If neither $\mathbf{u}$ nor $\mathbf{v}$ is constant, the **(Pearson) sample correlation** of $\mathbf{u}$ and $\mathbf{v}$ is

$$\operatorname{corr}(\mathbf{u}, \mathbf{v}) = \frac{S_{uv}}{\sqrt{S_{uu}S_{vv}}} = \frac{S_{uv}}{(n-1)s_u s_v}.$$

In the simple linear regression model we write $r = \operatorname{corr}(\mathbf{x}, \mathbf{y})$, $S_{yy} = \sum_i (y_i - \bar{y})^2$ and $s_y^2 = S_{yy}/(n-1)$.

Lemma 2.16 (Cauchy–Schwarz Inequality)**.** For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, $(\mathbf{u}'\mathbf{v})^2 \leqslant \|\mathbf{u}\|^2\|\mathbf{v}\|^2$, with equality if and only if one of the two vectors is a scalar multiple of the other.

Proof. If $\mathbf{u} = \mathbf{0}$, both sides vanish and $\mathbf{u} = 0 \cdot \mathbf{v}$. Let $\mathbf{u} \neq \mathbf{0}$, put $c = \mathbf{u}'\mathbf{v}/\|\mathbf{u}\|^2$ and $\mathbf{w} = \mathbf{v} - c\mathbf{u}$. Then $\mathbf{u}'\mathbf{w} = \mathbf{u}'\mathbf{v} - c\|\mathbf{u}\|^2 = 0$, so Pythagoras applied to $\mathbf{v} = c\mathbf{u} + \mathbf{w}$ gives

$$\|\mathbf{v}\|^2 = c^2\|\mathbf{u}\|^2 + \|\mathbf{w}\|^2 = \frac{(\mathbf{u}'\mathbf{v})^2}{\|\mathbf{u}\|^2} + \|\mathbf{w}\|^2 \;\geqslant\; \frac{(\mathbf{u}'\mathbf{v})^2}{\|\mathbf{u}\|^2},$$

which is the inequality, with equality iff $\mathbf{w} = \mathbf{0}$, i.e. $\mathbf{v} = c\mathbf{u}$. Conversely, if $\mathbf{v} = a\mathbf{u}$ for some a, then $c = a$ and $\mathbf{w} = \mathbf{0}$; and if $\mathbf{u} = a\mathbf{v}$ with $\mathbf{u} \neq \mathbf{0}$, then $a \neq 0$ and $\mathbf{v} = a^{-1}\mathbf{u}$. □

Proposition 2.17 (Properties of the Sample Correlation)**.** Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ be non-constant. Then:

(i) $-1 \leqslant \operatorname{corr}(\mathbf{u}, \mathbf{v}) \leqslant 1$;

(ii) $\operatorname{corr}(\mathbf{u}, \mathbf{v}) = 1$ (respectively -1) iff there are constants α and $\beta > 0$ (respectively $\beta < 0$) with $v_i = \alpha + \beta u_i$ for all i, i.e. iff the points (u_i, v_i) lie on a line of positive (respectively negative) slope;

(iii) for constants a, c and $b, d \neq 0$, replacing u_i by $a + bu_i$ and v_i by $c + dv_i$ for all i leaves $\operatorname{corr}(\mathbf{u}, \mathbf{v})$ unchanged if $bd > 0$ and changes its sign if $bd < 0$.

Proof. Put $\tilde{\mathbf{u}} = \mathbf{u} - \bar{u}\mathbf{1}$ and $\tilde{\mathbf{v}} = \mathbf{v} - \bar{v}\mathbf{1}$. Then $S_{uv} = \tilde{\mathbf{u}}'\tilde{\mathbf{v}}$, $S_{uu} = \|\tilde{\mathbf{u}}\|^2 > 0$ and $S_{vv} = \|\tilde{\mathbf{v}}\|^2 > 0$, so $\operatorname{corr}(\mathbf{u}, \mathbf{v}) = \tilde{\mathbf{u}}'\tilde{\mathbf{v}}/(\|\tilde{\mathbf{u}}\|\,\|\tilde{\mathbf{v}}\|)$.

(i) Lemma 2.16 applied to $\tilde{\mathbf{u}}$ and $\tilde{\mathbf{v}}$ gives $\operatorname{corr}(\mathbf{u}, \mathbf{v})^2 \leqslant 1$.

(ii) By the equality case of Lemma 2.16, $|\operatorname{corr}(\mathbf{u},\mathbf{v})| = 1$ iff one of $\tilde{\mathbf{u}}, \tilde{\mathbf{v}}$ is a multiple of the other; as both are nonzero, this means $\tilde{\mathbf{v}} = \beta\tilde{\mathbf{u}}$ with $\beta \neq 0$, i.e. $v_i = (\bar{v} - \beta\bar{u}) + \beta u_i$ for all i. In that case $\operatorname{corr}(\mathbf{u},\mathbf{v}) = \beta\|\tilde{\mathbf{u}}\|^2/(|\beta|\,\|\tilde{\mathbf{u}}\|^2) = \beta/|\beta|$, which is 1 for $\beta > 0$ and -1 for $\beta < 0$. Conversely, if $v_i = \alpha + \beta u_i$ for all i, then $\beta \neq 0$ (otherwise $\mathbf{v}$ would be constant), $\bar{v} = \alpha + \beta\bar{u}$, and $\tilde{\mathbf{v}} = \beta\tilde{\mathbf{u}}$.

(iii) The centered transformed vectors are $b\tilde{\mathbf{u}}$ and $d\tilde{\mathbf{v}}$ (the constants a and c cancel on centering), so the numerator S_{uv} is multiplied by bd and the denominator $\sqrt{S_{uu}S_{vv}}$ by $|b|\,|d|$. □

Corollary 2.18 (OLS in Simple Linear Regression)**.** Let $k = 1$ and $\mathbf{X} = [\mathbf{1}\ \ \mathbf{x}]$.

(i) (A4) holds if and only if $\mathbf{x}$ is not constant, i.e. $S_{xx} > 0$.

(ii) Under (A4),
$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{nS_{xx}}\begin{pmatrix}\sum_i x_i^2 & -n\bar{x}\\ -n\bar{x} & n\end{pmatrix} = \begin{pmatrix}\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} & -\frac{\bar{x}}{S_{xx}}\\ -\frac{\bar{x}}{S_{xx}} & \frac{1}{S_{xx}}\end{pmatrix}.$$

(iii) $\hat{\beta}_1 = S_{xy}/S_{xx}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$; if $\mathbf{y}$ is not constant, $\hat{\beta}_1 = r\,s_y/s_x$.

(iv) The fitted line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ passes through the point of means $(\bar{x},\bar{y})$, and $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$ for every i.

Proof. **(i)** Since $\mathbf{1} \neq \mathbf{0}$, the columns $\mathbf{1}$ and $\mathbf{x}$ are linearly dependent iff $\mathbf{x} = c\mathbf{1}$ for some c, i.e. iff $\mathbf{x}$ is constant, i.e. (Definition 2.15) iff $S_{xx} = 0$.

(ii) $\mathbf{X}'\mathbf{X} = \begin{pmatrix} n & n\bar{x}\\ n\bar{x} & \sum_i x_i^2\end{pmatrix}$. Multiplying it by the displayed matrix and using $\sum_i x_i^2 - n\bar{x}^2 = S_{xx}$,
$$\frac{1}{nS_{xx}}\begin{pmatrix} n\sum_i x_i^2 - n^2\bar{x}^2 & -n^2\bar{x} + n^2\bar{x}\\ n\bar{x}\sum_i x_i^2 - n\bar{x}\sum_i x_i^2 & -n^2\bar{x}^2 + n\sum_i x_i^2\end{pmatrix} = \frac{1}{nS_{xx}}\begin{pmatrix} nS_{xx} & 0\\ 0 & nS_{xx}\end{pmatrix} = \mathbf{I}_2,$$
so the displayed matrix is $(\mathbf{X}'\mathbf{X})^{-1}$. The second form uses $\sum_i x_i^2/(nS_{xx}) = (S_{xx} + n\bar{x}^2)/(nS_{xx})$.

(iii) Here $\mathbf{X}'\mathbf{y} = (n\bar{y}, \sum_i x_i y_i)'$. The first row of the normal equation (6) reads $n\hat{\beta}_0 + n\bar{x}\hat{\beta}_1 = n\bar{y}$, i.e. $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$. The second coordinate of $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is, by (ii),
$$\hat{\beta}_1 = \frac{-n\bar{x}\cdot n\bar{y} + n\sum_i x_i y_i}{nS_{xx}} = \frac{\sum_i x_i y_i - n\bar{x}\bar{y}}{S_{xx}} = \frac{S_{xy}}{S_{xx}}.$$
If $\mathbf{y}$ is not constant, $S_{xy} = r\sqrt{S_{xx}S_{yy}}$ by Definition 2.15, so $\hat{\beta}_1 = r\sqrt{S_{yy}/S_{xx}} = r\,s_y/s_x$.

(iv) By (iii), $\hat{\beta}_0 + \hat{\beta}_1\bar{x} = \bar{y}$, and $\hat{y}_i - \bar{y} = (\hat{\beta}_0 + \hat{\beta}_1 x_i) - (\hat{\beta}_0 + \hat{\beta}_1\bar{x}) = \hat{\beta}_1(x_i - \bar{x})$. □

Proposition 2.19 (The Slope as a Weighted Sum of the Responses)**.** Let $k = 1$ and $S_{xx} > 0$. Then $\hat{\beta}_1 = \sum_{i=1}^n w_i y_i$ with $w_i = (x_i - \bar{x})/S_{xx}$, and the weights satisfy
$$\sum_{i=1}^n w_i = 0, \qquad \sum_{i=1}^n w_i x_i = 1, \qquad \sum_{i=1}^n w_i^2 = \frac{1}{S_{xx}}.$$
(For general k, $\hat{\boldsymbol{\beta}} = \mathbf{C}\mathbf{y}$ with the constant matrix $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ of Theorem 2.4, so every $\hat{\beta}_j$ is a fixed linear combination of the responses.)

Proof. $\sum_i w_i = \sum_i (x_i - \bar{x})/S_{xx} = 0$. Hence $\sum_i w_i y_i = \sum_i w_i(y_i - \bar{y}) + \bar{y}\sum_i w_i = S_{xy}/S_{xx} = \hat{\beta}_1$ by Corollary 2.18(iii). Next, $\sum_i (x_i - \bar{x})x_i = \sum_i (x_i - \bar{x})^2 + \bar{x}\sum_i (x_i - \bar{x}) = S_{xx}$, so $\sum_i w_i x_i = 1$. Finally, $\sum_i w_i^2 = \sum_i (x_i - \bar{x})^2/S_{xx}^2 = 1/S_{xx}$. □

Corollary 2.20 (Regression Through the Origin)**.** Theorem 2.4 and its proof do not use the intercept column, so they apply to any design of full column rank. For the model $y_i = \beta x_i + \varepsilon_i$ without intercept, i.e. the single-column design $\mathbf{X} = \mathbf{x}$, which has full column rank iff $\mathbf{x} \neq \mathbf{0}$, the OLS estimator is

$$\hat{\beta} = \frac{\sum_i x_i y_i}{\sum_i x_i^2}, \qquad \hat{y}_i = \hat{\beta} x_i.$$

Proof. For $\mathbf{X} = \mathbf{x}$, $\mathbf{X}'\mathbf{X} = \sum_i x_i^2$, which is positive iff $\mathbf{x} \neq \mathbf{0}$, and $\mathbf{X}'\mathbf{y} = \sum_i x_i y_i$; Theorem 2.4 gives the ratio. Theorem 2.12(i) and (iii) still hold for this design, but (ii) does not (its proof reads $\mathbf{1}'\mathbf{e} = 0$ off the intercept row), so in general $\sum_i e_i \neq 0$. □

3 Finite-Sample Properties Without Normality

In this section, only assumptions (A1)–(A4) are used. No distributional assumption on $\boldsymbol{\varepsilon}$ is needed.

Before deriving the finite-sample properties of $\hat{\boldsymbol{\beta}}$, we collect the notation for the sum-of-squares and mean-square quantities used throughout the remaining sections.

Definition 3.1 (Sums of Squares and Mean Squares)**.** Given the fitted model $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$, define:

- **Total sum of squares**: $\mathrm{SST} = \sum_{i=1}^{n} (y_i - \bar{y})^2$, where $\bar{y} = n^{-1} \sum_i y_i$.
- **Regression sum of squares**: $\mathrm{SSR} = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$.
- **Error (residual) sum of squares**: $\mathrm{SSE} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \mathbf{e}'\mathbf{e}$, where $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ is the residual vector of Definition 2.6.
- **Mean squared error**: $\mathrm{MSE} = s^2 = \mathrm{SSE}/(n - k - 1)$, defined when $n > k + 1$.
- **Mean squared regression**: $\mathrm{MSR} = \mathrm{SSR}/k$, defined when $k \geqslant 1$ (at least one non-intercept regressor).

The decomposition $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$ is proved in Theorem 6.4 (§6); the unbiasedness $\mathsf{E}[\mathrm{MSE}] = \sigma^2$ is proved in Theorem 3.7 below.

We also record a tool used repeatedly in this and later sections.

Lemma 3.2 (Variance and covariance under linear transformations)**.** Let $\mathbf{z}$ be an $n \times 1$ random vector with mean $\boldsymbol{\mu} = \mathsf{E}[\mathbf{z}]$ and covariance matrix $\mathrm{var}(\mathbf{z}) = \mathsf{E}[(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})']$. For two random vectors $\mathbf{u}$ $(m \times 1)$ and $\mathbf{v}$ $(l \times 1)$ with finite second moments, the **cross-covariance matrix** is the $m \times l$ matrix

$$\mathrm{cov}(\mathbf{u}, \mathbf{v}) = \mathsf{E}\big[(\mathbf{u} - \mathsf{E}[\mathbf{u}])(\mathbf{v} - \mathsf{E}[\mathbf{v}])'\big],$$

so that $\mathrm{var}(\mathbf{z}) = \mathrm{cov}(\mathbf{z}, \mathbf{z})$. (As with $\mathbf{y}$, bold lower-case letters such as $\mathbf{u}$, $\mathbf{v}$ denote random vectors here and fixed vectors elsewhere, e.g. in Definition 2.15; each use is quantified where it occurs. The population quantity $\mathrm{cov}(\mathbf{u}, \mathbf{v})$ is not the sample quantity $\mathrm{corr}(\mathbf{u}, \mathbf{v})$ of Definition 2.15.) Let $\mathbf{A}$ $(m \times n)$ and $\mathbf{B}$ $(l \times n)$ be constant matrices and $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{d} \in \mathbb{R}^l$ constant vectors. Then

(i) $\mathsf{E}[\mathbf{A}\mathbf{z} + \mathbf{b}] = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$ and

$$\mathrm{var}(\mathbf{A}\mathbf{z} + \mathbf{b}) = \mathbf{A}\ \mathrm{var}(\mathbf{z})\ \mathbf{A}'; \tag{10}$$

(ii) $\mathrm{cov}(\mathbf{A}\mathbf{z} + \mathbf{b},\ \mathbf{B}\mathbf{z} + \mathbf{d}) = \mathbf{A}\ \mathrm{var}(\mathbf{z})\ \mathbf{B}'$.

In particular, under assumption (A3), $\mathrm{var}(\mathbf{y}) = \sigma^2 \mathbf{I}_n$.

Proof. **(i)** Since $\mathbf{A}$ and $\mathbf{b}$ are constant, $\mathsf{E}[\mathbf{A}\mathbf{z} + \mathbf{b}] = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$. Therefore

$$\begin{aligned}
\mathrm{var}(\mathbf{A}\mathbf{z} + \mathbf{b}) &= \mathsf{E}\big[(\mathbf{A}\mathbf{z} + \mathbf{b} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})(\mathbf{A}\mathbf{z} + \mathbf{b} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})'\big] \\
&= \mathsf{E}\big[\mathbf{A}(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})'\mathbf{A}'\big] \\
&= \mathbf{A}\,\mathsf{E}[(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})']\,\mathbf{A}' = \mathbf{A}\ \mathrm{var}(\mathbf{z})\ \mathbf{A}',
\end{aligned}$$

since constant matrices pull outside the expectation.

(ii) By the first part of (i), $\mathbf{Az}+\mathbf{b}-\mathsf{E}[\mathbf{Az}+\mathbf{b}] = \mathbf{A}(\mathbf{z}-\boldsymbol{\mu})$ and $\mathbf{Bz}+\mathbf{d}-\mathsf{E}[\mathbf{Bz}+\mathbf{d}] = \mathbf{B}(\mathbf{z}-\boldsymbol{\mu})$, so

$$\operatorname{cov}(\mathbf{Az}+\mathbf{b},\ \mathbf{Bz}+\mathbf{d}) = \mathsf{E}\big[\mathbf{A}(\mathbf{z}-\boldsymbol{\mu})(\mathbf{z}-\boldsymbol{\mu})'\mathbf{B}'\big] = \mathbf{A}\,\mathsf{E}[(\mathbf{z}-\boldsymbol{\mu})(\mathbf{z}-\boldsymbol{\mu})']\,\mathbf{B}' = \mathbf{A}\,\operatorname{var}(\mathbf{z})\,\mathbf{B}'.$$

(Taking $\mathbf{B}=\mathbf{A}$, $\mathbf{d}=\mathbf{b}$ recovers (10).)

Finally, applying (i) to $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}+\boldsymbol{\varepsilon}$ (with $\mathbf{A}=\mathbf{I}_n$, $\mathbf{b}=\mathbf{X}\boldsymbol{\beta}$ constant given the conditioning on $\mathbf{X}$) gives $\operatorname{var}(\mathbf{y}) = \operatorname{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$ by (A3). □

3.1 Unbiasedness

Theorem 3.3 (Unbiasedness of $\hat{\boldsymbol{\beta}}$)**.** Under (A2) and (A4) (the latter is needed for $\hat{\boldsymbol{\beta}}$ to exist), the OLS estimator is unbiased:

$$\boxed{\mathsf{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}.} \tag{11}$$

Proof. Substituting $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}+\boldsymbol{\varepsilon}$ into $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$:

$$\begin{aligned}\hat{\boldsymbol{\beta}} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta}+\boldsymbol{\varepsilon})\\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}\\ &= \mathbf{I}_{k+1}\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}\\ &= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}.\end{aligned} \tag{12}$$

Taking expectations, using (A2) ($\mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$):

$$\mathsf{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathsf{E}[\boldsymbol{\varepsilon}] = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\cdot\mathbf{0} = \boldsymbol{\beta}.$$ □

3.2 Variance–Covariance Matrix

Theorem 3.4 (Variance–Covariance Matrix of $\hat{\boldsymbol{\beta}}$)**.** Under (A3) and (A4):

$$\boxed{\operatorname{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.} \tag{13}$$

Proof. From (12), $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$, which is a linear transformation of $\boldsymbol{\varepsilon}$ (with a constant shift). Applying Lemma 3.2 with $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$:

$$\operatorname{var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\ \operatorname{var}(\boldsymbol{\varepsilon})\,\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}.$$

By (A3), $\operatorname{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$. Substituting:

$$\operatorname{var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1},$$

using $\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{I}_{k+1}$. The diagonal entry $[\operatorname{var}(\hat{\boldsymbol{\beta}})]_{jj} = \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$ gives $\operatorname{var}(\hat{\beta}_j)$. Here and throughout, the coordinates of a coefficient vector and the rows and columns of a matrix indexed by coefficients, such as $(\mathbf{X}'\mathbf{X})^{-1}$, are numbered from 0, matching $\beta_0, \beta_1, \ldots$; thus $[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$ is the diagonal entry belonging to β_j. Coordinates of vectors in $\mathbb{R}^n$ are numbered $1, \ldots, n$. □

3.3 The Gauss–Markov Theorem

Theorem 3.5 (Gauss–Markov Theorem)**.** Under (A1)–(A4), the OLS estimator $\hat{\boldsymbol{\beta}}$ is the **Best Linear Unbiased Estimator** (BLUE) of $\boldsymbol{\beta}$. That is, for any other linear unbiased estimator $\tilde{\boldsymbol{\beta}} = \mathbf{Cy}$ (where $\mathbf{C}$ is a $(k+1)\times n$ constant matrix), the difference $\operatorname{var}(\tilde{\boldsymbol{\beta}}) - \operatorname{var}(\hat{\boldsymbol{\beta}})$ is positive semi-definite. In particular, $\operatorname{var}(\tilde{\beta}_j) \geqslant \operatorname{var}(\hat{\beta}_j)$ for every j.

Proof. Step 1: Necessary condition for unbiasedness. By (A1), $\mathsf{E}[\tilde{\boldsymbol{\beta}}] = \mathbf{C}\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}$ for all $\boldsymbol{\beta} \in \mathbb{R}^{k+1}$ forces $\mathbf{C}\mathbf{X} = \mathbf{I}_{k+1}$ (setting $\mathbf{v} = \boldsymbol{\iota}_j$ in $\mathbf{C}\mathbf{X}\mathbf{v} = \mathbf{v}$ reads off the j-th column; here and below $\boldsymbol{\iota}_j$ denotes the standard basis vector with 1 in coordinate j and 0 elsewhere, i.e. the column of the identity matrix of the relevant dimension belonging to coordinate j, with coordinates numbered as in Theorem 3.4; so $\boldsymbol{\iota}_0 \in \mathbb{R}^{k+1}$ belongs to the intercept).

Step 2: Decompose $\mathbf{C}$. Write $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}$, where $\mathbf{D}$ is the excess beyond the OLS coefficient matrix. From $\mathbf{C}\mathbf{X} = \mathbf{I}_{k+1}$:

$$\mathbf{D}\mathbf{X} = \mathbf{C}\mathbf{X} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{I}_{k+1} - \mathbf{I}_{k+1} = \mathbf{0}.$$

Step 3: Compute $\operatorname{var}(\tilde{\boldsymbol{\beta}})$. Since $\tilde{\boldsymbol{\beta}} = \mathbf{C}\mathbf{y}$ with $\mathbf{C}$ constant, Lemma 3.2 gives

$$\operatorname{var}(\tilde{\boldsymbol{\beta}}) = \mathbf{C}\operatorname{var}(\mathbf{y})\mathbf{C}' = \sigma^2\mathbf{C}\mathbf{C}' = \sigma^2\left[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}\right]\left[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}'\right].$$

Expanding yields four terms:

$$\begin{aligned}
\sigma^2\mathbf{C}\mathbf{C}' = {} & \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} && \text{(term A)}\\
& + \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}' && \text{(term B)}\\
& + \sigma^2\mathbf{D}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} && \text{(term C)}\\
& + \sigma^2\mathbf{D}\mathbf{D}'. && \text{(term D)}
\end{aligned}$$

Term A simplifies to $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ because $\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{I}_{k+1}$. Term C vanishes because $\mathbf{D}\mathbf{X} = \mathbf{0}$ (Step 2). Term B equals (term C)$'$, so it also vanishes. Hence:

$$\operatorname{var}(\tilde{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} + \sigma^2\mathbf{D}\mathbf{D}' = \operatorname{var}(\hat{\boldsymbol{\beta}}) + \sigma^2\mathbf{D}\mathbf{D}'.$$

Since $\mathbf{D}\mathbf{D}'$ is positive semi-definite (for any $\mathbf{v} \in \mathbb{R}^{k+1}$, $\mathbf{v}'\mathbf{D}\mathbf{D}'\mathbf{v} = \|\mathbf{D}'\mathbf{v}\|^2 \geqslant 0$):

$$\operatorname{var}(\tilde{\boldsymbol{\beta}}) - \operatorname{var}(\hat{\boldsymbol{\beta}}) = \sigma^2\mathbf{D}\mathbf{D}' \succeq \mathbf{0}.$$

Equality holds if and only if $\mathbf{D}'\mathbf{v} = \mathbf{0}$ for all $\mathbf{v}$, i.e., $\mathbf{D} = \mathbf{0}$; equivalently, $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$. □

3.4 Estimation of σ^2

Lemma 3.6 (Expected Value of a Quadratic Form)**.** Let $\mathbf{z}$ be an $n \times 1$ random vector with $\mathsf{E}[\mathbf{z}] = \boldsymbol{\mu}$ and $\operatorname{var}(\mathbf{z}) = \boldsymbol{\Sigma}$. Let $\mathbf{A}$ be an $n \times n$ constant matrix. Then:

$$\mathsf{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}. \tag{14}$$

In particular, if $\mathsf{E}[\mathbf{z}] = \mathbf{0}$, then $\mathsf{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma})$.

Proof. The quadratic form is a scalar: $\mathbf{z}'\mathbf{A}\mathbf{z} = \sum_{i,j} a_{ij}z_iz_j$. Taking expectations:

$$\mathsf{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \sum_{i,j} a_{ij}\mathsf{E}[z_iz_j].$$

From $\operatorname{cov}(z_i, z_j) = \mathsf{E}[z_iz_j] - \mu_i\mu_j$, we get $\mathsf{E}[z_iz_j] = \Sigma_{ij} + \mu_i\mu_j$. Substituting:

$$\mathsf{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \sum_{i,j} a_{ij}\Sigma_{ij} + \sum_{i,j} a_{ij}\mu_i\mu_j.$$

The first sum equals $\operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma})$: indeed, $\operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}) = \sum_i(\mathbf{A}\boldsymbol{\Sigma})_{ii} = \sum_i\sum_j a_{ij}\Sigma_{ji} = \sum_{i,j} a_{ij}\Sigma_{ij}$, where the last step uses the symmetry $\Sigma_{ji} = \Sigma_{ij}$. The second sum equals $\boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$: indeed, $\boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu} = \sum_i \mu_i(\mathbf{A}\boldsymbol{\mu})_i = \sum_i \mu_i\sum_j a_{ij}\mu_j = \sum_{i,j} a_{ij}\mu_i\mu_j$. Combining yields (14). □

Theorem 3.7 (Unbiasedness of MSE)**.** Under (A2)–(A4), the **mean squared error**

$$s^2 = \text{MSE} = \frac{\text{SSE}}{n-k-1} = \frac{\mathbf{e}'\mathbf{e}}{n-k-1} \tag{15}$$

is an unbiased estimator of σ^2: $\mathsf{E}[\text{MSE}] = \sigma^2$.

Proof. Step 1: Express SSE **as a quadratic form in** $\boldsymbol{\varepsilon}$**.** Since $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ (Remark 2.11):

$$\text{SSE} = \mathbf{e}'\mathbf{e} = (\mathbf{M}\boldsymbol{\varepsilon})'(\mathbf{M}\boldsymbol{\varepsilon}) = \boldsymbol{\varepsilon}'\mathbf{M}'\mathbf{M}\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon},$$

where the last step uses $\mathbf{M}'\mathbf{M} = \mathbf{M}\mathbf{M} = \mathbf{M}$ (symmetric idempotent).

Step 2: Apply Lemma 3.6. With $\mathbf{z} = \boldsymbol{\varepsilon}$, $\mathbf{A} = \mathbf{M}$, $\boldsymbol{\mu} = \mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ (by (A2)), and $\boldsymbol{\Sigma} = \text{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$ (by (A3)):

$$\mathsf{E}[\text{SSE}] = \mathsf{E}[\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}] = \text{tr}(\mathbf{M}\cdot\sigma^2\mathbf{I}) + \mathbf{0}'\mathbf{M}\mathbf{0} = \sigma^2\,\text{tr}(\mathbf{M}).$$

Step 3: Evaluate the trace. By Theorem 2.9(iv), $\text{tr}(\mathbf{M}) = n-k-1$. Hence $\mathsf{E}[\text{SSE}] = \sigma^2(n-k-1)$.

Step 4: Conclude.

$$\mathsf{E}[\text{MSE}] = \mathsf{E}\left[\frac{\text{SSE}}{n-k-1}\right] = \frac{\sigma^2(n-k-1)}{n-k-1} = \sigma^2. \qquad \square$$

3.5 Variance of Individual Residuals

Theorem 3.8 (Variance of Residuals)**.** Under (A3) and (A4):

$$\text{var}(\mathbf{e}) = \sigma^2\mathbf{M}, \qquad \text{and in particular, } \text{var}(e_i) = \sigma^2(1-h_{ii}). \tag{16}$$

Proof. Since $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ and $\mathbf{M}$ is symmetric ($\mathbf{M}' = \mathbf{M}$) and idempotent ($\mathbf{M}^2 = \mathbf{M}$):

$$\text{var}(\mathbf{e}) = \mathbf{M}\,\text{var}(\boldsymbol{\varepsilon})\mathbf{M}' = \mathbf{M}(\sigma^2\mathbf{I})\mathbf{M}' = \sigma^2\mathbf{M}\mathbf{M} = \sigma^2\mathbf{M}^2 = \sigma^2\mathbf{M}.$$

The (i,i) entry gives $\text{var}(e_i) = \sigma^2 m_{ii}$. Since $\mathbf{M} = \mathbf{I} - \mathbf{H}$, we have $m_{ii} = 1 - h_{ii}$, so $\text{var}(e_i) = \sigma^2(1-h_{ii})$. $\square$

Remark 3.9. By Theorem 2.14(i), $0 \leqslant h_{ii} \leqslant 1$, so $\text{var}(e_i) \leqslant \sigma^2$. High-leverage observations (large h_{ii}) produce residuals with smaller variance: the fitted line is pulled toward them, leaving less room for the residual to vary.

Corollary 3.10 (Variance of the Fitted Values)**.** Under (A3) and (A4), the fitted vector has $\text{var}(\hat{\mathbf{y}}) = \sigma^2\mathbf{H}$, and $\text{cov}(\hat{\mathbf{y}}, \mathbf{e}) = \mathbf{0}$. In particular $\text{var}(\hat{y}_i) = \sigma^2 h_{ii}$ and

$$\text{var}(\hat{y}_i) + \text{var}(e_i) = \sigma^2 \qquad \text{for every } i.$$

Proof. Since $\mathbf{H}\mathbf{X} = \mathbf{X}$ (Theorem 2.9(ii)), $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{H}\boldsymbol{\varepsilon}$, and $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ (Remark 2.11). By Lemma 3.2(i) and (A3), $\text{var}(\hat{\mathbf{y}}) = \mathbf{H}(\sigma^2\mathbf{I})\mathbf{H}' = \sigma^2\mathbf{H}^2 = \sigma^2\mathbf{H}$ ($\mathbf{H}$ symmetric idempotent); by Lemma 3.2(ii), $\text{cov}(\hat{\mathbf{y}}, \mathbf{e}) = \mathbf{H}(\sigma^2\mathbf{I})\mathbf{M}' = \sigma^2\mathbf{H}\mathbf{M} = \mathbf{0}$ (Theorem 2.9(iii)). The (i,i) entry gives $\text{var}(\hat{y}_i) = \sigma^2 h_{ii}$, and adding $\text{var}(e_i) = \sigma^2(1-h_{ii})$ (Theorem 3.8) gives σ^2. Thus h_{ii} is the share of σ^2 carried by the fitted value of observation i, and $1-h_{ii}$ the share left in its residual. $\square$

3.6 Standardized Residuals and Unusual Observations

Definition 3.11 (Standardized Residuals, Outliers, and High-Leverage Points)**.** Assume $n > k+1$ and $s > 0$. For an observation with $h_{ii} < 1$, the **standardized residual** is

$$\tilde{e}_i = \frac{e_i}{s\sqrt{1-h_{ii}}},$$

the residual divided by the estimate $s\sqrt{1-h_{ii}}$ of its standard deviation $\sigma\sqrt{1-h_{ii}}$ (Theorems 3.7 and 3.8). By the usual conventions, observation i is flagged as an **outlier** (unusual in the direction of the response) if $|\tilde{e}_i| > 2$, and as a **high-leverage point** (unusual in the regressors) if $h_{ii} > 2(k+1)/n$. An observation can be both.

Remark 3.12 (Rationale for the conventions)**.** (a) Under (A2) and (A3), $e_i/(\sigma\sqrt{1-h_{ii}})$ has mean 0 (as $\mathsf{E}[\mathbf{e}] = \mathbf{M}\mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$) and variance 1 (Theorem 3.8) whatever the leverage, whereas the raw residuals have the unequal variances $\sigma^2(1-h_{ii})$; $\tilde{e}_i$ replaces σ by s. Standardized residuals are dimensionless, so a single cutoff can be used for every observation and every data set. (b) By Theorem 2.14, the average leverage is $(k+1)/n$, and the second rule flags leverages above twice the average. Since the leverages are non-negative and sum to $k+1$, fewer than $n/2$ observations can be flagged: if $m \geqslant 1$ of them exceed $2(k+1)/n$, then $k+1 = \sum_i h_{ii} > 2m(k+1)/n$, i.e. $m < n/2$. (c) The restriction $h_{ii} < 1$ is needed. If $h_{ii} = 1$, the identity $h_{ii} = h_{ii}^2 + \sum_{j\neq i} h_{ij}^2$ from the proof of Theorem 2.14(i) forces $h_{ij} = 0$ for all $j \neq i$, so $e_i = y_i - \sum_j h_{ij} y_j = y_i - y_i = 0$ for every $\mathbf{y}$: the fit passes through observation i whatever its response, and its residual carries no information about it.

3.7 The Case $k = 1$

Corollary 3.13 (Variances in Simple Linear Regression)**.** In the simple linear regression model of §2.8, under (A3) and (A4),

$$\operatorname{var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}, \qquad \operatorname{var}(\hat{\beta}_0) = \sigma^2\Big(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\Big), \qquad \operatorname{cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2\bar{x}}{S_{xx}},$$

and for every $x_0 \in \mathbb{R}$, with $\mathbf{x}_0 = (1, x_0)'$,

$$\operatorname{var}(\hat{\beta}_0 + \hat{\beta}_1 x_0) = \sigma^2 \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = \sigma^2\Big(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\Big).$$

Proof. By Theorem 3.4, $\operatorname{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$, whose entries are read off Corollary 2.18(ii). For the last statement, $\hat{\beta}_0 + \hat{\beta}_1 x_0 = \mathbf{x}_0'\hat{\boldsymbol{\beta}}$, so Lemma 3.2(i) gives

$$\operatorname{var}(\mathbf{x}_0'\hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = \sigma^2\Big(\frac{1}{n} + \frac{\bar{x}^2 - 2x_0\bar{x} + x_0^2}{S_{xx}}\Big) = \sigma^2\Big(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\Big).$$

(The slope's variance also follows from Proposition 2.19: $\hat{\beta}_1 = \sum_i w_i y_i$ with $\operatorname{var}(\mathbf{y}) = \sigma^2\mathbf{I}$, so $\operatorname{var}(\hat{\beta}_1) = \sigma^2 \sum_i w_i^2 = \sigma^2/S_{xx}$ by Lemma 3.2.) □

4 Multivariate Normal Tools

This section collects the distributional results needed for exact inference in Section 5.

4.1 Prerequisites: the Multivariate Normal Distribution

For a random vector $\mathbf{z} \in \mathbb{R}^n$, the **moment-generating function** (MGF) is $M_{\mathbf{z}}(\mathbf{t}) = \mathsf{E}[\exp(\mathbf{t}'\mathbf{z})]$, $\mathbf{t} \in \mathbb{R}^n$, whenever the expectation is finite.

Definition 4.1 (Multivariate normal distribution)**.** Let $\boldsymbol{\mu} \in \mathbb{R}^n$ and let $\boldsymbol{\Sigma}$ be a symmetric positive semi-definite $n \times n$ matrix. A random vector $\mathbf{z}$ has the **multivariate normal distribution** $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ if

$$M_{\mathbf{z}}(\mathbf{t}) = \exp\big(\mathbf{t}'\boldsymbol{\mu} + \tfrac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}\mathbf{t}\big) \quad \text{for all } \mathbf{t} \in \mathbb{R}^n. \tag{17}$$

Its mean vector is $\boldsymbol{\mu}$ and its covariance matrix is $\boldsymbol{\Sigma}$: the gradient of the right-hand side of (17) is $(\boldsymbol{\mu} + \boldsymbol{\Sigma}\mathbf{t})\, M_{\mathbf{z}}(\mathbf{t})$ and its Hessian is $\big[\boldsymbol{\Sigma} + (\boldsymbol{\mu} + \boldsymbol{\Sigma}\mathbf{t})(\boldsymbol{\mu} + \boldsymbol{\Sigma}\mathbf{t})'\big] M_{\mathbf{z}}(\mathbf{t})$, so evaluating at $\mathbf{t} = \mathbf{0}$ and using fact (N6) below gives $\mathsf{E}[\mathbf{z}] = \boldsymbol{\mu}$ and $\mathsf{E}[\mathbf{z}\mathbf{z}'] = \boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}'$, whence $\operatorname{var}(\mathbf{z}) = \mathsf{E}[\mathbf{z}\mathbf{z}'] - \boldsymbol{\mu}\boldsymbol{\mu}' = \boldsymbol{\Sigma}$. For $n = 1$ this is the usual $N(\mu, \sigma^2)$, $\sigma^2 \geqslant 0$.

The definition is stated through the MGF rather than through a density because $\boldsymbol{\Sigma}$ is allowed to be singular: several normal vectors below, such as the residual vector $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ with $\operatorname{var}(\mathbf{e}) = \sigma^2\mathbf{M}$ (Theorems 3.8 and 5.2), have singular covariance matrices and no density. The following facts about MGFs and distributions are used without proof; together with (L1)–(L4) they are the only results imported from outside these notes.

(N1) *Uniqueness.* If two random vectors have MGFs that are finite and equal on an open neighborhood of $\mathbf{0}$, they have the same distribution. (Used in Theorem 4.2.)

(N2) *Independence and factorization.* Random vectors $\mathbf{u}_1, \dots, \mathbf{u}_m$ (with MGFs finite near $\mathbf{0}$) are mutually independent if and only if their joint MGF factorizes: $M_{(\mathbf{u}_1,\dots,\mathbf{u}_m)}(\mathbf{t}_1, \dots, \mathbf{t}_m) = \prod_{i=1}^{m} M_{\mathbf{u}_i}(\mathbf{t}_i)$ near $\mathbf{0}$. (Used in Theorems 4.3, 4.5 and 9.2.)

(N3) *Functions of independent vectors.* If $\mathbf{u}$ and $\mathbf{v}$ are independent (written $\mathbf{u} \perp\!\!\!\perp \mathbf{v}$; the symbol $\perp$ is reserved for orthogonality of vectors and subspaces), then $f(\mathbf{u})$ and $g(\mathbf{v})$ are independent for any (measurable) functions f, g. (Used whenever "a function of $\hat{\boldsymbol{\beta}}$ is independent of SSE" is asserted, e.g. in Theorems 4.6, 5.2, 7.1, 7.4, 7.9 and 9.2.)

(N4) *Density in the nonsingular case.* If $\boldsymbol{\Sigma}$ is positive definite, $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ has the density $f(\mathbf{z}) = (2\pi)^{-n/2}\,|\boldsymbol{\Sigma}|^{-1/2} \exp\!\big(-\tfrac{1}{2}(\mathbf{z}-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{z}-\boldsymbol{\mu})\big)$ on $\mathbb{R}^n$. (Used in Theorem 5.3.)

(N5) *Distributions derived from the normal.* The **chi-square distribution** χ^2_r ($r \geqslant 1$) is the distribution of $\sum_{i=1}^{r} W_i^2$ for i.i.d. $W_i \sim N(0,1)$. If $Z \sim N(0,1)$ and $V \sim \chi^2_\nu$ are independent, $Z/\sqrt{V/\nu}$ has the t **distribution** t_ν. If $U \sim \chi^2_a$ and $V \sim \chi^2_b$ are independent, $(U/a)/(V/b)$ has the F **distribution** $F_{a,b}$. For $\tau \in (0,1)$, the τ **quantiles** $t_{\nu,\tau}$ and $F_{a,b,\tau}$ (degrees of freedom first) are the numbers with $\mathsf{P}(T \leqslant t_{\nu,\tau}) = \tau$ for $T \sim t_\nu$ and $\mathsf{P}(F \leqslant F_{a,b,\tau}) = \tau$ for $F \sim F_{a,b}$; all three have continuous distribution functions that are strictly increasing on their supports (the support of t_ν is $\mathbb{R}$, and that of χ^2_r and of $F_{a,b}$ is $[0,\infty)$), so these quantiles exist and are unique. In particular, for $\alpha \in (0,1)$, $\mathsf{P}(T > t_{\nu,1-\alpha}) = 1 - \mathsf{P}(T \leqslant t_{\nu,1-\alpha}) = \alpha$, and likewise $\mathsf{P}(F > F_{a,b,1-\alpha}) = \alpha$. (Used from Theorem 4.5 onward; the t and F quantiles enter in Theorems 7.9, 7.10, 9.1 and 9.2.)

(N6) *Moments from the MGF.* If $M_{\mathbf{z}}$ is finite on an open neighborhood of $\mathbf{0}$, then its first- and second-order partial derivatives exist there and, at $\mathbf{t} = \mathbf{0}$, $\partial M_{\mathbf{z}}/\partial t_i = \mathsf{E}[z_i]$ and $\partial^2 M_{\mathbf{z}}/\partial t_i\,\partial t_j = \mathsf{E}[z_i z_j]$; that is, the gradient of $M_{\mathbf{z}}$ at $\mathbf{0}$ is $\mathsf{E}[\mathbf{z}]$ and its Hessian (the matrix of second partial derivatives) is $\mathsf{E}[\mathbf{z}\mathbf{z}']$. (Used in Definition 4.1.)

(N7) *Independence implies uncorrelatedness.* If $\mathbf{u}$ and $\mathbf{v}$ are independent random vectors with finite second moments, then $\mathsf{E}[\mathbf{u}\mathbf{v}'] = \mathsf{E}[\mathbf{u}]\,\mathsf{E}[\mathbf{v}]'$, i.e. $\operatorname{cov}(\mathbf{u}, \mathbf{v}) = \mathbf{0}$ with cov as defined in Lemma 3.2. Consequently, for independent scalar random variables U and V, $\operatorname{var}(U \pm V) = \operatorname{var}(U) + \operatorname{var}(V)$: apply Lemma 3.2(i) with $\mathbf{z} = (U, V)'$ and $\mathbf{A} = (1, \pm 1)$, the off-diagonal entries of $\operatorname{var}(\mathbf{z})$ being $\operatorname{cov}(U, V) = 0$. (Used in Theorems 4.3 and 9.2 and in the solution to Exercise B6.)

The following consequence of (N3) and (N5) is used in Theorems 7.9 and 9.2: *the t distribution is symmetric about* 0. Let $T = Z/\sqrt{V/\nu}$ with $Z \sim N(0,1)$ and $V \sim \chi^2_\nu$ independent, as in (N5). Then $M_{-Z}(\theta) = \mathsf{E}[e^{\theta(-Z)}] = M_Z(-\theta) = e^{\theta^2/2} = M_Z(\theta)$ for all $\theta \in \mathbb{R}$, so $-Z \sim N(0,1)$ by Definition 4.1; $-Z$ is still independent of V ((N3)); hence $-T = (-Z)/\sqrt{V/\nu} \sim t_\nu$ by (N5), and $\mathsf{P}(T < -c) = \mathsf{P}(-T > c) = \mathsf{P}(T > c)$ for every $c \in \mathbb{R}$. Taking $c = t_{\nu,1-\alpha}$, for which $\mathsf{P}(T > c) = \alpha$ by (N5), gives the first identity below; the second follows from the first with $\alpha/2$ in place of α, because $\mathsf{P}(-t_{\nu,1-\alpha/2} \leqslant T \leqslant t_{\nu,1-\alpha/2}) = 1 - \mathsf{P}(T < -t_{\nu,1-\alpha/2}) - \mathsf{P}(T > t_{\nu,1-\alpha/2})$ and both subtracted probabilities equal $\alpha/2$:

$$\mathsf{P}(T < -t_{\nu,1-\alpha}) = \alpha \qquad \text{and} \qquad \mathsf{P}(-t_{\nu,1-\alpha/2} \leqslant T \leqslant t_{\nu,1-\alpha/2}) = 1-\alpha \qquad (T \sim t_\nu,\ \alpha \in (0,1)). \tag{18}$$

A second consequence, used in Proposition 7.12 and Corollary 7.17: *the square of a t_ν variable has the $F_{1,\nu}$ distribution.* With $T = Z/\sqrt{V/\nu}$ as above, $T^2 = (Z^2/1)/(V/\nu)$, where $Z^2 \sim \chi^2_1$ by the definition in (N5) (with $r = 1$) and Z^2 is independent of V by (N3); hence $T^2 \sim F_{1,\nu}$ by (N5). In particular $\mathsf{P}(T^2 \leqslant t^2_{\nu,1-\alpha/2}) = \mathsf{P}(|T| \leqslant t_{\nu,1-\alpha/2}) = 1-\alpha$ by (18), so $t^2_{\nu,1-\alpha/2} = F_{1,\nu,1-\alpha}$.

4.2 Linear Transformations, Independence, and Quadratic Forms

Theorem 4.2 (Closure Under Linear Transformations)**.** If $\mathbf{z} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is an $n \times 1$ normal random vector, $\mathbf{A}$ is an $m \times n$ constant matrix, and $\mathbf{b} \in \mathbb{R}^m$ is a constant vector, then

$$\mathbf{A}\mathbf{z} + \mathbf{b} \sim N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b},\ \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}'). \tag{19}$$

Proof. By Definition 4.1, the MGF of $\mathbf{z}$ is $M_{\mathbf{z}}(\mathbf{t}) = \exp(\mathbf{t}'\boldsymbol{\mu} + \frac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}\mathbf{t})$. The MGF of $\mathbf{w} = \mathbf{A}\mathbf{z} + \mathbf{b}$ is:

$$\begin{aligned} M_{\mathbf{w}}(\mathbf{t}) &= \mathsf{E}\left[e^{\mathbf{t}'(\mathbf{A}\mathbf{z}+\mathbf{b})}\right] = e^{\mathbf{t}'\mathbf{b}} \cdot \mathsf{E}\left[e^{(\mathbf{A}'\mathbf{t})'\mathbf{z}}\right] = e^{\mathbf{t}'\mathbf{b}} \cdot M_{\mathbf{z}}(\mathbf{A}'\mathbf{t}) \\ &= e^{\mathbf{t}'\mathbf{b}} \cdot \exp\big((\mathbf{A}'\mathbf{t})'\boldsymbol{\mu} + \tfrac{1}{2}(\mathbf{A}'\mathbf{t})'\boldsymbol{\Sigma}(\mathbf{A}'\mathbf{t})\big) \\ &= \exp\big(\mathbf{t}'(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}) + \tfrac{1}{2}\mathbf{t}'\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}'\mathbf{t}\big). \end{aligned}$$

This is the MGF (17) of $N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b},\ \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$ (note that $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}'$ is symmetric positive semi-definite). By uniqueness of the MGF, fact (N1), the result follows. □

Theorem 4.3 (Uncorrelatedness ⇔ Independence Under Joint Normality)**.** Let $(\mathbf{u}', \mathbf{v}')'$ be jointly normal:

$$\begin{pmatrix}\mathbf{u}\\ \mathbf{v}\end{pmatrix} \sim N\left(\begin{pmatrix}\boldsymbol{\mu}_u\\ \boldsymbol{\mu}_v\end{pmatrix}, \begin{pmatrix}\boldsymbol{\Sigma}_{uu} & \boldsymbol{\Sigma}_{uv}\\ \boldsymbol{\Sigma}_{vu} & \boldsymbol{\Sigma}_{vv}\end{pmatrix}\right).$$

Then $\boldsymbol{\Sigma}_{uv} = \mathbf{0}$ if and only if $\mathbf{u}$ and $\mathbf{v}$ are independent.

Proof. ($\Leftarrow$): Independence implies uncorrelatedness for any distribution (fact (N7)), and $\boldsymbol{\Sigma}_{uv} = \operatorname{cov}(\mathbf{u}, \mathbf{v})$ because $\boldsymbol{\Sigma}$ is the covariance matrix of $(\mathbf{u}', \mathbf{v}')'$ (Definition 4.1); hence $\boldsymbol{\Sigma}_{uv} = \mathbf{0}$.

($\Rightarrow$): With $\boldsymbol{\Sigma}_{uv} = \mathbf{0}$, the covariance is block-diagonal. The joint MGF is

$$M_{(\mathbf{u},\mathbf{v})}(\mathbf{s}, \mathbf{t}) = \exp\big((\mathbf{s}', \mathbf{t}')\boldsymbol{\mu} + \tfrac{1}{2}(\mathbf{s}', \mathbf{t}')\boldsymbol{\Sigma}(\mathbf{s}', \mathbf{t}')'\big).$$

Block expansion gives

$$(\mathbf{s}', \mathbf{t}')\boldsymbol{\mu} = \mathbf{s}'\boldsymbol{\mu}_u + \mathbf{t}'\boldsymbol{\mu}_v, \qquad (\mathbf{s}', \mathbf{t}')\boldsymbol{\Sigma}\begin{pmatrix}\mathbf{s}\\ \mathbf{t}\end{pmatrix} = \mathbf{s}'\boldsymbol{\Sigma}_{uu}\mathbf{s} + \mathbf{t}'\boldsymbol{\Sigma}_{vv}\mathbf{t},$$

using $\boldsymbol{\Sigma}_{uv} = \boldsymbol{\Sigma}_{vu}' = \mathbf{0}$. The joint MGF factors:

$$M_{(\mathbf{u},\mathbf{v})}(\mathbf{s}, \mathbf{t}) = \exp\big(\mathbf{s}'\boldsymbol{\mu}_u + \tfrac{1}{2}\mathbf{s}'\boldsymbol{\Sigma}_{uu}\mathbf{s}\big) \cdot \exp\big(\mathbf{t}'\boldsymbol{\mu}_v + \tfrac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}_{vv}\mathbf{t}\big) = M_{\mathbf{u}}(\mathbf{s})M_{\mathbf{v}}(\mathbf{t}),$$

where the last step identifies the two factors as the marginal MGFs by setting $\mathbf{t} = \mathbf{0}$, respectively $\mathbf{s} = \mathbf{0}$, in the joint MGF. By fact (N2), this factorization implies independence. □

Remark 4.4. For general (non-normal) distributions, uncorrelatedness does *not* imply independence. For example, let U be uniform on $\{-1, 0, 1\}$ and $V = U^2$: then $\operatorname{cov}(U, V) = \mathsf{E}[U^3] - \mathsf{E}[U]\,\mathsf{E}[U^2] = 0$, yet $\mathsf{P}(U = 0, V = 1) = 0 \neq \mathsf{P}(U = 0)\,\mathsf{P}(V = 1) = \frac{1}{3} \cdot \frac{2}{3}$. The equivalence is a special property of the multivariate normal.

Theorem 4.5 (Chi-Square Distribution of Quadratic Forms)**.** Let $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$ and $\mathbf{A}$ be an $n \times n$ symmetric idempotent matrix with $\mathbf{A} \neq \mathbf{0}$ (so that $r = \operatorname{rank}(\mathbf{A}) \geqslant 1$, as the definition of χ^2_r in (N5) requires). Then

$$\mathbf{z}'\mathbf{A}\mathbf{z} \sim \chi^2_r, \quad \text{where } r = \operatorname{rank}(\mathbf{A}) = \operatorname{tr}(\mathbf{A}). \tag{20}$$

Proof. Step 1: Spectral decomposition. Since $\mathbf{A}$ is symmetric, there exists an orthogonal matrix $\mathbf{P}$ ($\mathbf{P}'\mathbf{P} = \mathbf{P}\mathbf{P}' = \mathbf{I}$) such that $\mathbf{A} = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}'$ ((L3)), where $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. By idempotency, each $\lambda_i \in \{0, 1\}$ (Step 1 of the proof of Lemma 2.8). The number of eigenvalues equal to 1 is $\operatorname{tr}(\boldsymbol{\Lambda}) = \operatorname{tr}(\mathbf{P}'\mathbf{A}\mathbf{P}) = \operatorname{tr}(\mathbf{A}\mathbf{P}\mathbf{P}') = \operatorname{tr}(\mathbf{A})$ ((L2)), which equals $\operatorname{rank}(\mathbf{A}) = r$ by Lemma 2.8. Without loss of generality, let the first r eigenvalues be 1 and the rest be 0:

$$\boldsymbol{\Lambda} = \operatorname{diag}(\underbrace{1, \ldots, 1}_{r}, \underbrace{0, \ldots, 0}_{n-r}).$$

Step 2: Apply orthogonal transformation. Let $\mathbf{w} = \mathbf{P}'\mathbf{z}$. Since $\mathbf{P}$ is orthogonal:

$$\mathsf{E}[\mathbf{w}] = \mathbf{P}'\mathsf{E}[\mathbf{z}] = \mathbf{0}, \qquad \operatorname{var}(\mathbf{w}) = \mathbf{P}'\mathbf{I}\mathbf{P} = \mathbf{I}_n.$$

By Theorem 4.2, $\mathbf{w} \sim N(\mathbf{0}, \mathbf{I}_n)$. Its MGF $\exp(\frac{1}{2}\mathbf{t}'\mathbf{t}) = \prod_{i=1}^{n} \exp(\frac{1}{2}t_i^2)$ is the product of n standard normal MGFs, so by fact (N2) (the n-block form of the argument in Theorem 4.3) $w_1, \ldots, w_n$ are i.i.d. $N(0,1)$.

Step 3: Evaluate the quadratic form.

$$\mathbf{z}'\mathbf{A}\mathbf{z} = \mathbf{z}'\mathbf{P}\mathbf{\Lambda}\mathbf{P}'\mathbf{z} = (\mathbf{P}'\mathbf{z})'\mathbf{\Lambda}(\mathbf{P}'\mathbf{z}) = \mathbf{w}'\mathbf{\Lambda}\mathbf{w} = \sum_{i=1}^{n} \lambda_i w_i^2 = \sum_{i=1}^{r} w_i^2.$$

Step 4: Conclude. Since $w_1, \ldots, w_r$ are independent $N(0,1)$, by definition $\sum_{i=1}^{r} w_i^2 \sim \chi_r^2$. □

Theorem 4.6 (Independence of Quadratic Forms — a Special Case of Craig's Theorem)**.** Let $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$, and let $\mathbf{A}_1, \mathbf{A}_2$ be $n \times n$ symmetric idempotent matrices with $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$. Then $\mathbf{z}'\mathbf{A}_1\mathbf{z}$ and $\mathbf{z}'\mathbf{A}_2\mathbf{z}$ are independent.

Proof. Define $\mathbf{u} = \mathbf{A}_1\mathbf{z}$ and $\mathbf{v} = \mathbf{A}_2\mathbf{z}$. Stack them into a single vector obtained from $\mathbf{z}$ by one linear map:

$$\begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{pmatrix} \mathbf{z}.$$

Since $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$, Theorem 4.2 implies $(\mathbf{u}', \mathbf{v}')'$ is *jointly* normal. Their cross-covariance is, by Lemma 3.2(ii),

$$\operatorname{cov}(\mathbf{u}, \mathbf{v}) = \mathbf{A}_1 \operatorname{var}(\mathbf{z}) \mathbf{A}_2' = \mathbf{A}_1\mathbf{I}\mathbf{A}_2 = \mathbf{A}_1\mathbf{A}_2 = \mathbf{0},$$

where $\mathbf{A}_2' = \mathbf{A}_2$ by symmetry. By Theorem 4.3, $\mathbf{u}$ and $\mathbf{v}$ are independent. By symmetric idempotency of $\mathbf{A}_1$, $\mathbf{z}'\mathbf{A}_1\mathbf{z} = \mathbf{z}'\mathbf{A}_1'\mathbf{A}_1\mathbf{z} = \|\mathbf{u}\|^2$ is a function of $\mathbf{u}$ alone; similarly $\mathbf{z}'\mathbf{A}_2\mathbf{z} = \|\mathbf{v}\|^2$ depends only on $\mathbf{v}$. The two quadratic forms are therefore independent (fact (N3)). □

Remark 4.7. In Lemma 7.2 (Section 7), Theorem 4.6 is applied with $\mathbf{A}_1 = \mathbf{H} - n^{-1}\mathbf{1}\mathbf{1}'$ and $\mathbf{A}_2 = \mathbf{M}$ to prove SSR ⫫ SSE.

5 Exact Inference Under Normality

In this section and in Sections 7 and 9, all five assumptions (A1)–(A5) are in force, with exceptions noted where they occur: the identities of Corollaries 7.6 and 7.8 are algebraic and, like Theorem 10.3, use only (A4) and $\operatorname{rank}(\mathbf{R}) = q$, and the results of §7.5 that are algebraic or concern only moments say so. Sections 6, 8 and 10, by contrast, consist of algebraic identities that hold for every realization of $\mathbf{y}$ and use only the standing assumption (A4) (Theorem 8.3 also uses (A2), and the statements about expectations and variances at the end of Section 8 use (A1)–(A3) as stated there); this is restated at the start of each of those sections. Section 11 is likewise algebraic except where a distribution is asserted.

5.1 Sampling Distribution of $\hat{\boldsymbol{\beta}}$

Theorem 5.1 (Normality of $\hat{\boldsymbol{\beta}}$)**.** Under (A1)–(A5):

$$\boxed{\hat{\boldsymbol{\beta}} \sim N\big(\boldsymbol{\beta},\ \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\big)\,.} \tag{21}$$

Proof. From (12), $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$. This is an affine transformation of $\boldsymbol{\varepsilon}$. Under (A5), $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$. By Theorem 4.2 with $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, $\mathbf{z} = \boldsymbol{\varepsilon}$, $\mathbf{b} = \boldsymbol{\beta}$:

$$\begin{aligned} \hat{\boldsymbol{\beta}} &\sim N\big(\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \cdot \mathbf{0},\ (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\big) \\ &= N\big(\boldsymbol{\beta},\ \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\big). \end{aligned}$$

□

5.2 Distribution of SSE/σ^2 and Independence from $\hat{\boldsymbol{\beta}}$

Theorem 5.2 (Chi-Square Distribution of SSE/σ^2)**.** Under (A1)–(A5):

$$\frac{\mathrm{SSE}}{\sigma^2} \sim \chi^2_{n-k-1}, \tag{22}$$

and SSE is independent of $\hat{\boldsymbol{\beta}}$.

Proof. Step 1: Standardize. Let $\mathbf{z} = \boldsymbol{\varepsilon}/\sigma$. Under (A5), $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$. Since $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$:

$$\frac{\mathrm{SSE}}{\sigma^2} = \frac{\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}}{\sigma^2} = \mathbf{z}'\mathbf{M}\mathbf{z}.$$

Step 2: Apply Theorem 4.5. The matrix $\mathbf{M}$ is symmetric and idempotent (Theorem 2.9(i)), with $\mathrm{tr}(\mathbf{M}) = n - k - 1$ (Theorem 2.9(iv)). Hence $\mathbf{z}'\mathbf{M}\mathbf{z} \sim \chi^2_{n-k-1}$.

Step 3: Prove independence. Stack $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$ and $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ into a single vector obtained from $\boldsymbol{\varepsilon}$ by one linear map:

$$\begin{pmatrix} \hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \\ \mathbf{e} \end{pmatrix} = \begin{pmatrix} (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ \mathbf{M} \end{pmatrix} \boldsymbol{\varepsilon}.$$

Since $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$, Theorem 4.2 makes this stacked vector *jointly* normal (individual normality alone would not suffice; joint normality requires that both pieces come from a single normal vector). Their cross-covariance is, by Lemma 3.2(ii),

$$\begin{aligned} \mathrm{cov}\big((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon},\ \mathbf{M}\boldsymbol{\varepsilon}\big) &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\mathrm{var}(\boldsymbol{\varepsilon})\mathbf{M}' \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{M} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{M} = \mathbf{0}, \end{aligned}$$

using $\mathbf{X}'\mathbf{M} = (\mathbf{M}\mathbf{X})' = \mathbf{0}$ (Theorem 2.9(ii)). By Theorem 4.3, $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$ and $\mathbf{e}$ are independent. Since $\hat{\boldsymbol{\beta}}$ is $(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ plus a constant and $\mathrm{SSE} = \mathbf{e}'\mathbf{e}$ is a function of $\mathbf{e}$, $\hat{\boldsymbol{\beta}}$ is independent of $\mathbf{e}$ (hence of SSE). □

5.3 OLS as Maximum Likelihood Under Normality

Theorem 5.3 (OLS Equals MLE Under Normality)**.** Under (A1)–(A5), the OLS estimator $\hat{\boldsymbol{\beta}}$ is also the maximum likelihood estimator of $\boldsymbol{\beta}$. The MLE of σ^2 is $\hat{\sigma}^2_{\mathrm{MLE}} = \mathrm{SSE}/n$, which is biased.

Proof. Under (A5), $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$, so by Theorem 4.2 applied to the affine transformation $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$. Since $\sigma^2\mathbf{I}$ is positive definite, $\mathbf{y}$ has the density of fact (N4) with $|\sigma^2\mathbf{I}| = (\sigma^2)^n$, and the log-likelihood is:

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

Maximizing over $\boldsymbol{\beta}$: In $\ell(\boldsymbol{\beta}, \sigma^2)$, the first two terms do not involve $\boldsymbol{\beta}$, and the last term is $-(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})/(2\sigma^2)$ with $\sigma^2 > 0$ treated as fixed. Since $-1/(2\sigma^2) < 0$, maximizing ℓ over $\boldsymbol{\beta}$ is equivalent to minimizing $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = Q(\boldsymbol{\beta})$. This is exactly the OLS problem, so $\hat{\boldsymbol{\beta}}_{\mathrm{MLE}} = \hat{\boldsymbol{\beta}}_{\mathrm{OLS}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.

Maximizing over σ^2: Substituting $\hat{\boldsymbol{\beta}}$ and noting that $(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathrm{SSE}$, the profile log-likelihood becomes

$$\ell(\hat{\boldsymbol{\beta}}, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{\mathrm{SSE}}{2\sigma^2}.$$

Differentiating with respect to σ^2:

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\mathrm{SSE}}{2(\sigma^2)^2} = 0 \implies \hat{\sigma}^2_{\mathrm{MLE}} = \frac{\mathrm{SSE}}{n}.$$

Second-order condition: Writing $u = \sigma^2$ for brevity,

$$\frac{\partial^2 \ell}{\partial u^2} = \frac{n}{2u^2} - \frac{\mathrm{SSE}}{u^3}.$$

At the critical point $\hat{u} = \mathrm{SSE}/n$, so $u^2 = \mathrm{SSE}^2/n^2$ and $u^3 = \mathrm{SSE}^3/n^3$:

$$\left.\frac{\partial^2 \ell}{\partial u^2}\right|_{\hat{u}} = \frac{n \cdot n^2}{2\mathrm{SSE}^2} - \frac{\mathrm{SSE} \cdot n^3}{\mathrm{SSE}^3} = \frac{n^3}{2\mathrm{SSE}^2} - \frac{n^3}{\mathrm{SSE}^2} = -\frac{n^3}{2\mathrm{SSE}^2} < 0,$$

confirming a local maximum. It is the global maximum over $u \in (0, \infty)$: here $\mathrm{SSE} > 0$ (this holds with probability one, since $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$ by Theorem 5.2 is a continuous distribution; if $\mathrm{SSE} = 0$ the likelihood is unbounded and no MLE of σ^2 exists), so $\ell(\hat{\boldsymbol{\beta}}, u) \to -\infty$ both as $u \to 0^+$ (the term $-\mathrm{SSE}/(2u)$ dominates) and as $u \to \infty$ (the term $-\frac{n}{2} \ln u$ dominates). A continuous function on $(0, \infty)$ that tends to $-\infty$ at both ends attains its supremum at a critical point, and $\hat{u}$ is the only one. Finally, since for every fixed u the maximizer over $\boldsymbol{\beta}$ is $\hat{\boldsymbol{\beta}}$ (the unique global minimizer of Q, Theorem 2.4), $(\hat{\boldsymbol{\beta}}, \hat{u})$ maximizes ℓ jointly.

Bias of $\hat{\sigma}^2_{\mathrm{MLE}}$. Using Theorem 3.7 ($\mathsf{E}[\mathrm{SSE}] = (n-k-1)\sigma^2$):

$$\mathsf{E}[\hat{\sigma}^2_{\mathrm{MLE}}] = \mathsf{E}\left[\frac{\mathrm{SSE}}{n}\right] = \frac{(n-k-1)\sigma^2}{n} = \frac{n-k-1}{n}\sigma^2 \;<\; \sigma^2,$$

so $\hat{\sigma}^2_{\mathrm{MLE}}$ underestimates σ^2 on average by a factor of $(n-k-1)/n$. The unbiased estimator is $\mathrm{MSE} = \mathrm{SSE}/(n-k-1)$, which rescales by this factor. □

5.4 Distribution of the Residuals

Corollary 5.4 (Normality of the Residuals)**.** Under (A1)–(A5), $\mathbf{e} \sim N(\mathbf{0}, \sigma^2\mathbf{M})$, and for every i with $h_{ii} < 1$,

$$\frac{e_i}{\sigma\sqrt{1-h_{ii}}} \sim N(0,1).$$

Proof. By Remark 2.11, $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$, so Theorem 4.2 with $\mathbf{A} = \mathbf{M}$ gives $\mathbf{e} \sim N(\mathbf{0}, \mathbf{M}(\sigma^2\mathbf{I})\mathbf{M}') = N(\mathbf{0}, \sigma^2\mathbf{M})$. Applying Theorem 4.2 again with $\mathbf{A} = \boldsymbol{\iota}_i'/(\sigma\sqrt{1-h_{ii}})$, the variance of $e_i/(\sigma\sqrt{1-h_{ii}})$ is $\sigma^2(1-h_{ii})/(\sigma^2(1-h_{ii})) = 1$ and its mean is 0. □

Remark 5.5. The standardized residual of Definition 3.11 is the $N(0,1)$ variable of Corollary 5.4 multiplied by σ/s. It is neither exactly normal nor exactly t-distributed, because it is bounded: $e_i = \boldsymbol{\iota}_i'\mathbf{e} = \boldsymbol{\iota}_i'\mathbf{M}\mathbf{e} = (\mathbf{M}\boldsymbol{\iota}_i)'\mathbf{e}$ (as $\mathbf{M}\mathbf{e} = \mathbf{M}^2\mathbf{y} = \mathbf{e}$), so Lemma 2.16 gives $e_i^2 \leqslant \|\mathbf{M}\boldsymbol{\iota}_i\|^2\|\mathbf{e}\|^2 = \boldsymbol{\iota}_i'\mathbf{M}\boldsymbol{\iota}_i \cdot \mathrm{SSE} = (1-h_{ii})\mathrm{SSE}$, hence $\tilde{e}_i^2 \leqslant \mathrm{SSE}/s^2 = n-k-1$. A bounded variable is not $N(0,1)$, since $\mathsf{P}(Z > c) > 0$ for $Z \sim N(0,1)$ and every $c > 0$; nor is it t_ν, since for $T = Z/\sqrt{V/\nu}$ as in (N5) we have $T \geqslant Z$ on $\{Z > 0,\ V \leqslant \nu\}$, so $\mathsf{P}(T > c) \geqslant \mathsf{P}(Z > c)\,\mathsf{P}(V \leqslant \nu) > 0$, where $\mathsf{P}(V \leqslant \nu) \geqslant \mathsf{P}(|W_1| \leqslant 1, \ldots, |W_\nu| \leqslant 1) > 0$. The cutoff $|\tilde{e}_i| > 2$ is a rule of thumb, which we do not prove: when $n-k-1$ is large, s is close to σ with high probability, $\tilde{e}_i$ then behaves approximately like the $N(0,1)$ variable of Corollary 5.4, and the cutoff flags, in a correctly specified model, roughly the 5% of observations in the two tails of the standard normal distribution.

6 Sum of Squares and the Coefficient of Determination

The results of this section are algebraic identities about the fitted values and residuals of a given data set. They use only the standing assumption (A4) (applied to every design matrix involved) and, where stated, the presence of an intercept column; none of (A1)–(A3) or (A5) is used. (Theorem 6.10 refers to the t statistic of Section 7 only as a name for an algebraic ratio.)

6.1 Variance Decomposition

Definition 6.1 (Centering Matrix)**.** Let $\mathbf{J} = \frac{1}{n}\mathbf{1}\mathbf{1}' = \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}'$, the hat matrix (Definition 2.6) of the intercept-only design $\mathbf{X} = \mathbf{1}$; by Remark 2.10 it is the orthogonal projection onto $\mathcal{C}(\mathbf{1})$, and $\mathbf{J}\mathbf{v} = \bar{v}\mathbf{1}$ for any vector $\mathbf{v}$, where $\bar{v} = \frac{1}{n}\mathbf{1}'\mathbf{v}$. The **centering matrix** is the corresponding residual maker $\mathbf{M}_0 = \mathbf{I}_n - \mathbf{J} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}'$. For any vector $\mathbf{v}$, $\mathbf{M}_0\mathbf{v} = \mathbf{v} - \bar{v}\mathbf{1}$.

Remark 6.2 (Notation: $\mathbf{M}_0$ versus $\mathbf{M}$)**.** The subscript "0" in $\mathbf{M}_0$ signals that it is the residual maker with respect to the *intercept only* (the single column $\mathbf{1}$), whereas $\mathbf{M} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ from Definition 2.6 is the residual maker with respect to the *full* design matrix $\mathbf{X}$. Conceptually,

$$\mathbf{M}_0 \text{ centers a vector,} \qquad \mathbf{M} \text{ regresses out all of } \mathbf{X}.$$

Both are symmetric idempotent, but they project onto different subspaces:

$$\mathbf{M}_0 : \mathbb{R}^n \to \mathbf{1}^\perp, \qquad \mathbf{M} : \mathbb{R}^n \to \mathcal{C}(\mathbf{X})^\perp.$$

If the model includes an intercept then $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, hence $\mathcal{C}(\mathbf{X})^\perp \subseteq \mathbf{1}^\perp$, and so $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0\mathbf{M} = \mathbf{M}$ (this follows from the projection identity $\mathbf{H}\mathbf{J} = \mathbf{J}\mathbf{H} = \mathbf{J}$ proved in Lemma 7.2, Step 2, by taking complements). Moreover, $\mathbf{M}$ and $\mathbf{M}_0$ are not equal, nor is one a simple scalar multiple of the other; they coincide only in the trivial case $\mathbf{X} = \mathbf{1}$. Indeed, $\mathbf{M} \neq \mathbf{0}$ because $\operatorname{tr}(\mathbf{M}) = n - k - 1 \geqslant 1$ (Theorem 2.9(iv)), so $\mathbf{M} = c\mathbf{M}_0$ forces $c \neq 0$, and idempotency of both gives $c^2\mathbf{M}_0 = c\mathbf{M}_0$, hence $c = 1$; then $\mathcal{C}(\mathbf{X})^\perp = \mathbf{1}^\perp$, i.e. $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{1})$, which by (A4) means $k = 0$, i.e. $\mathbf{X} = \mathbf{1}$.

Remark 6.3. $\mathbf{M}_0$ is symmetric and idempotent with $\operatorname{tr}(\mathbf{M}_0) = n - 1$: it is the residual maker $\mathbf{M}$ of the intercept-only design $\mathbf{X} = \mathbf{1}$, so Theorem 2.9(i) and (iv) apply with $k = 0$.

Theorem 6.4 (Variance Decomposition)**.** In the model with an intercept, the sums of squares introduced in Definition 3.1 satisfy:

$$\boxed{\text{SST} = \text{SSR} + \text{SSE},} \tag{23}$$

with matrix expressions:

- $\text{SST} = \sum(y_i - \bar{y})^2 = \mathbf{y}'\mathbf{M}_0\mathbf{y}$ (Total Sum of Squares),
- $\text{SSR} = \sum(\hat{y}_i - \bar{y})^2 = \hat{\mathbf{y}}'\mathbf{M}_0\hat{\mathbf{y}}$ (Regression Sum of Squares),
- $\text{SSE} = \sum(y_i - \hat{y}_i)^2 = \mathbf{e}'\mathbf{e}$ (Error/Residual Sum of Squares).

Proof. Step 1: Verify the matrix expressions for SST **and** SSR**.** For any vector $\mathbf{v} \in \mathbb{R}^n$, Definition 6.1 gives $\mathbf{M}_0\mathbf{v} = \mathbf{v} - \bar{v}\mathbf{1}$, so

$$\mathbf{v}'\mathbf{M}_0\mathbf{v} = (\mathbf{M}_0\mathbf{v})'(\mathbf{M}_0\mathbf{v}) = \|\mathbf{v} - \bar{v}\mathbf{1}\|^2 = \sum_{i=1}^{n}(v_i - \bar{v})^2,$$

using $\mathbf{M}_0'\mathbf{M}_0 = \mathbf{M}_0^2 = \mathbf{M}_0$ (symmetric idempotent). Applying this with $\mathbf{v} = \mathbf{y}$ gives $\text{SST} = \mathbf{y}'\mathbf{M}_0\mathbf{y}$; with $\mathbf{v} = \hat{\mathbf{y}}$ gives $\text{SSR} = \hat{\mathbf{y}}'\mathbf{M}_0\hat{\mathbf{y}}$, provided $\bar{\hat{y}} = \bar{y}$—which follows from $\mathbf{1}'\mathbf{e} = 0$ (Theorem 2.12(ii)) since $\mathbf{1}'\hat{\mathbf{y}} = \mathbf{1}'(\mathbf{y} - \mathbf{e}) = \mathbf{1}'\mathbf{y}$.

Step 2: Decompose $\mathbf{M}_0\mathbf{y}$ into orthogonal components. Write

$$\mathbf{M}_0\mathbf{y} = \mathbf{M}_0(\hat{\mathbf{y}} + \mathbf{e}) = \mathbf{M}_0\hat{\mathbf{y}} + \mathbf{M}_0\mathbf{e} = \mathbf{M}_0\hat{\mathbf{y}} + \mathbf{e}, \tag{24}$$

where the last equality uses $\mathbf{M}_0\mathbf{e} = \mathbf{e} - \bar{e}\mathbf{1} = \mathbf{e}$ because $\bar{e} = n^{-1}\mathbf{1}'\mathbf{e} = 0$ (Theorem 2.12(ii)).

Step 3: Expand SST**.** Using Step 2 and $\mathbf{M}_0^2 = \mathbf{M}_0$:

$$\begin{aligned}\text{SST} = \mathbf{y}'\mathbf{M}_0\mathbf{y} = (\mathbf{M}_0\mathbf{y})'(\mathbf{M}_0\mathbf{y}) &= (\mathbf{M}_0\hat{\mathbf{y}} + \mathbf{e})'(\mathbf{M}_0\hat{\mathbf{y}} + \mathbf{e}) \\ &= \text{SSR} + 2\mathbf{e}'\mathbf{M}_0\hat{\mathbf{y}} + \text{SSE}.\end{aligned}$$

Step 4: The cross term vanishes. Using $\mathbf{e}'\hat{\mathbf{y}} = 0$ and $\mathbf{e}'\mathbf{1} = 0$ (Theorem 2.12(ii), (iii)):

$$\mathbf{e}'\mathbf{M}_0\hat{\mathbf{y}} = \mathbf{e}'\hat{\mathbf{y}} - \bar{y}\,\mathbf{e}'\mathbf{1} = 0.$$

Hence SST = SSR + SSE. □

Remark 6.5 (Geometric view)**.** Equation (24) is an orthogonal decomposition of $\mathbf{M}_0\mathbf{y}$ into two pieces: the centered fitted values $\mathbf{M}_0\hat{\mathbf{y}}$, which lie in the subspace $\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$ (the non-intercept directions of the column space), and the residuals $\mathbf{e}$, which lie in $\mathcal{C}(\mathbf{X})^\perp$. Pythagoras applied to these two orthogonal pieces is precisely SST = SSR + SSE.

6.2 Coefficient of Determination

Definition 6.6 (R^2 and Adjusted R^2)**.** Assume the model has an intercept and SST > 0, i.e. $\mathbf{y}$ is not a constant vector (SST $= \|\mathbf{M}_0\mathbf{y}\|^2 = 0$ iff $\mathbf{y} = \bar{y}\mathbf{1}$). The **coefficient of determination** is

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}. \tag{25}$$

It measures the proportion of total variation in $\mathbf{y}$ explained by the regression. The **adjusted** R^2 is

$$\bar{R}^2 = 1 - \frac{\text{SSE}/(n-k-1)}{\text{SST}/(n-1)} = 1 - \frac{n-1}{n-k-1}(1-R^2) = 1 - \frac{\text{MSE}}{s_y^2}, \tag{26}$$

where $s_y^2 = \text{SST}/(n-1)$. The adjusted R^2 can be negative.

Proposition 6.7 (Bounds on R^2)**.** In any model with an intercept and SST > 0, $0 \leqslant R^2 \leqslant 1$.

Proof. $R^2 = \text{SSR}/\text{SST} \geqslant 0$ because $\text{SSR} = \sum(\hat{y}_i - \bar{y})^2 \geqslant 0$ and $\text{SST} = \sum(y_i - \bar{y})^2 > 0$ by hypothesis. For the upper bound, Theorem 6.4 gives SST = SSR + SSE with SSE $\geqslant 0$, so SSR $\leqslant$ SST, i.e., $R^2 \leqslant 1$. □

Theorem 6.8 (Monotonicity of R^2)**.** Adding a regressor to the model cannot decrease R^2. That is, if R_k^2 is the coefficient of determination with k regressors and R_{k+1}^2 is that with an additional regressor, then $R_{k+1}^2 \geqslant R_k^2$.

Proof. Let $\mathcal{C}_k \subseteq \mathcal{C}_{k+1}$ denote the column spaces of the smaller and larger design matrices (the inclusion holds because every column of $\mathbf{X}_k$ is also a column of $\mathbf{X}_{k+1}$). Write $\hat{\mathbf{y}}_k, \hat{\mathbf{y}}_{k+1}$ for the corresponding OLS fits.

By Theorem 2.5, $\hat{\mathbf{y}}_{k+1}$ is the point in $\mathcal{C}_{k+1}$ closest to $\mathbf{y}$, so $\|\mathbf{y} - \hat{\mathbf{y}}_{k+1}\| \leqslant \|\mathbf{y} - \mathbf{v}\|$ for every $\mathbf{v} \in \mathcal{C}_{k+1}$. In particular, $\hat{\mathbf{y}}_k \in \mathcal{C}_k \subseteq \mathcal{C}_{k+1}$, so

$$\text{SSE}_{k+1} = \|\mathbf{y} - \hat{\mathbf{y}}_{k+1}\|^2 \;\leqslant\; \|\mathbf{y} - \hat{\mathbf{y}}_k\|^2 = \text{SSE}_k.$$

Since SST does not depend on the regressors,

$$R_{k+1}^2 = 1 - \frac{\text{SSE}_{k+1}}{\text{SST}} \;\geqslant\; 1 - \frac{\text{SSE}_k}{\text{SST}} = R_k^2.$$ □

Lemma 6.9 (Reduction in SSE from adding one regressor)**.** Write the enlarged design as $\mathbf{X}_{k+1} = [\mathbf{X}_k \;\; \mathbf{x}]$, where $\mathbf{x}$ is the new regressor column, and assume $\mathbf{x}'\mathbf{M}_k\mathbf{x} > 0$ (where $\mathbf{M}_k$ is the residual maker for $\mathbf{X}_k$), which holds whenever $\mathbf{x}$ is not in $\mathcal{C}(\mathbf{X}_k)$. Then the OLS coefficient on $\mathbf{x}$ in the enlarged model is

$$\hat{\beta}_{k+1} = \frac{\mathbf{x}'\mathbf{M}_k\mathbf{y}}{\mathbf{x}'\mathbf{M}_k\mathbf{x}}, \tag{27}$$

and the reduction in residual sum of squares is

$$\text{SSE}_k - \text{SSE}_{k+1} = \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x} = \frac{(\mathbf{x}'\mathbf{M}_k\mathbf{y})^2}{\mathbf{x}'\mathbf{M}_k\mathbf{x}} \;\geqslant\; 0. \tag{28}$$

This identity uses only normal equations (no forward dependence on Section 8).

Proof. The partitioned normal equations $\mathbf{X}'_{k+1}\mathbf{X}_{k+1}\hat{\boldsymbol{\beta}}_{k+1} = \mathbf{X}'_{k+1}\mathbf{y}$ read

$$\mathbf{X}'_k\mathbf{X}_k\hat{\boldsymbol{\beta}}^*_k + \mathbf{X}'_k\mathbf{x}\hat{\beta}_{k+1} = \mathbf{X}'_k\mathbf{y}, \tag{a}$$
$$\mathbf{x}'\mathbf{X}_k\hat{\boldsymbol{\beta}}^*_k + \mathbf{x}'\mathbf{x}\,\hat{\beta}_{k+1} = \mathbf{x}'\mathbf{y}, \tag{b}$$

where $\hat{\boldsymbol{\beta}}^*_k$ denotes the $\mathbf{X}_k$-block of the enlarged estimator (its first $k+1$ coordinates, the coefficients on the columns of $\mathbf{X}_k$; generally different from the short-regression estimator $\hat{\boldsymbol{\beta}}_k$). From (a), $\hat{\boldsymbol{\beta}}^*_k = (\mathbf{X}'_k\mathbf{X}_k)^{-1}\mathbf{X}'_k(\mathbf{y} - \mathbf{x}\hat{\beta}_{k+1})$. Substituting into (b):

$$\mathbf{x}'\mathbf{X}_k(\mathbf{X}'_k\mathbf{X}_k)^{-1}\mathbf{X}'_k(\mathbf{y} - \mathbf{x}\hat{\beta}_{k+1}) + \mathbf{x}'\mathbf{x}\,\hat{\beta}_{k+1} = \mathbf{x}'\mathbf{y}.$$

Writing $\mathbf{H}_k = \mathbf{X}_k(\mathbf{X}'_k\mathbf{X}_k)^{-1}\mathbf{X}'_k = \mathbf{I} - \mathbf{M}_k$ and rearranging:

$$\mathbf{x}'\mathbf{H}_k\mathbf{y} - \mathbf{x}'\mathbf{H}_k\mathbf{x}\,\hat{\beta}_{k+1} + \mathbf{x}'\mathbf{x}\,\hat{\beta}_{k+1} = \mathbf{x}'\mathbf{y} \iff \mathbf{x}'(\mathbf{I} - \mathbf{H}_k)\mathbf{x}\,\hat{\beta}_{k+1} = \mathbf{x}'(\mathbf{I} - \mathbf{H}_k)\mathbf{y},$$

which gives (27).

For (28), relate $\hat{\boldsymbol{\beta}}^*_k$ to the short-regression estimator $\hat{\boldsymbol{\beta}}_k = (\mathbf{X}'_k\mathbf{X}_k)^{-1}\mathbf{X}'_k\mathbf{y}$: from (a),

$$\hat{\boldsymbol{\beta}}^*_k = \hat{\boldsymbol{\beta}}_k - (\mathbf{X}'_k\mathbf{X}_k)^{-1}\mathbf{X}'_k\mathbf{x}\,\hat{\beta}_{k+1}.$$

Multiplying by $\mathbf{X}_k$ gives $\mathbf{X}_k\hat{\boldsymbol{\beta}}^*_k = \hat{\mathbf{y}}_k - \mathbf{H}_k\mathbf{x}\,\hat{\beta}_{k+1}$, so

$$\hat{\mathbf{y}}_{k+1} = \mathbf{X}_k\hat{\boldsymbol{\beta}}^*_k + \mathbf{x}\hat{\beta}_{k+1} = \hat{\mathbf{y}}_k + (\mathbf{I} - \mathbf{H}_k)\mathbf{x}\,\hat{\beta}_{k+1} = \hat{\mathbf{y}}_k + \mathbf{M}_k\mathbf{x}\,\hat{\beta}_{k+1},$$

and the residuals satisfy $\mathbf{e}_{k+1} = \mathbf{e}_k - \mathbf{M}_k\mathbf{x}\,\hat{\beta}_{k+1}$. Taking squared norms, using $\mathbf{e}_k = \mathbf{M}_k\mathbf{y}$ and $\mathbf{M}_k^2 = \mathbf{M}_k$:

$$\mathrm{SSE}_{k+1} = \mathrm{SSE}_k - 2\hat{\beta}_{k+1}\cdot\mathbf{y}'\mathbf{M}_k\mathbf{x} + \hat{\beta}^2_{k+1}\cdot\mathbf{x}'\mathbf{M}_k\mathbf{x}.$$

Substituting $\hat{\beta}_{k+1} = (\mathbf{x}'\mathbf{M}_k\mathbf{y})/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$ makes the middle term $-2\hat{\beta}^2_{k+1}\cdot\mathbf{x}'\mathbf{M}_k\mathbf{x}$, so

$$\mathrm{SSE}_{k+1} = \mathrm{SSE}_k - \hat{\beta}^2_{k+1}\cdot\mathbf{x}'\mathbf{M}_k\mathbf{x},$$

which is (28). The non-negativity reflects Theorem 6.8. □

Theorem 6.10 ($\bar{R}^2$ Increases iff $|t| > 1$)**.** Let $\mathbf{X}_{k+1} = [\mathbf{X}_k\ \ \mathbf{x}]$ as in Lemma 6.9, with $\mathbf{x} \notin \mathcal{C}(\mathbf{X}_k)$ (so that $\mathbf{x}'\mathbf{M}_k\mathbf{x} > 0$ and $\mathbf{X}_{k+1}$ has full column rank), and assume $n > k+2$, $\mathrm{SST} > 0$ and $\mathrm{SSE}_{k+1} > 0$, so that both adjusted R^2's and the t statistic below are defined. Then adding the regressor $\mathbf{x}$ increases $\bar{R}^2$ if and only if its t-statistic in the enlarged model exceeds 1 in absolute value.

Proof. Let the model with k regressors have $\mathrm{MSE}_k = \mathrm{SSE}_k/(n-k-1)$, and the model with an additional regressor have $\mathrm{MSE}_{k+1} = \mathrm{SSE}_{k+1}/(n-k-2)$. Since $\bar{R}^2 = 1 - \mathrm{MSE}/s_y^2$ with the same $s_y^2 = \mathrm{SST}/(n-1) > 0$ in both models, $\bar{R}^2_{k+1} > \bar{R}^2_k$ if and only if $\mathrm{MSE}_{k+1} < \mathrm{MSE}_k$, i.e.,

$$\frac{\mathrm{SSE}_{k+1}}{n-k-2} < \frac{\mathrm{SSE}_k}{n-k-1}. \tag{29}$$

Step 1: Apply Lemma 6.9. By (28), $\mathrm{SSE}_k = \mathrm{SSE}_{k+1} + \hat{\beta}^2_{k+1}\cdot\mathbf{x}'\mathbf{M}_k\mathbf{x}$. Substitute this into (29):

$$\frac{\mathrm{SSE}_{k+1}}{n-k-2} < \frac{\mathrm{SSE}_{k+1} + \hat{\beta}^2_{k+1}\cdot\mathbf{x}'\mathbf{M}_k\mathbf{x}}{n-k-1}$$
$$(n-k-1)\mathrm{SSE}_{k+1} < (n-k-2)\big[\mathrm{SSE}_{k+1} + \hat{\beta}^2_{k+1}\cdot\mathbf{x}'\mathbf{M}_k\mathbf{x}\big]$$
$$\mathrm{SSE}_{k+1} < (n-k-2)\cdot\hat{\beta}^2_{k+1}\cdot\mathbf{x}'\mathbf{M}_k\mathbf{x}.$$

Dividing by $\mathrm{SSE}_{k+1} = (n-k-2)\mathrm{MSE}_{k+1} > 0$:

$$1 < \frac{\hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}}{\mathrm{MSE}_{k+1}}. \tag{30}$$

Step 2: Identify the right-hand side as t^2. The t-statistic for $\hat{\beta}_{k+1}$ in the enlarged model is $t_{k+1} = \hat{\beta}_{k+1}/\mathrm{SE}(\hat{\beta}_{k+1})$ with $\mathrm{SE}(\hat{\beta}_{k+1})^2 = \mathrm{MSE}_{k+1} \cdot [(\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}]_{k+1,k+1}$ (defined formally in Theorem 7.1; here $[\cdot]_{k+1,k+1}$ is the diagonal entry belonging to β_{k+1}, the last one; the subscript of the statistic t_{k+1}, like that of t_j in Corollary 7.17, is the index of the coefficient tested, whereas the subscript of the distribution t_ν is its degrees of freedom). We claim

$$[(\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}]_{k+1,k+1} = (\mathbf{x}'\mathbf{M}_k\mathbf{x})^{-1}. \tag{31}$$

Indeed, let $\mathbf{G} = (\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}\mathbf{X}'_{k+1}$, so that $\hat{\boldsymbol{\beta}}_{k+1} = \mathbf{G}\mathbf{y}$ for every $\mathbf{y} \in \mathbb{R}^n$. Lemma 6.9 says that the last coordinate of $\mathbf{G}\mathbf{y}$ is $\mathbf{x}'\mathbf{M}_k\mathbf{y}/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$ for every $\mathbf{y}$, so the last row of $\mathbf{G}$ is $\mathbf{g}' := \mathbf{x}'\mathbf{M}_k/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$. On the other hand $\mathbf{G}\mathbf{G}' = (\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}\mathbf{X}'_{k+1}\mathbf{X}_{k+1}(\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1} = (\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}$, whose last diagonal entry is the squared norm of the last row of $\mathbf{G}$:

$$[(\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}]_{k+1,k+1} = \mathbf{g}'\mathbf{g} = \frac{\mathbf{x}'\mathbf{M}_k\mathbf{M}_k\mathbf{x}}{(\mathbf{x}'\mathbf{M}_k\mathbf{x})^2} = \frac{1}{\mathbf{x}'\mathbf{M}_k\mathbf{x}},$$

using $\mathbf{M}'_k = \mathbf{M}_k = \mathbf{M}_k^2$. (Equivalently: $\mathrm{var}(\hat{\beta}_{k+1})$ computed from (27) via Lemma 3.2 is $\sigma^2/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$, and computed from Theorem 3.4 it is $\sigma^2[(\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}]_{k+1,k+1}$; this is the block-inverse, or Schur complement, formula for the last diagonal entry, obtained here from Lemma 6.9 alone.) Hence $\mathrm{SE}(\hat{\beta}_{k+1})^2 = \mathrm{MSE}_{k+1}/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$, so

$$t_{k+1}^2 = \frac{\hat{\beta}_{k+1}^2}{\mathrm{SE}(\hat{\beta}_{k+1})^2} = \frac{\hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}}{\mathrm{MSE}_{k+1}}.$$

Substituting into (30) gives $t_{k+1}^2 > 1$, i.e., $|t_{k+1}| > 1$. □

Remark 6.11 (Logical status of the forward reference)**.** The proof refers to one later statement: the form of $\mathrm{SE}(\hat{\beta}_{k+1})$ in Theorem 7.1, which is a *definition* of the t statistic rather than a result whose proof is needed here. It is used only to *interpret* the ratio $\hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}/\mathrm{MSE}_{k+1}$ as t_{k+1}^2; equation (30) and the identity (31) are self-contained algebraic consequences of Lemma 6.9. A reader seeking strict linearity can defer Theorem 6.10 until after Section 7.

6.3 R^2 as a Squared Correlation

Proposition 6.12 (R^2 as a Squared Correlation)**.** In a model with an intercept and $\mathrm{SST} > 0$:

(i) if $\mathrm{SSR} > 0$, then $\hat{\mathbf{y}}$ is not constant and $R^2 = \mathrm{corr}(\mathbf{y}, \hat{\mathbf{y}})^2$;

(ii) $\mathrm{corr}(\mathbf{y}, \mathbf{v})^2 \leqslant R^2$ for every non-constant $\mathbf{v} \in \mathcal{C}(\mathbf{X})$; by (i), equality holds for $\mathbf{v} = \hat{\mathbf{y}}$ when $\mathrm{SSR} > 0$.

Thus $\hat{\mathbf{y}}$ is the linear combination of the regressors that is most highly correlated with $\mathbf{y}$, and R^2 is the square of that correlation.

Proof. Step 1: Centred cross-products. For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, Definition 6.1 gives $\mathbf{M}_0\mathbf{u} = \mathbf{u} - \bar{u}\mathbf{1}$, so $S_{uv} = (\mathbf{M}_0\mathbf{u})'(\mathbf{M}_0\mathbf{v})$ in the notation of Definition 2.15. Let $\mathbf{v} \in \mathcal{C}(\mathbf{X})$. Then $\mathbf{M}_0\mathbf{v} = \mathbf{v} - \bar{v}\mathbf{1} \in \mathcal{C}(\mathbf{X})$ because $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, so $\mathbf{e}'\mathbf{M}_0\mathbf{v} = 0$ (Theorem 2.12(i)); together with $\mathbf{M}_0\mathbf{e} = \mathbf{e}$ (equation (24)) this gives

$$S_{yv} = (\mathbf{M}_0\hat{\mathbf{y}} + \mathbf{e})'\mathbf{M}_0\mathbf{v} = (\mathbf{M}_0\hat{\mathbf{y}})'(\mathbf{M}_0\mathbf{v}) = S_{\hat{y}v}.$$

Step 2: The correlation with $\hat{\mathbf{y}}$. By Theorem 6.4, $S_{\hat{y}\hat{y}} = \|\mathbf{M}_0\hat{\mathbf{y}}\|^2 = \mathrm{SSR} > 0$ and $S_{yy} = \mathrm{SST}$; in particular $\hat{\mathbf{y}}$ is not constant. Step 1 with $\mathbf{v} = \hat{\mathbf{y}}$ gives $S_{y\hat{y}} = \mathrm{SSR}$, so

$$\mathrm{corr}(\mathbf{y}, \hat{\mathbf{y}})^2 = \frac{S_{y\hat{y}}^2}{S_{yy}S_{\hat{y}\hat{y}}} = \frac{\mathrm{SSR}^2}{\mathrm{SST} \cdot \mathrm{SSR}} = \frac{\mathrm{SSR}}{\mathrm{SST}} = R^2.$$

Step 3: The maximum. Let $\mathbf{v} \in \mathcal{C}(\mathbf{X})$ be non-constant. By Step 1 and Lemma 2.16 applied to $\mathbf{M}_0\hat{\mathbf{y}}$ and $\mathbf{M}_0\mathbf{v}$,

$$S_{yv}^2 = \big((\mathbf{M}_0\hat{\mathbf{y}})'(\mathbf{M}_0\mathbf{v})\big)^2 \leqslant \|\mathbf{M}_0\hat{\mathbf{y}}\|^2\,\|\mathbf{M}_0\mathbf{v}\|^2 = \mathrm{SSR}\cdot S_{vv},$$

so $\mathrm{corr}(\mathbf{y},\mathbf{v})^2 = S_{yv}^2/(\mathrm{SST}\cdot S_{vv}) \leqslant \mathrm{SSR}/\mathrm{SST} = R^2$. □

Corollary 6.13 (R^2 in Simple Linear Regression)**.** In the simple linear regression model of §2.8, with $S_{xx} > 0$ and $\mathrm{SST} > 0$,

$$R^2 = r^2, \qquad \mathrm{SSR} = \hat{\beta}_1^2 S_{xx} = \hat{\beta}_1^2 (n-1)s_x^2,$$

and, if $n > 2$, $s_y^2 = \hat{\beta}_1^2 s_x^2 + s^2\,\dfrac{n-2}{n-1}$, where $s^2 = \mathrm{SSE}/(n-2)$ is the mean squared error of Definition 3.1.

Proof. By Corollary 2.18(iv), $\mathrm{SSR} = \sum_i(\hat{y}_i - \bar{y})^2 = \hat{\beta}_1^2\sum_i(x_i-\bar{x})^2 = \hat{\beta}_1^2 S_{xx}$. Since $S_{yy} = \mathrm{SST} > 0$, $\mathbf{y}$ is not constant, and Corollary 2.18(iii) gives $\hat{\beta}_1^2 = r^2 S_{yy}/S_{xx}$, so $\mathrm{SSR} = r^2\,\mathrm{SST}$, i.e. $R^2 = r^2$. Finally, dividing $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$ (Theorem 6.4) by $n-1$, with $\mathrm{SST}/(n-1) = s_y^2$, $\mathrm{SSR}/(n-1) = \hat{\beta}_1^2 s_x^2$ and $\mathrm{SSE} = (n-2)s^2$, gives the last identity. □

7 Hypothesis Testing and Confidence Regions

7.1 Individual Coefficient t Test

Theorem 7.1 (Individual t Test)**.** Under (A1)–(A5), for testing $H_0\colon \beta_j = \beta_j^0$:

$$t = \frac{\hat{\beta}_j - \beta_j^0}{\mathrm{SE}(\hat{\beta}_j)} \sim t_{n-k-1} \quad \text{under } H_0, \tag{32}$$

where $\mathrm{SE}(\hat{\beta}_j) = s\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$ and $s = \sqrt{\mathrm{MSE}}$.

Proof. Step 1: Marginal distribution of $\hat{\beta}_j$. By Theorem 5.1, $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$. The j-th coordinate of a multivariate normal vector is itself normal, with mean β_j and variance equal to the (j,j)-entry of the covariance matrix (this is Theorem 4.2 applied with $\mathbf{A} = \boldsymbol{\iota}_j'$, the transpose of the j-th standard basis vector):

$$\hat{\beta}_j = \boldsymbol{\iota}_j'\hat{\boldsymbol{\beta}} \sim N\big(\beta_j,\ \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}\big).$$

Standardizing:

$$Z = \frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \sim N(0,1).$$

Step 2: Chi-square quantity independent of Z. By Theorem 5.2, $V = \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$ and is independent of $\hat{\boldsymbol{\beta}}$. Since Z is a (deterministic) function of $\hat{\beta}_j$, which is a coordinate of $\hat{\boldsymbol{\beta}}$, and a function of an independent random variable remains independent, Z is independent of V.

Step 3. By definition, if $Z \sim N(0,1)$ and $V \sim \chi^2_\nu$ are independent, then $Z/\sqrt{V/\nu} \sim t_\nu$. Applying this:

$$\begin{aligned}\frac{Z}{\sqrt{V/(n-k-1)}} &= \frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \cdot \frac{1}{\sqrt{\mathrm{SSE}/[\sigma^2(n-k-1)]}} \\ &= \frac{\hat{\beta}_j - \beta_j}{\sqrt{\mathrm{MSE}}\cdot\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} = \frac{\hat{\beta}_j - \beta_j}{s\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \sim t_{n-k-1}.\end{aligned}$$

Under $H_0\colon \beta_j = \beta_j^0$, replace β_j by β_j^0. □

7.2 Overall F Test

Lemma 7.2 (Distribution of SSR/σ^2 Under H_0)**.** Let $k \geqslant 1$, and let $\mathbf{J} = \frac{1}{n}\mathbf{1}\mathbf{1}'$ be the projection onto $\mathcal{C}(\mathbf{1})$ of Definition 6.1. Under H_0: $\beta_1 = \cdots = \beta_k = 0$ (with (A1)–(A5)):

$$\frac{\text{SSR}}{\sigma^2} \sim \chi^2_k, \qquad \text{and SSR is independent of SSE.}$$

Proof. Step 1: Express SSR/σ^2 **as a quadratic form in standard normals.** Two matrix observations are useful:

- $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ by definition of the hat matrix.
- $\bar{y}\mathbf{1} = \mathbf{J}\mathbf{y}$, because $\mathbf{J}\mathbf{y} = n^{-1}\mathbf{1}(\mathbf{1}'\mathbf{y}) = n^{-1}\mathbf{1} \cdot n\bar{y} = \bar{y}\mathbf{1}$.

Therefore

$$\hat{\mathbf{y}} - \bar{y}\mathbf{1} = \mathbf{H}\mathbf{y} - \mathbf{J}\mathbf{y} = (\mathbf{H} - \mathbf{J})\mathbf{y},$$

and $\text{SSR} = (\hat{\mathbf{y}} - \bar{y}\mathbf{1})'(\hat{\mathbf{y}} - \bar{y}\mathbf{1}) = \mathbf{y}'(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J})\mathbf{y}$.

Under H_0, $\boldsymbol{\beta} = (\beta_0, 0, \ldots, 0)'$, so $\mathbf{X}\boldsymbol{\beta} = \beta_0\mathbf{1}$ and $\mathbf{y} = \beta_0\mathbf{1} + \boldsymbol{\varepsilon}$. We claim $(\mathbf{H} - \mathbf{J})\mathbf{y} = (\mathbf{H} - \mathbf{J})\boldsymbol{\varepsilon}$: indeed, since $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ we have $\mathbf{H}\mathbf{1} = \mathbf{1}$ (by Theorem 2.9(ii) applied to the first column of $\mathbf{X}$), and $\mathbf{J}\mathbf{1} = n^{-1}\mathbf{1}(\mathbf{1}'\mathbf{1}) = \mathbf{1}$, so

$$(\mathbf{H} - \mathbf{J})(\beta_0\mathbf{1}) = \beta_0(\mathbf{H}\mathbf{1} - \mathbf{J}\mathbf{1}) = \beta_0(\mathbf{1} - \mathbf{1}) = \mathbf{0},$$

and hence $(\mathbf{H} - \mathbf{J})\mathbf{y} = (\mathbf{H} - \mathbf{J})(\beta_0\mathbf{1} + \boldsymbol{\varepsilon}) = (\mathbf{H} - \mathbf{J})\boldsymbol{\varepsilon}$.

Let $\mathbf{z} = \boldsymbol{\varepsilon}/\sigma$. Under (A5), $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$. Then

$$\frac{\text{SSR}}{\sigma^2} = \frac{\boldsymbol{\varepsilon}'(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J})\boldsymbol{\varepsilon}}{\sigma^2} = \mathbf{z}'(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J})\mathbf{z}.$$

This reduces to $\mathbf{z}'(\mathbf{H} - \mathbf{J})\mathbf{z}$ once we establish in Step 2 that $\mathbf{H} - \mathbf{J}$ is symmetric idempotent, which implies $(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J}) = (\mathbf{H} - \mathbf{J})^2 = \mathbf{H} - \mathbf{J}$.

Step 2: $\mathbf{H}-\mathbf{J}$ is symmetric idempotent with rank k. Symmetry is immediate. For idempotency, we first show

$$\mathbf{H}\mathbf{J} = \mathbf{J} = \mathbf{J}\mathbf{H}. \tag{33}$$

Using $\mathbf{H}\mathbf{1} = \mathbf{1}$: $\mathbf{H}\mathbf{J} = n^{-1}\mathbf{H}\mathbf{1}\mathbf{1}' = n^{-1}\mathbf{1}\mathbf{1}' = \mathbf{J}$; by symmetry, $\mathbf{J}\mathbf{H} = (\mathbf{H}\mathbf{J})' = \mathbf{J}$. Also $\mathbf{J}^2 = n^{-2}\mathbf{1}(\mathbf{1}'\mathbf{1})\mathbf{1}' = \mathbf{J}$. Then

$$(\mathbf{H} - \mathbf{J})^2 = \mathbf{H}^2 - \mathbf{H}\mathbf{J} - \mathbf{J}\mathbf{H} + \mathbf{J}^2 = \mathbf{H} - \mathbf{J} - \mathbf{J} + \mathbf{J} = \mathbf{H} - \mathbf{J}.$$

For the rank, Lemma 2.8 gives

$$\text{rank}(\mathbf{H} - \mathbf{J}) = \text{tr}(\mathbf{H}) - \text{tr}(\mathbf{J}) = (k + 1) - 1 = k,$$

with $\text{tr}(\mathbf{J}) = n^{-1}\,\text{tr}(\mathbf{1}\mathbf{1}') = 1$.

The Step 1 expression simplifies: $(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J}) = \mathbf{H} - \mathbf{J}$, so $\text{SSR}/\sigma^2 = \mathbf{z}'(\mathbf{H} - \mathbf{J})\mathbf{z} \sim \chi^2_k$ by Theorem 4.5.

Step 3: Independence of SSR **and** SSE**.** Apply Theorem 4.6 to $\mathbf{A}_1 = \mathbf{H} - \mathbf{J}$ and $\mathbf{A}_2 = \mathbf{M}$. The required $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$ follows from $\mathbf{H}\mathbf{M} = \mathbf{0}$ (Theorem 2.9(iii)) and $\mathbf{J}\mathbf{M} = \mathbf{J} - \mathbf{J}\mathbf{H} = \mathbf{0}$ by (33):

$$(\mathbf{H} - \mathbf{J})\mathbf{M} = \mathbf{0}.$$

Hence $\text{SSR}/\sigma^2 = \mathbf{z}'(\mathbf{H} - \mathbf{J})\mathbf{z}$ and $\text{SSE}/\sigma^2 = \mathbf{z}'\mathbf{M}\mathbf{z}$ are independent, and so are SSR and SSE. □

Theorem 7.3 (Overall F Test)**.** Let $k \geqslant 1$. Under (A1)–(A5), for testing $H_0\colon \beta_1 = \beta_2 = \cdots = \beta_k = 0$ (the third expression below presupposes $\mathrm{SST} > 0$, so that R^2 is defined; under (A5) this holds with probability one, see the end of the proof):

$$\boxed{F = \frac{\mathrm{MSR}}{\mathrm{MSE}} = \frac{\mathrm{SSR}/k}{\mathrm{SSE}/(n-k-1)} = \frac{R^2/k}{(1-R^2)/(n-k-1)} \sim F_{k,\,n-k-1} \quad \text{under } H_0.} \tag{34}$$

Proof. By Lemma 7.2 and Theorem 5.2, under H_0: $\mathrm{SSR}/\sigma^2 \sim \chi^2_k$, $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$, and they are independent. By the definition of the F distribution (fact (N5)):

$$F = \frac{(\mathrm{SSR}/\sigma^2)/k}{(\mathrm{SSE}/\sigma^2)/(n-k-1)} = \frac{\mathrm{SSR}/k}{\mathrm{SSE}/(n-k-1)} = \frac{\mathrm{MSR}}{\mathrm{MSE}} \sim F_{k,n-k-1}.$$

The equivalence $F = \frac{R^2/k}{(1-R^2)/(n-k-1)}$ follows from $R^2 = \mathrm{SSR}/\mathrm{SST}$ and $1 - R^2 = \mathrm{SSE}/\mathrm{SST}$, which require $\mathrm{SST} > 0$ (Definition 6.6). Under (A5) this holds with probability one: pick a unit vector $\mathbf{v} \perp \mathbf{1}$ (one exists since $n \geqslant 2$), so that $\mathbf{M}_0\mathbf{v} = \mathbf{v}$ and hence $\mathbf{v}'\mathbf{M}_0 = \mathbf{v}'$; if $\mathrm{SST} = \|\mathbf{M}_0\mathbf{y}\|^2 = 0$ then $\mathbf{v}'\mathbf{y} = \mathbf{v}'\mathbf{M}_0\mathbf{y} = 0$, whereas $\mathbf{v}'\mathbf{y} \sim N(\mathbf{v}'\mathbf{X}\boldsymbol{\beta}, \sigma^2)$ (Theorem 4.2 applied to $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$, as in the proof of Theorem 5.3) with $\sigma^2 > 0$ by (A3) is a continuous distribution, so $\mathsf{P}(\mathbf{v}'\mathbf{y} = 0) = 0$. □

7.3 General Linear Hypothesis

Theorem 7.4 (General F Test)**.** Under (A1)–(A5), consider $H_0\colon \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ where $\mathbf{R}$ is a $q \times (k+1)$ matrix of rank q and $\mathbf{r} \in \mathbb{R}^q$. The test statistic

$$\boxed{F = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{q \cdot s^2} \sim F_{q,\,n-k-1} \quad \text{under } H_0.} \tag{35}$$

Proof. Step 1. By Theorem 5.1, $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$. By Theorem 4.2:

$$\mathbf{R}\hat{\boldsymbol{\beta}} \sim N\big(\mathbf{R}\boldsymbol{\beta},\ \sigma^2\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big).$$

Under H_0, $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$, so $\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r} \sim N(\mathbf{0}, \sigma^2\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}')$.

Step 2: A quadratic form in $\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}$ is chi-square. Let $\boldsymbol{\Sigma}_R = \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$. $\boldsymbol{\Sigma}_R$ is positive definite: for nonzero $\mathbf{v} \in \mathbb{R}^q$,

$$\mathbf{v}'\boldsymbol{\Sigma}_R\mathbf{v} = (\mathbf{R}'\mathbf{v})'(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{R}'\mathbf{v}) > 0,$$

since $\mathrm{rank}(\mathbf{R}) = q$ makes $\mathbf{R}'\mathbf{v} \neq \mathbf{0}$, and $(\mathbf{X}'\mathbf{X})^{-1}$ is positive definite (Theorem 2.4, Step 4). Let $\boldsymbol{\Sigma}_R^{1/2}$ be its positive definite symmetric square root ((L4)).

Set $\mathbf{w} = \sigma^{-1}\boldsymbol{\Sigma}_R^{-1/2}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})$. Theorem 4.2 gives $\mathbf{w} \sim N(\mathbf{0}, \mathbf{I}_q)$, so $\mathbf{w}'\mathbf{w} \sim \chi^2_q$ by Theorem 4.5:

$$W \ :=\ \mathbf{w}'\mathbf{w} = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'\boldsymbol{\Sigma}_R^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{\sigma^2} \sim \chi^2_q.$$

Step 3. By Theorem 5.2, $V = \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$, independent of $\hat{\boldsymbol{\beta}}$; since W is a function of $\hat{\boldsymbol{\beta}}$, W is independent of V. Then:

$$F = \frac{W/q}{V/(n-k-1)} = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'\boldsymbol{\Sigma}_R^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})/(q\sigma^2)}{\mathrm{SSE}/[\sigma^2(n-k-1)]} = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'\boldsymbol{\Sigma}_R^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{q \cdot s^2} \sim F_{q,n-k-1}.$$ □

Remark 7.5 (Special cases)**.** The individual t test (Theorem 7.1) is the case $q = 1$, $\mathbf{R} = \boldsymbol{\iota}_j'$ (the transpose of the j-th standard basis vector), $\mathbf{r} = \beta_j^0$, where $t^2 = F \sim F_{1,n-k-1}$. The overall F test (Theorem 7.3) is the case $\mathbf{R} = [\mathbf{0}_{k\times 1} \mid \mathbf{I}_k]$ (of rank $q = k$), $\mathbf{r} = \mathbf{0}_k$: by Corollary 7.6 below, the statistic (35) then equals $\frac{(\mathrm{SSE}_R - \mathrm{SSE}_U)/k}{\mathrm{SSE}_U/(n-k-1)}$, where the reduced model minimizes $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ over $\boldsymbol{\beta} = (\beta_0, 0, \ldots, 0)'$, i.e. $\|\mathbf{y} - \beta_0\mathbf{1}\|^2$ over β_0; its normal equation $\mathbf{1}'\mathbf{1}\,\beta_0 = \mathbf{1}'\mathbf{y}$ (Theorem 2.4 with $\mathbf{X} = \mathbf{1}$) gives $\hat{\beta}_0 = \bar{y}$, so the reduced fit is $\bar{y}\mathbf{1}$, $\mathrm{SSE}_R = \|\mathbf{y} - \bar{y}\mathbf{1}\|^2 = \mathrm{SST}$, and $\mathrm{SSE}_R - \mathrm{SSE}_U = \mathrm{SST} - \mathrm{SSE} = \mathrm{SSR}$ by Theorem 6.4. The statistic is therefore $(\mathrm{SSR}/k)/(\mathrm{SSE}/(n-k-1)) = \mathrm{MSR}/\mathrm{MSE}$, as in (34).

Corollary 7.6 (Nested F Test via Residual Sum of Squares)**.** Let SSE_U and SSE_R be the residual sums of squares from the full and reduced models under $H_0 \colon \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ (q constraints, $\mathrm{rank}(\mathbf{R}) = q$). Then the F statistic in Theorem 7.4 is equivalently expressed as

$$F = \frac{(\mathrm{SSE}_R - \mathrm{SSE}_U)/q}{\mathrm{SSE}_U/(n-k-1)}. \tag{36}$$

(This is an algebraic identity between the two expressions for F: like Theorem 10.3, on which its proof rests, it uses only (A4) and $\mathrm{rank}(\mathbf{R}) = q$. Assumptions (A1)–(A5) enter only through the distribution $F \sim F_{q,\,n-k-1}$ under H_0 in Theorem 7.4.)

Proof. By Theorem 10.3, the constrained estimator satisfies

$$\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big]^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}). \tag{37}$$

Step 1: Decompose SSE_R**.** Since $\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R = (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) + \mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R) = \mathbf{e} + \mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)$,

$$\begin{aligned}\mathrm{SSE}_R = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R\|^2 &= \|\mathbf{e}\|^2 + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\|^2 + 2\mathbf{e}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\\ &= \mathrm{SSE}_U + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\|^2,\end{aligned}$$

since the cross term vanishes: $\mathbf{e}'\mathbf{X} = \mathbf{0}'$ by Theorem 2.12(i).

Step 2: Simplify the squared norm. Let $\mathbf{a} = [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})$ denote the trailing factor (a $q \times 1$ vector) in (37), so that $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{a}$. Then

$$\begin{aligned}\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\|^2 &= (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)'\mathbf{X}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\\ &= \big[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{a}\big]'\mathbf{X}'\mathbf{X}\big[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{a}\big]\\ &= \mathbf{a}'\mathbf{R}\underbrace{(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}}_{=(\mathbf{X}'\mathbf{X})^{-1}}\mathbf{R}'\mathbf{a}\\ &= \mathbf{a}'\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{a}.\end{aligned}$$

Expanding $\mathbf{a}$:

$$\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\|^2 = (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'\big[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big]^{-1}\underbrace{\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big]^{-1}}_{=\mathbf{I}_q}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}).$$

The bracketed product collapses to $\mathbf{I}_q$, leaving

$$\mathrm{SSE}_R - \mathrm{SSE}_U = \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\|^2 = (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'\big[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big]^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}). \tag{38}$$

Step 3: Compare with Theorem 7.4. The numerator in Theorem 7.4's F statistic equals the right-hand side of (38) divided by q. Since the denominator $s^2 = \mathrm{SSE}_U/(n-k-1)$ is the same, substituting yields

$$F = \frac{(\mathrm{SSE}_R - \mathrm{SSE}_U)/q}{\mathrm{SSE}_U/(n-k-1)}.$$

□

Remark 7.7 (Logical status of the forward reference)**.** The proof relies on the constrained estimator formula (37) from Theorem 10.3 in Section 10; Theorem 10.3 is purely algebraic and does not depend on anything in this section, so no circularity arises. A reader seeking strict linearity can defer this corollary until after Section 10.

In the special case of *homogeneous* constraints, $\mathbf{r} = \mathbf{0}$, there is also a derivation without constrained OLS: the feasible set $\{\boldsymbol{\beta} : \mathbf{R}\boldsymbol{\beta} = \mathbf{0}\}$ is then a subspace, $\mathbf{X}\{\boldsymbol{\beta} : \mathbf{R}\boldsymbol{\beta} = \mathbf{0}\}$ is a subspace of $\mathcal{C}(\mathbf{X})$ of dimension $k+1-q$ (by (L1) and (A4)), and SSE_R is the squared distance from $\mathbf{y}$ to it. Writing $\mathbf{H}_R$ for the orthogonal projection onto this subspace (Remark 2.10; it is symmetric idempotent with column space the subspace), $\mathbf{H}\mathbf{H}_R = \mathbf{H}_R$ (the subspace lies in $\mathcal{C}(\mathbf{X})$, on which $\mathbf{H}$ acts as the identity), so $\mathbf{H} - \mathbf{H}_R$ is symmetric idempotent of rank q, $\mathbf{y} = \mathbf{H}_R\mathbf{y} + (\mathbf{H} - \mathbf{H}_R)\mathbf{y} + \mathbf{M}\mathbf{y}$ is an orthogonal decomposition, and $\mathrm{SSE}_R - \mathrm{SSE}_U = \mathbf{y}'(\mathbf{H} - \mathbf{H}_R)\mathbf{y}$ by Pythagoras; Theorem 10.3 with $\mathbf{r} = \mathbf{0}$ gives the explicit form $\mathbf{H}_R = \mathbf{H} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. For $\mathbf{r} \neq \mathbf{0}$ the feasible set is an affine set, not a subspace, and this identity is *false*: $\mathbf{y}'(\mathbf{H} - \mathbf{H}_R)\mathbf{y}$ with the same $\mathbf{H}_R$ equals $(\mathbf{R}\hat{\boldsymbol{\beta}})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}})$, whereas the correct value (38) is centered at $\mathbf{r}$. The constrained-OLS route of the proof above covers general $\mathbf{r}$.

Corollary 7.8 ($t^2 = F$ for a Single Linear Constraint)**.** For testing $H_0\colon \mathbf{c}'\boldsymbol{\beta} = r_0$ (a single constraint, $q = 1$, $\mathbf{R} = \mathbf{c}'$ with $\mathbf{c} \neq \mathbf{0}$), the F statistic from Theorem 7.4 equals the square of the t statistic:

$$t = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}}, \qquad F = t^2,$$

and $F = t^2 \sim F_{1,n-k-1}$ under H_0 by Theorem 7.4. The identity $F = t^2$ itself is algebraic: like Corollary 7.6, it uses only (A4) (through $\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c} > 0$ for $\mathbf{c} \neq \mathbf{0}$, Theorem 2.4, Step 4), and (A1)–(A5) enter only through the distribution.

Proof. With $q = 1$, $\mathbf{R} = \mathbf{c}'$, $\mathbf{r} = r_0$, the general F statistic becomes:

$$F = \frac{(\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0)[\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}]^{-1}(\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0)}{1 \cdot s^2} = \left(\frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}}\right)^2 = t^2. \qquad \square$$

7.4 Confidence Intervals and Regions

Theorem 7.9 (Confidence Interval for $\mathbf{c}'\boldsymbol{\beta}$)**.** Under (A1)–(A5), for $\mathbf{c} \neq \mathbf{0}$ (so that $\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c} > 0$ by Theorem 2.4, Step 4), a $(1-\alpha) \times 100\%$ confidence interval for $\mathbf{c}'\boldsymbol{\beta}$ is

$$\mathbf{c}'\hat{\boldsymbol{\beta}} \pm t_{n-k-1,\,1-\alpha/2} \cdot s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}, \tag{39}$$

where $t_{n-k-1,1-\alpha/2}$ is the $(1-\alpha/2)$ quantile of t_{n-k-1} ((N5)).

Proof. By Theorem 4.2 applied to $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$ with $\mathbf{A} = \mathbf{c}'$ (a $1 \times (k+1)$ row vector), $\mathbf{c}'\hat{\boldsymbol{\beta}} \sim N(\mathbf{c}'\boldsymbol{\beta},\ \sigma^2\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c})$. Standardizing:

$$Z = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{\sigma\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \sim N(0,1).$$

By Theorem 5.2, $V = \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$ and is independent of $\hat{\boldsymbol{\beta}}$; since Z is a function of $\hat{\boldsymbol{\beta}}$ alone, Z is independent of V. By definition of the t distribution:

$$\frac{Z}{\sqrt{V/(n-k-1)}} = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{\sigma\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \cdot \frac{1}{\sqrt{\mathrm{SSE}/[\sigma^2(n-k-1)]}} = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \sim t_{n-k-1},$$

where the last step uses $s^2 = \mathrm{SSE}/(n-k-1)$. The pivot gives, by the symmetry of the t distribution about 0, equation (18),

$$\mathsf{P}\left(-t_{n-k-1,1-\alpha/2} \leqslant \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \leqslant t_{n-k-1,1-\alpha/2}\right) = 1-\alpha,$$

and rearranging for $\mathbf{c}'\boldsymbol{\beta}$ yields the interval. $\square$

Theorem 7.10 (Joint Confidence Ellipsoid)**.** Under (A1)–(A5), a $(1-\alpha)\times 100\%$ joint confidence region for $\boldsymbol{\beta}$ is

$$\left\{\boldsymbol{\beta}\in\mathbb{R}^{k+1} : (\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}) \leqslant (k+1)s^2 F_{k+1,n-k-1,1-\alpha}\right\}. \tag{40}$$

Proof. Step 1: A chi-square quadratic form in $\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}$. By Theorem 5.1,

$$\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}\sim N(\mathbf{0},\sigma^2(\mathbf{X}'\mathbf{X})^{-1}).$$

Let $(\mathbf{X}'\mathbf{X})^{1/2}$ be the positive definite symmetric square root ((L4)) and define $\mathbf{w}=\sigma^{-1}(\mathbf{X}'\mathbf{X})^{1/2}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})$. Theorem 4.2 gives $\mathbf{w}\sim N(\mathbf{0},\mathbf{I}_{k+1})$, so $\mathbf{w}'\mathbf{w}\sim\chi^2_{k+1}$ by Theorem 4.5:

$$\mathbf{w}'\mathbf{w}=\frac{(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})}{\sigma^2}\sim\chi^2_{k+1}.$$

Step 2: Form the F ratio. This is a function of $\hat{\boldsymbol{\beta}}$, hence independent of $\mathrm{SSE}/\sigma^2\sim\chi^2_{n-k-1}$ (Theorem 5.2). By definition of F,

$$\frac{(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})/[(k+1)\sigma^2]}{\mathrm{SSE}/[\sigma^2(n-k-1)]}=\frac{(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})}{(k+1)s^2}\sim F_{k+1,n-k-1}.$$

The confidence region collects the $\boldsymbol{\beta}$ for which this statistic is at most $F_{k+1,n-k-1,1-\alpha}$, an event of probability $1-\alpha$ by (N5). □

7.5 Testing Conventions and Further Consequences

Throughout this subsection $\nu=n-k-1$, $\mathbf{c}\neq\mathbf{0}$, $s>0$ (under (A5) this holds with probability one, since $\mathrm{SSE}/\sigma^2\sim\chi^2_\nu$ has a continuous distribution function by (N5)), and t denotes the statistic of Corollary 7.8 for $H_0\colon \mathbf{c}'\boldsymbol{\beta}=r_0$,

$$t=\frac{\mathbf{c}'\hat{\boldsymbol{\beta}}-r_0}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}},$$

evaluated at the observed data; for $\mathbf{c}=\boldsymbol{\iota}_j$ and $r_0=\beta_j^0$ it is the statistic of Theorem 7.1. The proof of Theorem 7.9 shows that, under (A1)–(A5), $(\mathbf{c}'\hat{\boldsymbol{\beta}}-\mathbf{c}'\boldsymbol{\beta})/(s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}})\sim t_\nu$; hence $t\sim t_\nu$ under H_0.

Definition 7.11 (One- and Two-Sided t Tests and p-Values)**.** Let $\alpha\in(0,1)$ and $T\sim t_\nu$. The level-α t tests of $H_0\colon \mathbf{c}'\boldsymbol{\beta}=r_0$ against the three alternatives, and their ***p*-values**, are:

Alternative H_1	Reject H_0 if	p-value
$\mathbf{c}'\boldsymbol{\beta}>r_0$	$t>t_{\nu,1-\alpha}$	$\mathsf{P}(T\geqslant t)$
$\mathbf{c}'\boldsymbol{\beta}<r_0$	$t<-t_{\nu,1-\alpha}$	$\mathsf{P}(T\leqslant t)$
$\mathbf{c}'\boldsymbol{\beta}\neq r_0$	$\lvert t\rvert>t_{\nu,1-\alpha/2}$	$\mathsf{P}(\lvert T\rvert\geqslant\lvert t\rvert)$

In each p-value, t is the observed value and T is an independent t_ν variable: the p-value is the probability, under the null distribution of t, of a value at least as extreme, in the direction of H_1, as the one observed. The most common case is $H_0\colon \beta_j=0$ against $\beta_j\neq 0$, with $t=\hat{\beta}_j/\mathrm{SE}(\hat{\beta}_j)$.

Proposition 7.12 (Exact Level and p-Values)**.** Under (A1)–(A5):

(i) if $H_0\colon \mathbf{c}'\boldsymbol{\beta}=r_0$ holds, each of the three tests of Definition 7.11 rejects H_0 with probability exactly α; and, whether or not H_0 holds, the interval of Theorem 7.9 contains $\mathbf{c}'\boldsymbol{\beta}$ with probability exactly $1-\alpha$;

(ii) each test rejects H_0 if and only if its p-value is smaller than α;

(iii) the two-sided p-value equals $\mathsf{P}(F\geqslant t^2)$ for $F\sim F_{1,\nu}$, the p-value of the F test of Theorem 7.4 with $\mathbf{R}=\mathbf{c}'$ and $\mathbf{r}=r_0$ (Corollary 7.8), where the p-value of an F test with q restrictions and observed statistic F_{obs} is $\mathsf{P}(F\geqslant F_{\mathrm{obs}})$ for $F\sim F_{q,\nu}$, its null distribution.

Proof. **(i)** Under H_0, $t \sim t_\nu$. By (N5), $\mathsf{P}(t > t_{\nu,1-\alpha}) = \alpha$; by (18), $\mathsf{P}(t < -t_{\nu,1-\alpha}) = \alpha$ and $\mathsf{P}(|t| > t_{\nu,1-\alpha/2}) = 1 - \mathsf{P}(-t_{\nu,1-\alpha/2} \leqslant t \leqslant t_{\nu,1-\alpha/2}) = \alpha$. The coverage probability is the display at the end of the proof of Theorem 7.9, which holds for the true $\mathbf{c}'\boldsymbol{\beta}$.

(ii) Let $G(a) = \mathsf{P}(T \geqslant a)$ for $a \in \mathbb{R}$. By (N5), the distribution function of t_ν is continuous and strictly increasing on $\mathbb{R}$, so G is continuous and strictly decreasing, with $G(t_{\nu,1-\alpha}) = \alpha$. *Upper alternative:* $p = G(t) < \alpha = G(t_{\nu,1-\alpha})$ iff $t > t_{\nu,1-\alpha}$. *Lower alternative:* since $-T \sim t_\nu$ (§4.1), $p = \mathsf{P}(T \leqslant t) = \mathsf{P}(-T \geqslant -t) = G(-t)$, which is $< \alpha$ iff $-t > t_{\nu,1-\alpha}$. *Two-sided:* for $a \geqslant 0$, $\mathsf{P}(|T| \geqslant a) = \mathsf{P}(T \geqslant a) + \mathsf{P}(T \leqslant -a) = 2G(a)$ (the two events are disjoint for $a > 0$, and for $a = 0$ both sides equal 1 because $G(0) = \frac{1}{2}$ by symmetry and continuity), so $p = 2G(|t|) < \alpha$ iff $G(|t|) < \alpha/2 = G(t_{\nu,1-\alpha/2})$ iff $|t| > t_{\nu,1-\alpha/2}$.

(iii) $\mathsf{P}(|T| \geqslant |t|) = \mathsf{P}(T^2 \geqslant t^2)$, and $T^2 \sim F_{1,\nu}$ (§4.1). By Corollary 7.8, t^2 is the observed value of the statistic of Theorem 7.4 with $\mathbf{R} = \mathbf{c}'$, $\mathbf{r} = r_0$, whose null distribution is $F_{1,\nu}$. □

Proposition 7.13 (Duality of Tests and Confidence Intervals)**.** Assume $s > 0$ (under (A5) this holds with probability one, since $\mathrm{SSE}/\sigma^2 \sim \chi^2_\nu$ has a continuous distribution function by (N5)). For every $r_0 \in \mathbb{R}$, the two-sided level-α test of $H_0\colon \mathbf{c}'\boldsymbol{\beta} = r_0$ rejects if and only if r_0 lies outside the $(1-\alpha)\times 100\%$ confidence interval of Theorem 7.9. Hence that interval is the set of hypothesized values r_0 that the test does not reject; in particular, $H_0\colon \beta_j = 0$ is rejected at level α iff the interval for β_j excludes 0. (This is an algebraic statement about the two decision rules; it uses only (A4) and $s > 0$.)

Proof. Let $c = t_{\nu,1-\alpha/2}$ and $\mathrm{SE} = s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}$, which is positive because $s > 0$ and $\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c} > 0$ (Theorem 2.4, Step 4). The test rejects iff $|\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0| > c\,\mathrm{SE}$, i.e. iff $r_0 < \mathbf{c}'\hat{\boldsymbol{\beta}} - c\,\mathrm{SE}$ or $r_0 > \mathbf{c}'\hat{\boldsymbol{\beta}} + c\,\mathrm{SE}$, i.e. iff $r_0 \notin [\mathbf{c}'\hat{\boldsymbol{\beta}} - c\,\mathrm{SE},\ \mathbf{c}'\hat{\boldsymbol{\beta}} + c\,\mathrm{SE}]$, which is the interval of Theorem 7.9. The last statement is the case $\mathbf{c} = \boldsymbol{\iota}_j$, $r_0 = 0$. □

Proposition 7.14 (Expected Regression Mean Square)**.** Write $\mathbf{X} = [\mathbf{1}\ \ \mathbf{X}_*]$ and $\boldsymbol{\beta} = (\beta_0, \boldsymbol{\beta}_*')'$, where $\boldsymbol{\beta}_* = (\beta_1, \dots, \beta_k)'$ collects the slopes. Under (A1)–(A3) (normality is not needed),

$$\mathsf{E}[\mathrm{SSR}] = k\sigma^2 + \|\mathbf{M}_0\mathbf{X}_*\boldsymbol{\beta}_*\|^2, \qquad \text{so} \qquad \mathsf{E}[\mathrm{MSR}] = \sigma^2 + \frac{\|\mathbf{M}_0\mathbf{X}_*\boldsymbol{\beta}_*\|^2}{k} \ \geqslant\ \sigma^2 = \mathsf{E}[\mathrm{MSE}],$$

with equality if and only if $\boldsymbol{\beta}_* = \mathbf{0}$, i.e. iff the null hypothesis of the overall F test (Theorem 7.3) holds.

Proof. Step 1: SSR as a quadratic form. Step 1 of the proof of Lemma 7.2 shows, without using H_0 or (A5), that $\mathrm{SSR} = \mathbf{y}'(\mathbf{H}-\mathbf{J})'(\mathbf{H}-\mathbf{J})\mathbf{y}$ and $(\mathbf{H}-\mathbf{J})\mathbf{1} = \mathbf{0}$; Step 2 shows, again without them, that $\mathbf{H}-\mathbf{J}$ is symmetric idempotent of trace k. Hence $\mathrm{SSR} = \mathbf{y}'(\mathbf{H}-\mathbf{J})\mathbf{y}$.

Step 2: Expectation. By (A1) and Lemma 3.2 (with (A3)), $\mathsf{E}[\mathbf{y}] = \mathbf{X}\boldsymbol{\beta}$ and $\mathrm{var}(\mathbf{y}) = \sigma^2\mathbf{I}$, so Lemma 3.6 gives

$$\mathsf{E}[\mathrm{SSR}] = \mathrm{tr}\big((\mathbf{H}-\mathbf{J})\sigma^2\mathbf{I}\big) + \boldsymbol{\beta}'\mathbf{X}'(\mathbf{H}-\mathbf{J})\mathbf{X}\boldsymbol{\beta} = k\sigma^2 + \|(\mathbf{H}-\mathbf{J})\mathbf{X}\boldsymbol{\beta}\|^2,$$

using $\mathbf{H}-\mathbf{J} = (\mathbf{H}-\mathbf{J})'(\mathbf{H}-\mathbf{J})$.

Step 3: The non-centrality term. Since $(\mathbf{H}-\mathbf{J})\mathbf{1} = \mathbf{0}$ and $\mathbf{H}\mathbf{X}_* = \mathbf{X}_*$ (Theorem 2.9(ii)), $(\mathbf{H}-\mathbf{J})\mathbf{X}\boldsymbol{\beta} = (\mathbf{H}-\mathbf{J})\mathbf{X}_*\boldsymbol{\beta}_* = \mathbf{X}_*\boldsymbol{\beta}_* - \mathbf{J}\mathbf{X}_*\boldsymbol{\beta}_* = \mathbf{M}_0\mathbf{X}_*\boldsymbol{\beta}_*$. This vanishes iff $\mathbf{X}_*\boldsymbol{\beta}_* = a\mathbf{1}$ for some a, i.e. iff $\mathbf{X}(-a, \boldsymbol{\beta}_*')' = \mathbf{0}$, which by (A4) forces $a = 0$ and $\boldsymbol{\beta}_* = \mathbf{0}$ (and conversely $\boldsymbol{\beta}_* = \mathbf{0}$ gives $\mathbf{M}_0\mathbf{X}_*\boldsymbol{\beta}_* = \mathbf{0}$). Dividing by k and recalling $\mathsf{E}[\mathrm{MSE}] = \sigma^2$ (Theorem 3.7) completes the proof. □

The overall F statistic is thus a ratio of two estimators of σ^2 when H_0 holds, and its numerator has the larger expectation otherwise; values of F well above 1 are evidence against H_0, and how far above 1 is decided by the $F_{k,\nu}$ quantile.

Corollary 7.15 (R^2 from the Overall F Statistic)**.** If SST > 0 and SSE > 0, the overall F statistic of Theorem 7.3 determines R^2:

$$R^2 = \frac{kF}{kF + n - k - 1}.$$

This is an algebraic identity; it uses only (A4) and the two stated conditions. Under (A5) both conditions hold with probability one.

Proof. SSE > 0 gives $R^2 < 1$, so by (34), $F = (n-k-1)R^2/\{k(1-R^2)\}$, i.e. $kF(1-R^2) = (n-k-1)R^2$, i.e. $R^2(kF + n - k - 1) = kF$; the bracket is positive since $F \geqslant 0$ and $n > k+1$. Under (A5), SST > 0 with probability one by the proof of Theorem 7.3, and $\mathsf{P}(\text{SSE} = 0) = 0$ because $\text{SSE}/\sigma^2 \sim \chi^2_\nu$ (Theorem 5.2) has a continuous distribution function ((N5)). □

Corollary 7.16 (Nested F Test via R^2)**.** In the setting of Corollary 7.6, assume SST > 0 and $\text{SSE}_U > 0$, and put $R^2_U = 1 - \text{SSE}_U/\text{SST}$ and $R^2_R = 1 - \text{SSE}_R/\text{SST}$. Then

$$F = \frac{(R^2_U - R^2_R)/q}{(1 - R^2_U)/(n-k-1)}.$$

If the constraints set q of the coefficients to zero ($\mathbf{R}$ consists of the corresponding rows of $\mathbf{I}_{k+1}$, $\mathbf{r} = \mathbf{0}$), then SSE_R is the residual sum of squares of the OLS regression of $\mathbf{y}$ on the remaining columns of $\mathbf{X}$; if these include the intercept column (as when only slopes are set to zero), R^2_R is the coefficient of determination (Definition 6.6) of that regression. These are algebraic identities; they use only (A4), $\text{rank}(\mathbf{R}) = q$ and the two stated conditions.

Proof. Both fits have the same response, hence the same SST; so $\text{SSE}_R - \text{SSE}_U = \text{SST}(R^2_U - R^2_R)$ and $\text{SSE}_U = \text{SST}(1 - R^2_U)$, and substituting into Corollary 7.6 and cancelling SST gives the formula. For the second statement, $\mathbf{X}\boldsymbol{\beta}$ with $\beta_j = 0$ for the q constrained indices is $\mathbf{X}_R\boldsymbol{\beta}_R$, where $\mathbf{X}_R$ consists of the remaining columns and $\boldsymbol{\beta}_R$ of the remaining coefficients; minimizing $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ under the constraints is therefore minimizing $\|\mathbf{y} - \mathbf{X}_R\boldsymbol{\beta}_R\|^2$ over $\boldsymbol{\beta}_R$, the OLS problem for the design $\mathbf{X}_R$, which has full column rank by (A4); when $\mathbf{X}_R$ contains the intercept column, $1 - \text{SSE}_R/\text{SST}$ is its R^2. □

For the next result and for Section 8, write $\mathbf{X}_{\cdot j}$ for the column of $\mathbf{X}$ that multiplies β_j ($j = 0, \ldots, k$; $\mathbf{X}_{\cdot 0} = \mathbf{1}$), $\mathbf{X}_{(j)}$ for the $n \times k$ matrix obtained from $\mathbf{X}$ by deleting this column (here $k \geqslant 1$), which has full column rank by (A4), $\mathbf{M}_{(j)}$ for the residual maker of $\mathbf{X}_{(j)}$, and $\text{SSE}_{(j)}$ for the residual sum of squares of the regression of $\mathbf{y}$ on $\mathbf{X}_{(j)}$.

Corollary 7.17 (Dropping One Regressor: $F = t^2$)**.** Let $k \geqslant 1$, fix $j \in \{0, \ldots, k\}$, assume SSE > 0, and let $t_j = \hat\beta_j/\text{SE}(\hat\beta_j)$ be the statistic of Theorem 7.1 for $\beta_j^0 = 0$. Then

$$\text{SSE}_{(j)} - \text{SSE} = \hat\beta_j^2\, \mathbf{X}'_{\cdot j}\mathbf{M}_{(j)}\mathbf{X}_{\cdot j}, \qquad [(\mathbf{X}'\mathbf{X})^{-1}]_{jj} = \frac{1}{\mathbf{X}'_{\cdot j}\mathbf{M}_{(j)}\mathbf{X}_{\cdot j}}, \qquad \frac{(\text{SSE}_{(j)} - \text{SSE})/1}{\text{SSE}/(n-k-1)} = t_j^2.$$

Under (A1)–(A5) and $H_0\colon \beta_j = 0$, this statistic has the $F_{1,n-k-1}$ distribution. For $k = 1$ and $j = 1$, it is the overall F statistic of Theorem 7.3, so that $F = t_1^2$.

Proof. Step 1: Reordering the columns. Let $\mathbf{P}$ be the permutation matrix that moves column j to the last position, so that $\tilde{\mathbf{X}} = \mathbf{X}\mathbf{P} = [\mathbf{X}_{(j)} \;\; \mathbf{X}_{\cdot j}]$. Since $\mathbf{P}'\mathbf{P} = \mathbf{I}$, $(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1} = \mathbf{P}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{P}$, and the OLS estimator for $\tilde{\mathbf{X}}$ is $\mathbf{P}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{P}\mathbf{P}'\mathbf{X}'\mathbf{y} = \mathbf{P}'\hat{\boldsymbol{\beta}}$, the same coefficients in the new order; the fitted values $\tilde{\mathbf{X}}\mathbf{P}'\hat{\boldsymbol{\beta}} = \mathbf{X}\hat{\boldsymbol{\beta}}$, the residuals and SSE are unchanged; and the last diagonal entry of $(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}$ is $[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$.

Step 2: The identities. By (A4), $\mathbf{X}_{\cdot j} \notin \mathcal{C}(\mathbf{X}_{(j)})$, so Lemma 6.9 applies with $\mathbf{X}_{(j)}$ in place of $\mathbf{X}_k$ and $\mathbf{X}_{\cdot j}$ in place of $\mathbf{x}$; it gives the first identity, and identity (31), whose proof uses Lemma 6.9 alone, gives the second. Hence $\text{SE}(\hat\beta_j)^2 = s^2/(\mathbf{X}'_{\cdot j}\mathbf{M}_{(j)}\mathbf{X}_{\cdot j})$, and

$$\frac{\text{SSE}_{(j)} - \text{SSE}}{s^2} = \frac{\hat\beta_j^2\, \mathbf{X}'_{\cdot j}\mathbf{M}_{(j)}\mathbf{X}_{\cdot j}}{s^2} = \frac{\hat\beta_j^2}{\text{SE}(\hat\beta_j)^2} = t_j^2.$$

(Equivalently, by Corollary 7.16 the regression on $\mathbf{X}_{(j)}$ is the constrained fit for $\mathbf{R} = \boldsymbol{\iota}_j'$, $\mathbf{r} = 0$, so the identity also follows from Corollaries 7.6 and 7.8.)

Step 3: Distribution. Under H_0, $t_j \sim t_{n-k-1}$ by Theorem 7.1, so $t_j^2 \sim F_{1,n-k-1}$ (§4.1).

Step 4: The case $k = 1$. Then $\mathbf{X}_{(1)} = \mathbf{1}$, whose fit is $\bar{y}\mathbf{1}$ with residual sum of squares SST (Remark 7.5), so the statistic is $(\text{SST} - \text{SSE})/(\text{SSE}/(n-2)) = \text{SSR}/(\text{SSE}/(n-2))$ by Theorem 6.4, which is F of (34) with $k = 1$. □

8 Partitioned Regression

Theorem 8.1 is an algebraic identity that uses only the standing assumption (A4); Theorem 8.3 uses (A2) in addition. The results that follow Theorem 8.3 are algebraic except for their statements about expectations and variances, which use (A1)–(A3) as stated in each. No distributional assumption is used in this section.

8.1 The Frisch–Waugh–Lovell Theorem

Theorem 8.1 (Frisch–Waugh–Lovell)**.** Partition $\mathbf{X} = [\mathbf{X}_1 \;\; \mathbf{X}_2]$ and $\boldsymbol{\beta} = (\boldsymbol{\beta}_1', \boldsymbol{\beta}_2')'$, so the model is $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$. Let $\mathbf{M}_2 = \mathbf{I} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'$ be the residual maker for $\mathbf{X}_2$. Then:

(i) $\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{M}_2\mathbf{y}$.

(ii) The OLS coefficient vector from regressing $\mathbf{M}_2\mathbf{y}$ on $\mathbf{M}_2\mathbf{X}_1$ is the $\hat{\boldsymbol{\beta}}_1$ of (i), and the residual vector of that regression equals the residual vector $\mathbf{e}$ from the full regression.

Proof. (i). The normal equations for the full model are:

$$\begin{pmatrix} \mathbf{X}_1'\mathbf{X}_1 & \mathbf{X}_1'\mathbf{X}_2 \\ \mathbf{X}_2'\mathbf{X}_1 & \mathbf{X}_2'\mathbf{X}_2 \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}}_1 \\ \hat{\boldsymbol{\beta}}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{X}_1'\mathbf{y} \\ \mathbf{X}_2'\mathbf{y} \end{pmatrix}.$$

The second block equation reads $\mathbf{X}_2'\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}_2'\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = \mathbf{X}_2'\mathbf{y}$, which rearranges to

$$\hat{\boldsymbol{\beta}}_2 = (\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'(\mathbf{y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1).$$

Substituting this into the first block equation $\mathbf{X}_1'\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}_1'\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = \mathbf{X}_1'\mathbf{y}$:

$$\mathbf{X}_1'\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \underbrace{\mathbf{X}_1'\mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'}_{=\mathbf{X}_1'\mathbf{H}_2}(\mathbf{y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1) = \mathbf{X}_1'\mathbf{y}, \tag{41}$$

where we recognize the projection onto $\mathcal{C}(\mathbf{X}_2)$:

$$\mathbf{H}_2 := \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2' = \mathbf{I} - \mathbf{M}_2.$$

Expanding (41) using $\mathbf{X}_1'\mathbf{H}_2 = \mathbf{X}_1' - \mathbf{X}_1'\mathbf{M}_2$:

$$\mathbf{X}_1'\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}_1'(\mathbf{y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1) - \mathbf{X}_1'\mathbf{M}_2(\mathbf{y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1) = \mathbf{X}_1'\mathbf{y}.$$

The first two terms combine to $\mathbf{X}_1'\mathbf{y}$, cancelling the right-hand side, leaving $\mathbf{X}_1'\mathbf{M}_2(\mathbf{y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1) = \mathbf{0}$, or

$$\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 = \mathbf{X}_1'\mathbf{M}_2\mathbf{y}.$$

Since $\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1 = (\mathbf{M}_2\mathbf{X}_1)'(\mathbf{M}_2\mathbf{X}_1)$ is positive definite (the columns of $\mathbf{M}_2\mathbf{X}_1$ are linearly independent: if $\mathbf{M}_2\mathbf{X}_1\mathbf{v} = \mathbf{0}$ then $\mathbf{X}_1\mathbf{v} = \mathbf{H}_2\mathbf{X}_1\mathbf{v} \in \mathcal{C}(\mathbf{X}_2)$, say $\mathbf{X}_1\mathbf{v} = \mathbf{X}_2\mathbf{w}$, so $[\mathbf{X}_1 \;\; \mathbf{X}_2](\mathbf{v}', -\mathbf{w}')' = \mathbf{0}$ and (A4) forces $\mathbf{v} = \mathbf{0}$):

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{M}_2\mathbf{y}.$$

(ii). Let $\tilde{\mathbf{y}} = \mathbf{M}_2\mathbf{y}$ and $\tilde{\mathbf{X}}_1 = \mathbf{M}_2\mathbf{X}_1$. The columns of $\tilde{\mathbf{X}}_1$ are linearly independent (shown in (i)), so the OLS coefficient vector from regressing $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$ is defined, and, using $\mathbf{M}_2' = \mathbf{M}_2 = \mathbf{M}_2^2$ (Theorem 2.9(i) for the design $\mathbf{X}_2$, which has full column rank by (A4)) and then (i),

$$(\tilde{\mathbf{X}}_1'\tilde{\mathbf{X}}_1)^{-1}\tilde{\mathbf{X}}_1'\tilde{\mathbf{y}} = (\mathbf{X}_1'\mathbf{M}_2'\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{M}_2'\mathbf{M}_2\mathbf{y} = (\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{M}_2\mathbf{y} = \hat{\boldsymbol{\beta}}_1.$$

The residual from this regression is therefore

$$\tilde{\mathbf{e}} = \mathbf{M}_2\mathbf{y} - \mathbf{M}_2\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 = \mathbf{M}_2(\mathbf{y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1).$$

Substituting $\mathbf{y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1 = \mathbf{X}_2\hat{\boldsymbol{\beta}}_2 + \mathbf{e}$ from the full regression and using $\mathbf{M}_2\mathbf{X}_2 = \mathbf{0}$:

$$\tilde{\mathbf{e}} = \mathbf{M}_2\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 + \mathbf{M}_2\mathbf{e} = \mathbf{M}_2\mathbf{e}. \tag{42}$$

Finally, $\mathbf{M}_2\mathbf{e} = \mathbf{e}$: since $\mathbf{X}'\mathbf{e} = \mathbf{0}$ (Theorem 2.12(i)) includes $\mathbf{X}_2'\mathbf{e} = \mathbf{0}$,

$$\mathbf{M}_2\mathbf{e} = \mathbf{e} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{e} = \mathbf{e}.$$

Hence $\tilde{\mathbf{e}} = \mathbf{e}$. □

Remark 8.2 (The key identity $\mathbf{M}_2\mathbf{M} = \mathbf{M}$)**.** The computation above rests on the identity $\mathbf{M}_2\mathbf{e} = \mathbf{e}$, which can be written equivalently as $\mathbf{M}_2\mathbf{M} = \mathbf{M}$ (applying both sides to $\mathbf{y}$). This holds because $\mathcal{C}(\mathbf{X}_2) \subseteq \mathcal{C}(\mathbf{X})$, so the orthogonal complement $\mathcal{C}(\mathbf{X})^\perp \subseteq \mathcal{C}(\mathbf{X}_2)^\perp$: anything already orthogonal to $\mathbf{X}$ is automatically orthogonal to $\mathbf{X}_2$, and $\mathbf{M}_2$ leaves it unchanged. This identity is the abstract reason why "residualizing with respect to a subset of regressors" preserves the full-model residual.

8.2 Omitted Variable Bias

Theorem 8.3 (Omitted Variable Bias)**.** Suppose the true model is $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ with $\mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ (assumption (A2)), but the researcher estimates the "short" regression $\mathbf{y} = \mathbf{X}_1\boldsymbol{\gamma} + \mathbf{u}$, obtaining $\tilde{\boldsymbol{\gamma}} = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y}$. Then:

$$\mathsf{E}[\tilde{\boldsymbol{\gamma}}] = \boldsymbol{\beta}_1 + \underbrace{(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2}_{\text{auxiliary regression coefficients}}\boldsymbol{\beta}_2. \tag{43}$$

The estimator $\tilde{\boldsymbol{\gamma}}$ is unbiased for $\boldsymbol{\beta}_1$ if and only if $\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{0}$, which holds when either $\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$ ($\mathbf{X}_1$ and $\mathbf{X}_2$ are orthogonal) or $\boldsymbol{\beta}_2 = \mathbf{0}$ (the omitted variables are irrelevant).

Proof. Substituting the true model $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ into the short-regression estimator:

$$\begin{aligned}\tilde{\boldsymbol{\gamma}} &= (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y} = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'(\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}) \\ &= \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\boldsymbol{\varepsilon}.\end{aligned}$$

Taking expectations (using $\mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$):

$$\mathsf{E}[\tilde{\boldsymbol{\gamma}}] = \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2.$$

□

Remark 8.4 (Two-factor structure of the bias)**.** The bias term $(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\,\boldsymbol{\beta}_2$ is the product of two conceptually distinct quantities:

- $(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2$ is the matrix of OLS coefficients from regressing each column of $\mathbf{X}_2$ on $\mathbf{X}_1$—an *auxiliary regression* describing how the omitted variables relate to the included ones.
- $\boldsymbol{\beta}_2$ is the effect of the omitted variables on $\mathbf{y}$ in the true model.

The bias vanishes if either factor is zero: if $\mathbf{X}_1$ and $\mathbf{X}_2$ are orthogonal ($\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$), omission is harmless; if the omitted variables have no effect ($\boldsymbol{\beta}_2 = \mathbf{0}$), there is nothing to omit. Both ingredients must be nonzero to generate bias.

Remark 8.5 (Connection to FWL)**.** Theorem 8.1 shows that the correct (full-model) coefficient $\hat{\boldsymbol{\beta}}_1$ is obtained by regressing $\mathbf{M}_2\mathbf{y}$ on $\mathbf{M}_2\mathbf{X}_1$—that is, after partialling $\mathbf{X}_2$ out of both $\mathbf{y}$ and $\mathbf{X}_1$. The short regression instead regresses $\mathbf{y}$ directly on $\mathbf{X}_1$, effectively replacing $\mathbf{M}_2$ with $\mathbf{I}$. The two coefficient vectors differ by an exact sample identity: the first block of the normal equations of the full model, $\mathbf{X}_1'\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}_1'\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = \mathbf{X}_1'\mathbf{y} = \mathbf{X}_1'\mathbf{X}_1\tilde{\boldsymbol{\gamma}}$, gives

$$\tilde{\boldsymbol{\gamma}} - \hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\,\hat{\boldsymbol{\beta}}_2,$$

which has the same form as the bias term in (43) with $\hat{\boldsymbol{\beta}}_2$ in place of $\boldsymbol{\beta}_2$; taking expectations, with $\mathsf{E}[\hat{\boldsymbol{\beta}}_1] = \boldsymbol{\beta}_1$ and $\mathsf{E}[\hat{\boldsymbol{\beta}}_2] = \boldsymbol{\beta}_2$ (Theorem 3.3, under (A2)), recovers (43). OVB measures the cost of not partialling out the omitted variables.

Theorem 8.6 (Omitted Variables: Fitted Values and Error Variance)**.** In the setting of Theorem 8.3, assume (A1)–(A3) for the true model $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$, whose design $\mathbf{X} = [\mathbf{X}_1 \;\; \mathbf{X}_2]$ satisfies (A4), and let $\mathbf{X}_1$ have $p < n$ columns. Let $\mathbf{H}_1 = \mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'$ and $\mathbf{M}_1 = \mathbf{I} - \mathbf{H}_1$ be the hat matrix and residual maker of the short regression, $\hat{\mathbf{y}}_1 = \mathbf{H}_1\mathbf{y}$ its fitted values and $\tilde{s}^2 = \mathbf{y}'\mathbf{M}_1\mathbf{y}/(n-p)$ its mean squared error. Then:

(i) the fitted values are biased: $\mathsf{E}[\hat{\mathbf{y}}_1] - \mathsf{E}[\mathbf{y}] = -\mathbf{M}_1\mathbf{X}_2\boldsymbol{\beta}_2$;

(ii) $\mathsf{E}[\tilde{s}^2] = \sigma^2 + \dfrac{\|\mathbf{M}_1\mathbf{X}_2\boldsymbol{\beta}_2\|^2}{n-p} = \sigma^2 + \dfrac{1}{n-p}\displaystyle\sum_{i=1}^{n}\big(\mathsf{E}[\hat{y}_{1i}] - \mathsf{E}[y_i]\big)^2 \;\geqslant\; \sigma^2$, with equality if and only if $\boldsymbol{\beta}_2 = \mathbf{0}$.

Proof. The columns of $\mathbf{X}_1$ are among those of $\mathbf{X}$, so $\mathbf{X}_1$ has full column rank and Theorem 2.9 applies to $\mathbf{H}_1$ and $\mathbf{M}_1$ (with p in place of $k+1$).

(i) By (A1), $\mathsf{E}[\mathbf{y}] = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2$, and $\mathbf{H}_1\mathbf{X}_1 = \mathbf{X}_1$, so $\mathsf{E}[\mathbf{H}_1\mathbf{y}] = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{H}_1\mathbf{X}_2\boldsymbol{\beta}_2$; subtracting $\mathsf{E}[\mathbf{y}]$ leaves $-(\mathbf{I}-\mathbf{H}_1)\mathbf{X}_2\boldsymbol{\beta}_2 = -\mathbf{M}_1\mathbf{X}_2\boldsymbol{\beta}_2$.

(ii) By Lemma 3.6 with $\mathsf{E}[\mathbf{y}] = \mathbf{X}\boldsymbol{\beta}$ and $\operatorname{var}(\mathbf{y}) = \sigma^2\mathbf{I}$ (Lemma 3.2, (A3)),

$$\mathsf{E}[\mathbf{y}'\mathbf{M}_1\mathbf{y}] = \sigma^2\operatorname{tr}(\mathbf{M}_1) + (\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2)'\mathbf{M}_1(\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2) = \sigma^2(n-p) + \|\mathbf{M}_1\mathbf{X}_2\boldsymbol{\beta}_2\|^2,$$

using $\operatorname{tr}(\mathbf{M}_1) = n-p$, $\mathbf{M}_1\mathbf{X}_1 = \mathbf{0}$ and $\mathbf{M}_1 = \mathbf{M}_1'\mathbf{M}_1$. Dividing by $n-p$ gives the first expression, and by (i) $\|\mathbf{M}_1\mathbf{X}_2\boldsymbol{\beta}_2\|^2$ is the sum of the squared biases of the fitted values. It vanishes iff $\mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{H}_1\mathbf{X}_2\boldsymbol{\beta}_2$, i.e. iff $\mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{X}_1\mathbf{a}$ for some $\mathbf{a}$, i.e. iff $\mathbf{X}(-\mathbf{a}', \boldsymbol{\beta}_2')' = \mathbf{0}$, which by (A4) holds iff $\mathbf{a} = \mathbf{0}$ and $\boldsymbol{\beta}_2 = \mathbf{0}$. □

Proposition 8.7 (Extraneous Variables)**.** Suppose the true model is $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \boldsymbol{\varepsilon}$ with (A1)–(A3), where $\mathbf{X}_1$ has p columns, but the long regression on $\mathbf{X} = [\mathbf{X}_1 \;\; \mathbf{X}_2]$ is fitted, where $\mathbf{X}$ has full column rank and $n > k+1$, giving $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_1', \hat{\boldsymbol{\beta}}_2')'$ and s^2. Then $\mathsf{E}[\hat{\boldsymbol{\beta}}_1] = \boldsymbol{\beta}_1$, $\mathsf{E}[\hat{\boldsymbol{\beta}}_2] = \mathbf{0}$, $\mathsf{E}[\hat{\mathbf{y}}] = \mathsf{E}[\mathbf{y}]$ and $\mathsf{E}[s^2] = \sigma^2$; and $\operatorname{var}(\hat{\boldsymbol{\beta}}_1) - \operatorname{var}(\tilde{\boldsymbol{\gamma}})$ is positive semi-definite, where $\tilde{\boldsymbol{\gamma}} = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y}$ is the OLS estimator of the true model.

Proof. The true model is the model with design $\mathbf{X}$ and coefficient vector $\boldsymbol{\beta} = (\boldsymbol{\beta}_1', \mathbf{0}')'$, and (A1)–(A4) hold for it. Theorem 3.3 gives $\mathsf{E}[\hat{\boldsymbol{\beta}}] = (\boldsymbol{\beta}_1', \mathbf{0}')'$, hence $\mathsf{E}[\hat{\mathbf{y}}] = \mathbf{X}\mathsf{E}[\hat{\boldsymbol{\beta}}] = \mathbf{X}_1\boldsymbol{\beta}_1 = \mathsf{E}[\mathbf{y}]$, and Theorem 3.7 gives $\mathsf{E}[s^2] = \sigma^2$. For the variances, $\hat{\boldsymbol{\beta}}_1 = \mathbf{C}\mathbf{y}$, where $\mathbf{C}$ consists of the first p rows of $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$; since $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{I}_{k+1}$, $\mathbf{C}\mathbf{X}_1 = \mathbf{I}_p$, so $\mathsf{E}[\hat{\boldsymbol{\beta}}_1] = \mathbf{C}\mathbf{X}_1\boldsymbol{\beta}_1 = \boldsymbol{\beta}_1$ for every $\boldsymbol{\beta}_1$: $\hat{\boldsymbol{\beta}}_1$ is a linear unbiased estimator in the true model, whose design $\mathbf{X}_1$ has full column rank (its columns are among those of $\mathbf{X}$). The Gauss–Markov theorem (Theorem 3.5) applied to the true model, with $\mathbf{X}_1$ in place of $\mathbf{X}$, gives $\operatorname{var}(\hat{\boldsymbol{\beta}}_1) - \operatorname{var}(\tilde{\boldsymbol{\gamma}}) \succeq \mathbf{0}$. □

Remark 8.8 (Omitting versus including)**.** Theorems 8.3 and 8.6 and Proposition 8.7 compare the two errors of specification. Omitting a relevant regressor biases the coefficients (unless $\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{0}$) and the fitted values, and inflates $\mathsf{E}[\tilde{s}^2]$ by the sum of the squared biases of the fitted values divided by $n-p$ (Theorem 8.6(ii)), which tends to widen intervals computed from $\tilde{s}$. Including an irrelevant regressor keeps every estimator unbiased and costs only precision.

8.3 Added-Variable Plots, Partial Correlation, and Variance Inflation

In this subsection $j \in \{1, \dots, k\}$, and $\mathbf{X}_{\cdot j}$, $\mathbf{X}_{(j)}$, $\mathbf{M}_{(j)}$ and $\mathrm{SSE}_{(j)}$ are as defined before Corollary 7.17; $\mathbf{X}_{(j)}$ contains the intercept column $\mathbf{1}$.

Definition 8.9 (Added-Variable Plot and Partial Correlation)**.** Let $\mathbf{e}_{(j)} = \mathbf{M}_{(j)}\mathbf{y}$ be the residuals of the regression of $\mathbf{y}$ on the other columns, and $\tilde{\mathbf{x}}_j = \mathbf{M}_{(j)}\mathbf{X}_{\cdot j}$ the residuals of the regression of the j-th regressor on them. The **added-variable plot** (or partial regression plot) for regressor j is the scatter plot of the points $(\tilde{x}_{ij}, e_{(j)i})$, $i = 1, \dots, n$. When $\mathbf{e}_{(j)}$ is not constant, the **partial correlation** of $\mathbf{y}$ and regressor j given the other regressors is $r_{yj\cdot} = \operatorname{corr}(\tilde{\mathbf{x}}_j, \mathbf{e}_{(j)})$ (Definition 2.15; $\tilde{\mathbf{x}}_j$ is never constant, by Corollary 8.10(i) below).

Corollary 8.10 (What the Added-Variable Plot Shows)**.**

(i) $\mathbf{e}_{(j)}$ and $\tilde{\mathbf{x}}_j$ have mean zero, and $\tilde{\mathbf{x}}_j$ is not constant.

(ii) The OLS line through the added-variable plot (the regression of $\mathbf{e}_{(j)}$ on $[\mathbf{1}\ \ \tilde{\mathbf{x}}_j]$) has intercept 0 and slope

$$\frac{\tilde{\mathbf{x}}_j'\mathbf{e}_{(j)}}{\tilde{\mathbf{x}}_j'\tilde{\mathbf{x}}_j} = \frac{\mathbf{X}_{\cdot j}'\mathbf{M}_{(j)}\mathbf{y}}{\mathbf{X}_{\cdot j}'\mathbf{M}_{(j)}\mathbf{X}_{\cdot j}} = \hat{\beta}_j,$$

the coefficient of regressor j in the full regression.

(iii) The residuals of that line are the residuals $\mathbf{e}$ of the full regression; in particular its residual sum of squares is SSE.

(iv) Under (A3), $\operatorname{var}(\hat{\beta}_j) = \sigma^2/(\mathbf{X}_{\cdot j}'\mathbf{M}_{(j)}\mathbf{X}_{\cdot j}) = \sigma^2/\|\tilde{\mathbf{x}}_j\|^2$.

Proof. **(i)** Since $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$, $\mathbf{M}_{(j)}\mathbf{1} = \mathbf{0}$ (Theorem 2.9(ii) for the design $\mathbf{X}_{(j)}$), and by symmetry $\mathbf{1}'\mathbf{M}_{(j)} = \mathbf{0}'$; hence $\mathbf{1}'\mathbf{e}_{(j)} = 0$ and $\mathbf{1}'\tilde{\mathbf{x}}_j = 0$. If $\tilde{\mathbf{x}}_j = \mathbf{0}$, then $\mathbf{X}_{\cdot j} = (\mathbf{I} - \mathbf{M}_{(j)})\mathbf{X}_{\cdot j} \in \mathcal{C}(\mathbf{X}_{(j)})$, contradicting (A4); a nonzero vector with mean zero is not constant.

(ii) By (i), Corollary 2.18 applies to the regression of $\mathbf{e}_{(j)}$ on $[\mathbf{1}\ \ \tilde{\mathbf{x}}_j]$. Because both vectors have mean zero, its slope is $S_{\tilde{x}e}/S_{\tilde{x}\tilde{x}} = \tilde{\mathbf{x}}_j'\mathbf{e}_{(j)}/\tilde{\mathbf{x}}_j'\tilde{\mathbf{x}}_j$ and its intercept is $0 - \text{slope} \cdot 0 = 0$. Since $\mathbf{M}_{(j)}$ is symmetric idempotent, $\tilde{\mathbf{x}}_j'\mathbf{e}_{(j)} = \mathbf{X}_{\cdot j}'\mathbf{M}_{(j)}\mathbf{y}$ and $\tilde{\mathbf{x}}_j'\tilde{\mathbf{x}}_j = \mathbf{X}_{\cdot j}'\mathbf{M}_{(j)}\mathbf{X}_{\cdot j}$. By Theorem 8.1(i) with $\mathbf{X}_1 = \mathbf{X}_{\cdot j}$ and $\mathbf{X}_2 = \mathbf{X}_{(j)}$ (after reordering the columns, as in Step 1 of the proof of Corollary 7.17), this ratio is $\hat{\beta}_j$.

(iii) As the intercept is 0, the line's fitted values are $\hat{\beta}_j\tilde{\mathbf{x}}_j$ and its residuals are $\mathbf{M}_{(j)}\mathbf{y} - \mathbf{M}_{(j)}\mathbf{X}_{\cdot j}\hat{\beta}_j$, the residuals of the regression of $\mathbf{M}_{(j)}\mathbf{y}$ on $\mathbf{M}_{(j)}\mathbf{X}_{\cdot j}$; by Theorem 8.1(ii) they equal $\mathbf{e}$.

(iv) By (ii), $\hat{\beta}_j = \mathbf{a}'\mathbf{y}$ with the constant vector $\mathbf{a} = \tilde{\mathbf{x}}_j/\|\tilde{\mathbf{x}}_j\|^2$ (because $\tilde{\mathbf{x}}_j'\mathbf{y} = \mathbf{X}_{\cdot j}'\mathbf{M}_{(j)}\mathbf{y}$), so Lemma 3.2 and (A3) give $\operatorname{var}(\hat{\beta}_j) = \sigma^2\mathbf{a}'\mathbf{a} = \sigma^2/\|\tilde{\mathbf{x}}_j\|^2$. (This agrees with $\sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$ by Corollary 7.17.) □

Proposition 8.11 (Partial Correlation and the t Statistic)**.** Assume SSE > 0 and let t_j be as in Corollary 7.17. Then $\mathbf{e}_{(j)}$ is not constant, and

$$r_{yj\cdot}^2 = \frac{\text{SSE}_{(j)} - \text{SSE}}{\text{SSE}_{(j)}}, \qquad r_{yj\cdot} = \frac{t_j}{\sqrt{t_j^2 + n - k - 1}}.$$

Proof. Step 1: The correlation. By Corollary 7.17, $\text{SSE}_{(j)} \geqslant \text{SSE} > 0$, and $\text{SSE}_{(j)} = \|\mathbf{e}_{(j)}\|^2$; so $\mathbf{e}_{(j)} \neq \mathbf{0}$, and having mean zero (Corollary 8.10(i)) it is not constant. Since both vectors have mean zero, $r_{yj\cdot} = \tilde{\mathbf{x}}_j'\mathbf{e}_{(j)}/(\|\tilde{\mathbf{x}}_j\|\,\|\mathbf{e}_{(j)}\|)$.

Step 2: The first identity. By Corollaries 7.17 and 8.10(ii), $\text{SSE}_{(j)} - \text{SSE} = \hat{\beta}_j^2\|\tilde{\mathbf{x}}_j\|^2$, which equals $(\tilde{\mathbf{x}}_j'\mathbf{e}_{(j)})^2/\|\tilde{\mathbf{x}}_j\|^2$. Dividing by $\text{SSE}_{(j)} = \|\mathbf{e}_{(j)}\|^2$ gives $r_{yj\cdot}^2$.

Step 3: The second identity. Write $\Delta = \text{SSE}_{(j)} - \text{SSE}$. By Corollary 7.17, $\Delta/\text{SSE} = t_j^2/(n-k-1)$, so

$$r_{yj\cdot}^2 = \frac{\Delta}{\text{SSE} + \Delta} = \frac{\Delta/\text{SSE}}{1 + \Delta/\text{SSE}} = \frac{t_j^2}{t_j^2 + n - k - 1}.$$

The sign of $r_{yj\cdot}$ is that of $\tilde{\mathbf{x}}_j'\mathbf{e}_{(j)}$, which by Corollary 8.10(ii) is the sign of $\hat{\beta}_j$ and hence of t_j (as $\text{SE}(\hat{\beta}_j) > 0$). Taking the square root with this sign gives the second identity. □

Corollary 8.12 (Variance Inflation Factor)**.** Let R_j^2 be the coefficient of determination of the regression of $\mathbf{X}_{\cdot j}$ on $\mathbf{X}_{(j)}$, and $S_{jj} = \sum_i (x_{ij} - \bar{x}_j)^2 = (n-1)s_j^2$, where $\bar{x}_j$ and s_j^2 are the mean and sample variance of the j-th regressor. Then $S_{jj} > 0$ and $R_j^2 < 1$, and under (A3)

$$\operatorname{var}(\hat{\beta}_j) = \frac{\sigma^2}{S_{jj}(1 - R_j^2)} = \frac{\sigma^2\,\text{VIF}_j}{(n-1)s_j^2}, \qquad \text{VIF}_j = \frac{1}{1 - R_j^2},$$

where VIF_j is the **variance inflation factor** of regressor j; correspondingly,

$$\mathrm{SE}(\hat{\beta}_j) = \frac{s\sqrt{\mathrm{VIF}_j}}{s_j\sqrt{n-1}}.$$

For $k = 1$, $R_1^2 = 0$ and the formula reduces to $\mathrm{var}(\hat{\beta}_1) = \sigma^2/S_{xx}$ of Corollary 3.13.

Proof. $\mathbf{X}_{\cdot j}$ is not constant, for otherwise $\mathbf{X}_{\cdot j} \in \mathcal{C}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{X}_{(j)})$, contradicting (A4); so $S_{jj} > 0$, and S_{jj} is the total sum of squares of the auxiliary regression of $\mathbf{X}_{\cdot j}$ on $\mathbf{X}_{(j)}$, which has an intercept. Its residual sum of squares is $\|\mathbf{M}_{(j)}\mathbf{X}_{\cdot j}\|^2 = \|\tilde{\mathbf{x}}_j\|^2 > 0$ (Corollary 8.10(i)), so $R_j^2 = 1 - \|\tilde{\mathbf{x}}_j\|^2/S_{jj} < 1$ (Definition 6.6) and $\|\tilde{\mathbf{x}}_j\|^2 = S_{jj}(1-R_j^2)$. Substituting into Corollary 8.10(iv) gives $\mathrm{var}(\hat{\beta}_j)$; replacing $\sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj} = \sigma^2/\|\tilde{\mathbf{x}}_j\|^2$ by $s^2/\|\tilde{\mathbf{x}}_j\|^2$ gives $\mathrm{SE}(\hat{\beta}_j)^2$ (Theorem 7.1 and Corollary 7.17). For $k = 1$, $\mathbf{X}_{(1)} = \mathbf{1}$, whose fitted values are $\bar{x}\mathbf{1}$, so the residual sum of squares is S_{xx} and $R_1^2 = 0$. □

Remark 8.13 (Sources of imprecision)**.** By Corollary 8.12, $\mathrm{var}(\hat{\beta}_j)$ is large when the error variance σ^2 is large, when the j-th regressor varies little in the sample ($S_{jj} = (n-1)s_j^2$ small, through a small n or a small spread s_j), or when it is nearly a linear combination of the other regressors (R_j^2 close to 1, i.e. a large VIF_j: collinearity). $\mathrm{VIF}_j \geqslant 1$, with equality iff $R_j^2 = 0$, i.e. iff the fitted values of the auxiliary regression are constant.

9 Prediction

Let $\mathbf{x}_0 \in \mathbb{R}^{k+1}$ be a new observation's regressor vector. The predicted value is $\hat{y}_0 = \mathbf{x}_0'\hat{\boldsymbol{\beta}}$.

Theorem 9.1 (Confidence Interval for the Conditional Mean)**.** Under (A1)–(A5), a $(1-\alpha)\times 100\%$ confidence interval for $\mathsf{E}[y_0 \mid \mathbf{x}_0] = \mathbf{x}_0'\boldsymbol{\beta}$ is

$$\mathbf{x}_0'\hat{\boldsymbol{\beta}} \pm t_{n-k-1,\,1-\alpha/2} \cdot s\sqrt{\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}. \tag{44}$$

Proof. This is the special case $\mathbf{c} = \mathbf{x}_0$ of Theorem 7.9; its hypothesis $\mathbf{c} \neq \mathbf{0}$ holds because the first component of $\mathbf{x}_0$ is the intercept's 1 (Definition 1.1). □

Theorem 9.2 (Prediction Interval for a New Observation)**.** Under (A1)–(A5), let $y_0 = \mathbf{x}_0'\boldsymbol{\beta} + \varepsilon_0$ be a new observation whose error $\varepsilon_0 \sim N(0,\sigma^2)$ is independent of $\boldsymbol{\varepsilon}$ (that is, (A3) and (A5) extend to the new observation). Then a $(1-\alpha)\times 100\%$ prediction interval for y_0 is

$$\mathbf{x}_0'\hat{\boldsymbol{\beta}} \pm t_{n-k-1,\,1-\alpha/2} \cdot s\sqrt{1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}. \tag{45}$$

Proof. Step 1: Set up. The prediction error is

$$\hat{y}_0 - y_0 = \mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - \varepsilon_0.$$

Here $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$ depends on the training-sample errors $\boldsymbol{\varepsilon}$, whereas ε_0 is the new-observation error, which by hypothesis is independent of $\boldsymbol{\varepsilon}$; hence $\varepsilon_0 \perp\!\!\!\perp \mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ (fact (N3)). Moreover the stacked vector $(\boldsymbol{\varepsilon}', \varepsilon_0)'$ is jointly normal: by independence its MGF factorizes as $\mathsf{E}[e^{\mathbf{t}'\boldsymbol{\varepsilon}}]\,\mathsf{E}[e^{\theta_0\varepsilon_0}] = \exp(\frac{\sigma^2}{2}\mathbf{t}'\mathbf{t})\exp(\frac{\sigma^2}{2}\theta_0^2)$, which is the MGF (17) of $N(\mathbf{0}, \sigma^2\mathbf{I}_{n+1})$ (fact (N2) and Definition 4.1). Consequently

$$\begin{pmatrix} \hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \\ \mathbf{e} \\ \varepsilon_0 \end{pmatrix} = \begin{pmatrix} (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' & \mathbf{0} \\ \mathbf{M} & \mathbf{0} \\ \mathbf{0}' & 1 \end{pmatrix} \begin{pmatrix} \boldsymbol{\varepsilon} \\ \varepsilon_0 \end{pmatrix} \tag{46}$$

is jointly normal (Theorem 4.2), with $\mathrm{cov}(\mathbf{e}, \hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \sigma^2\mathbf{M}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{0}$ (as in Theorem 5.2, Step 3) and $\mathrm{cov}(\mathbf{e}, \varepsilon_0) = [\mathbf{M}\ \ \mathbf{0}]\,\sigma^2\mathbf{I}_{n+1}\,[\mathbf{0}'\ \ 1]' = \mathbf{0}$ (Lemma 3.2(ii)).

Step 2: Compute the variance. Using Step 1's independence (so that the variances of the two terms add, fact (N7)) and Lemma 3.2 (with $\mathbf{A} = \mathbf{x}_0'$):

$$\begin{aligned}\operatorname{var}(\hat{y}_0 - y_0) &= \operatorname{var}\big(\mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\big) + \operatorname{var}(\varepsilon_0) \\ &= \mathbf{x}_0' \operatorname{var}(\hat{\boldsymbol{\beta}})\mathbf{x}_0 + \sigma^2 \\ &= \sigma^2 \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 + \sigma^2 \\ &= \sigma^2\big(1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0\big).\end{aligned}$$

Step 3: Derive the t pivotal quantity. As a linear function of the jointly normal vector $(\boldsymbol{\varepsilon}', \varepsilon_0)'$ of Step 1, namely $\hat{y}_0 - y_0 = [\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \;\; -1]\,(\boldsymbol{\varepsilon}', \varepsilon_0)'$, the prediction error is normal (Theorem 4.2) with mean 0 and the variance of Step 2:

$$\hat{y}_0 - y_0 \sim N\big(0,\ \sigma^2(1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)\big).$$

For independence from SSE, note that "$\hat{\boldsymbol{\beta}} \perp\!\!\!\perp$ SSE and $\varepsilon_0 \perp\!\!\!\perp$ SSE" would *not* by itself imply that the function $\mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - \varepsilon_0$ of the pair is independent of SSE; what is needed is independence of $\mathbf{e}$ from the pair $(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}, \varepsilon_0)$ jointly. This follows from (46): the vector there is jointly normal and both cross-covariances of $\mathbf{e}$ with the other two blocks vanish, so Theorem 4.3 (with $\mathbf{u} = \mathbf{e}$ and $\mathbf{v} = ((\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})', \varepsilon_0)'$) gives $\mathbf{e} \perp\!\!\!\perp (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}, \varepsilon_0)$. Hence $\mathrm{SSE} = \mathbf{e}'\mathbf{e}$ is independent of $\hat{y}_0 - y_0$ (fact (N3)). Replacing σ with s gives a t_{n-k-1} pivot (fact (N5), exactly as in Theorem 7.9):

$$\frac{\hat{y}_0 - y_0}{s\sqrt{1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}} \sim t_{n-k-1},$$

and inverting, using (18) exactly as at the end of the proof of Theorem 7.9, yields the prediction interval. □

Remark 9.3. The prediction interval is always wider than the confidence interval for the mean, because the prediction variance includes the additional σ^2 term from the irreducible noise ε_0.

Corollary 9.4 (Intervals in Simple Linear Regression)**.** In the simple linear regression model of §2.8, under (A1)–(A5) and with the hypotheses of Theorem 9.2 for the new observation, at $x_0 \in \mathbb{R}$ the $(1-\alpha)\times 100\%$ confidence interval for $\beta_0 + \beta_1 x_0$ and prediction interval for y_0 are

$$\hat{\beta}_0 + \hat{\beta}_1 x_0 \pm t_{n-2,\,1-\alpha/2} \cdot s\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}, \qquad \hat{\beta}_0 + \hat{\beta}_1 x_0 \pm t_{n-2,\,1-\alpha/2} \cdot s\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}.$$

Both are narrowest at $x_0 = \bar{x}$ and widen as x_0 moves away from $\bar{x}$; their squared half-widths are quadratic in $x_0 - \bar{x}$, so the half-widths themselves grow approximately linearly for large $|x_0 - \bar{x}|$.

Proof. Apply Theorems 9.1 and 9.2 with $k = 1$, $\mathbf{x}_0 = (1, x_0)'$ and $n-k-1 = n-2$; by Corollary 3.13, $\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = 1/n + (x_0 - \bar{x})^2/S_{xx}$. □

10 Constrained Least Squares and Modeling Remarks

Theorem 10.3 below is a purely algebraic statement about a given $\mathbf{y}$ and $\mathbf{X}$; it uses only the standing assumption (A4) and no distributional assumption.

10.1 Remarks on Model Specification

Remark 10.1 (No-intercept model)**.** When the model omits the intercept, $\mathbf{X}$ has no column of ones. The OLS formula $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ still applies, but Theorem 2.12(ii) is no longer available (its proof reads $\mathbf{1}'\mathbf{e} = 0$ off the intercept row of $\mathbf{X}'\mathbf{e} = \mathbf{0}$), so $\sum e_i \neq 0$ in general; since Steps 1, 2 and 4 of the proof of Theorem 6.4 all use $\mathbf{1}'\mathbf{e} = 0$, the decomposition $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$ may fail. Consequently the two expressions in Definition 6.6 no longer agree and Proposition 6.7 does not apply: $1 - \mathrm{SSE}/\mathrm{SST}$

can be negative, while SSR/SST $\geqslant 0$ can exceed 1. (For $n = 2$ with the single regressor $\mathbf{X} = (1, 0)'$ and $\mathbf{y} = (1, 3)'$: $\hat{\boldsymbol{\beta}} = 1$, $\mathbf{e} = (0, 3)'$, SST $= 2$, SSR $= 5$, SSE $= 9$, so $1 - \text{SSE/SST} = -3.5$ and SSR/SST $= 2.5$.) Use the uncentered $R^2 = 1 - \text{SSE}/\sum y_i^2 = 1 - \text{SSE}/\mathbf{y}'\mathbf{y}$ instead (for $\mathbf{y} \neq \mathbf{0}$): it lies in $[0, 1]$ because $\text{SSE} = \|\mathbf{My}\|^2 = \mathbf{y}'\mathbf{My}$ and $\mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{My} = \mathbf{y}'\mathbf{Hy} = \|\mathbf{Hy}\|^2 \geqslant 0$ (Theorem 2.9(i)); in the example it equals $1 - 9/10 = 0.1$.

Remark 10.2 (Dummy variable trap)**.** A categorical variable with g levels requires $g-1$ dummy regressors when the model includes an intercept. Including all g dummies creates the linear dependence $\mathbf{1} = D_1 + \cdots + D_g$, violating (A4).

10.2 Constrained Least Squares

Theorem 10.3 (Constrained OLS)**.** Consider $\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ subject to $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ (q linear constraints, $\text{rank}(\mathbf{R}) = q$). The constrained estimator is:

$$\hat{\boldsymbol{\beta}}_R = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big]^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}), \tag{47}$$

and $\text{SSE}_R \geqslant \text{SSE}_U$ (where SSE_U is the unconstrained residual sum of squares).

Proof. The objective $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ is a convex quadratic function of $\boldsymbol{\beta}$ (its Hessian $2\mathbf{X}'\mathbf{X}$ is positive definite by (A4)), and the constraint $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ is affine. Rather than invoking the theory of constrained optimization (the KKT conditions), we use the Lagrangian only to *produce a candidate* (Steps 1–2) and then verify directly that the candidate is the unique minimizer over the feasible set (Step 3); nothing from outside these notes is needed.

Step 1: Lagrangian and first-order conditions. The Lagrangian is $\mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\lambda}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + 2\boldsymbol{\lambda}'(\mathbf{R}\boldsymbol{\beta} - \mathbf{r})$ (the factor 2 simplifies later algebra). By Lemma 2.2, differentiating $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{y}'\mathbf{y} - 2\mathbf{y}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}$ with respect to $\boldsymbol{\beta}'$ gives $-2\mathbf{y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}$. The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\beta}'} = -2\mathbf{y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X} + 2\boldsymbol{\lambda}'\mathbf{R} = \mathbf{0}', \tag{i}$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\lambda}'} = 2(\mathbf{R}\boldsymbol{\beta} - \mathbf{r})' = \mathbf{0}'. \tag{ii}$$

Transposing (i) gives $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}}_R + \mathbf{R}'\hat{\boldsymbol{\lambda}} = \mathbf{X}'\mathbf{y}$, so

$$\hat{\boldsymbol{\beta}}_R = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}} = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}}. \tag{48}$$

Step 2: Solve for $\hat{\boldsymbol{\lambda}}$. Left-multiply (48) by $\mathbf{R}$ and impose the constraint (ii), $\mathbf{R}\hat{\boldsymbol{\beta}}_R = \mathbf{r}$:

$$\mathbf{r} = \mathbf{R}\hat{\boldsymbol{\beta}}_R = \mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}}.$$

Since $\text{rank}(\mathbf{R}) = q$ and $(\mathbf{X}'\mathbf{X})^{-1}$ is positive definite (Theorem 2.4, Step 4), $\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$ is positive definite (hence invertible); see Step 2 of the proof of Theorem 7.4 for the detailed argument. Rearranging:

$$\hat{\boldsymbol{\lambda}} = [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}).$$

Substituting back into (48) gives (47).

Step 3: $\hat{\boldsymbol{\beta}}_R$ is the unique minimizer over the feasible set. Define $\hat{\boldsymbol{\beta}}_R$ by (47) and $\hat{\boldsymbol{\lambda}}$ as in Step 2. By construction, $\mathbf{R}\hat{\boldsymbol{\beta}}_R = \mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}} = \mathbf{R}\hat{\boldsymbol{\beta}} - (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}) = \mathbf{r}$ (feasibility), and, transposing (i),

$$\mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R) = \mathbf{X}'\mathbf{y} - \mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} + \mathbf{R}'\hat{\boldsymbol{\lambda}} = \mathbf{R}'\hat{\boldsymbol{\lambda}}, \tag{49}$$

using $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$. Now let $\boldsymbol{\beta}$ be any feasible vector, $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$. Writing $\mathbf{y} - \mathbf{X}\boldsymbol{\beta} = (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R) + \mathbf{X}(\hat{\boldsymbol{\beta}}_R - \boldsymbol{\beta})$, the cross term vanishes:

$$(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R)'\mathbf{X}(\hat{\boldsymbol{\beta}}_R - \boldsymbol{\beta}) = \hat{\boldsymbol{\lambda}}'\mathbf{R}(\hat{\boldsymbol{\beta}}_R - \boldsymbol{\beta}) = \hat{\boldsymbol{\lambda}}'(\mathbf{r} - \mathbf{r}) = 0$$

by (49). Hence

$$\|\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\|^2 = \|\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\beta}}_R\|^2 + \|\mathbf{X}(\hat{\boldsymbol{\beta}}_R-\boldsymbol{\beta})\|^2 \;\geqslant\; \|\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\beta}}_R\|^2,$$

with equality iff $\mathbf{X}(\hat{\boldsymbol{\beta}}_R-\boldsymbol{\beta})=\mathbf{0}$, i.e. (by (A4)) iff $\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}_R$. So $\hat{\boldsymbol{\beta}}_R$ is the unique constrained minimizer and $\mathrm{SSE}_R = \|\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\beta}}_R\|^2$.

Step 4: $\mathrm{SSE}_R \geqslant \mathrm{SSE}_U$. The constrained problem minimizes $\|\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\|^2$ over $\{\boldsymbol{\beta}:\mathbf{R}\boldsymbol{\beta}=\mathbf{r}\}\subseteq\mathbb{R}^{k+1}$. Since $\hat{\boldsymbol{\beta}}$ minimizes the same objective over the full space, the minimum over a subset cannot be smaller: $\mathrm{SSE}_R \geqslant \mathrm{SSE}_U$. Equality holds iff $\hat{\boldsymbol{\beta}}$ already satisfies the constraint, i.e., $\mathbf{R}\hat{\boldsymbol{\beta}}=\mathbf{r}$: if $\mathbf{R}\hat{\boldsymbol{\beta}}=\mathbf{r}$, then (47) gives $\hat{\boldsymbol{\beta}}_R=\hat{\boldsymbol{\beta}}$ and hence $\mathrm{SSE}_R=\mathrm{SSE}_U$; conversely, if $\mathrm{SSE}_R=\mathrm{SSE}_U$, then $Q(\hat{\boldsymbol{\beta}}_R)=\mathrm{SSE}_R=\mathrm{SSE}_U=Q(\hat{\boldsymbol{\beta}})$, so $\hat{\boldsymbol{\beta}}_R$ minimizes Q over all of $\mathbb{R}^{k+1}$, hence $\hat{\boldsymbol{\beta}}_R=\hat{\boldsymbol{\beta}}$ by the uniqueness in Theorem 2.4 (by (7) with $\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}_R$, $\mathbf{X}(\hat{\boldsymbol{\beta}}_R-\hat{\boldsymbol{\beta}})=\mathbf{0}$, so $\hat{\boldsymbol{\beta}}_R=\hat{\boldsymbol{\beta}}$ under (A4)), and therefore $\mathbf{R}\hat{\boldsymbol{\beta}}=\mathbf{R}\hat{\boldsymbol{\beta}}_R=\mathbf{r}$ by the feasibility established in Step 3. □

11 Qualitative Regressors, Reparametrization, and Interactions

This section develops the modeling remarks of Section 10 (Remark 10.2). Its results are algebraic identities about a given $\mathbf{y}$ and design matrix and use only the standing assumption (A4), except where a distribution is asserted; such statements use (A1)–(A5), as in Section 7.

11.1 Qualitative Regressors

Definition 11.1 (Factors and Indicator Coding)**.** A **qualitative regressor** (or **factor**) assigns each observation to exactly one of $g\geqslant 2$ **levels**. The **indicator** (or dummy variable) of level l is the vector $D_l\in\mathbb{R}^n$ with i-th entry 1 if observation i is at level l and 0 otherwise. **Baseline coding** enters the intercept and the indicators of all levels but one, the **baseline**; by Remark 10.2, entering all g indicators together with the intercept would violate (A4). A **binary** regressor is a factor with $g=2$, entered as a single indicator.

Proposition 11.2 (Coefficients of a Factor Are Differences of Level Means)**.** Let $\mathbf{y}$ be regressed on $\mathbf{X}=[\mathbf{1}\;\; D_2\;\; \cdots\;\; D_g]$ (baseline level 1; here the coefficient of D_l is indexed by l, so the coefficients are $\beta_0,\beta_2,\ldots,\beta_g$), and suppose every level is observed: level l has $n_l\geqslant 1$ observations, with mean response $\bar{y}_l=D_l'\mathbf{y}/n_l$. Then $\mathbf{X}$ has full column rank, and:

(i) the fitted value of each observation is the mean response of its level;

(ii) $\hat{\beta}_0=\bar{y}_1$ and $\hat{\beta}_l=\bar{y}_l-\bar{y}_1$ for $l=2,\ldots,g$;

(iii) $\mathrm{SSE}=\sum_{l=1}^{g}\sum_{i:\,D_{l,i}=1}(y_i-\bar{y}_l)^2$ is the within-level sum of squares, and for $n>g$, $s^2=\mathrm{SSE}/(n-g)$.

Proof. Step 1: Column space and rank. Every observation is at exactly one level, so $D_1+\cdots+D_g=\mathbf{1}$, i.e. $D_1=\mathbf{1}-D_2-\cdots-D_g$. Hence $\mathcal{C}(\mathbf{X})=\mathcal{C}([D_1\;\;\cdots\;\;D_g])$. The indicators have disjoint, non-empty supports, so $D_l'D_m=0$ for $l\neq m$ and $D_l'D_l=n_l>0$; if $\sum_l a_lD_l=\mathbf{0}$, multiplying by D_m' gives $a_mn_m=0$, so $D_1,\ldots,D_g$ are linearly independent. Thus $\dim\mathcal{C}(\mathbf{X})=g$, the number of columns of $\mathbf{X}$, and $\mathbf{X}$ has full column rank.

Step 2: The fitted values. Let $\mathbf{v}=\sum_l\bar{y}_lD_l\in\mathcal{C}(\mathbf{X})$. For each m, $D_m'(\mathbf{y}-\mathbf{v})=D_m'\mathbf{y}-n_m\bar{y}_m=0$, so $\mathbf{y}-\mathbf{v}$ is orthogonal to every D_m, hence to $\mathcal{C}(\mathbf{X})$. By Step 1 of the proof of Theorem 2.5, $\mathbf{v}$ is then the unique point of $\mathcal{C}(\mathbf{X})$ closest to $\mathbf{y}$, i.e. $\hat{\mathbf{y}}=\mathbf{v}$. This is (i), and $\mathrm{SSE}=\|\mathbf{y}-\mathbf{v}\|^2$ is the within-level sum of squares in (iii); with $k=g-1$ regressors, $n-k-1=n-g$.

Step 3: The coefficients. At level 1 all of $D_2,\ldots,D_g$ vanish, so the fitted value is $\hat{\beta}_0$, and at level $l\geqslant 2$ it is $\hat{\beta}_0+\hat{\beta}_l$. As every level is observed, (i) gives $\hat{\beta}_0=\bar{y}_1$ and $\hat{\beta}_0+\hat{\beta}_l=\bar{y}_l$. □

Corollary 11.3 (One Indicator: the Pooled Two-Sample t Test)**.** Let $g = 2$, $\mathbf{X} = [\mathbf{1} \;\; D_2]$, $n_1, n_2 \geqslant 1$ and $n = n_1 + n_2 \geqslant 3$. Then $\hat{\beta}_0 = \bar{y}_1$, $\hat{\beta}_1 = \bar{y}_2 - \bar{y}_1$,

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n - 2}, \qquad \mathrm{SE}(\hat{\beta}_1) = s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}},$$

where $(n_l - 1)s_l^2$ is the sum of squared deviations of the responses at level l from $\bar{y}_l$ (so s^2 is the pooled variance), and, when $s > 0$ (under (A5) this holds with probability one, as in Proposition 7.13), the statistic of Theorem 7.1 for $H_0\colon \beta_1 = 0$ is the pooled two-sample t statistic

$$t = \frac{\bar{y}_2 - \bar{y}_1}{s\sqrt{1/n_1 + 1/n_2}}.$$

Under (A1)–(A5) the responses are normal with common variance σ^2 and means $\mu_1 = \beta_0$ at level 1 and $\mu_2 = \beta_0 + \beta_1$ at level 2, and under $H_0\colon \mu_1 = \mu_2$, $t \sim t_{n-2}$; moreover the overall F statistic equals t^2 (Corollary 7.17).

Proof. The coefficients and s^2 are Proposition 11.2 with $g = 2$ (the coefficient of D_2, called β_2 there, is indexed here by its column, as β_1). For the standard error, apply Corollary 2.18 with $\mathbf{x} = D_2$: $\bar{x} = n_2/n$ and $x_i^2 = x_i$, so $S_{xx} = \sum_i x_i^2 - n\bar{x}^2 = n_2 - n_2^2/n = n_1 n_2/n$, and $[(\mathbf{X}'\mathbf{X})^{-1}]_{11} = 1/S_{xx} = n/(n_1 n_2) = 1/n_1 + 1/n_2$ (the entry belonging to β_1). Hence $\mathrm{SE}(\hat{\beta}_1) = s\sqrt{1/n_1 + 1/n_2}$ (Theorem 7.1). Under (A1), $\mathsf{E}[y_i] = \beta_0 + \beta_1 D_{2,i}$, which gives the two means, and $H_0\colon \beta_1 = 0$ is $\mu_1 = \mu_2$; Theorem 7.1 with $n - k - 1 = n - 2$ gives the distribution, and Corollary 7.17 with $k = 1$ gives $F = t^2$. □

Proposition 11.4 (Reparametrization)**.** Let $\mathbf{X}$ satisfy (A4) and let $\tilde{\mathbf{X}} = \mathbf{X}\mathbf{A}$ with $\mathbf{A}$ an invertible $(k+1) \times (k+1)$ matrix. Then:

(i) $\tilde{\mathbf{X}}$ has full column rank and $\mathcal{C}(\tilde{\mathbf{X}}) = \mathcal{C}(\mathbf{X})$;

(ii) the regressions of $\mathbf{y}$ on $\mathbf{X}$ and on $\tilde{\mathbf{X}}$ have the same fitted values, residuals, SSE, SSR and s^2, hence the same R^2 and overall F statistic;

(iii) $\hat{\tilde{\boldsymbol{\beta}}} = \mathbf{A}^{-1}\hat{\boldsymbol{\beta}}$, and under (A3), $\mathrm{var}(\hat{\tilde{\boldsymbol{\beta}}}) = \sigma^2\mathbf{A}^{-1}(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{A}^{-1})'$;

(iv) conversely, if $\tilde{\mathbf{X}}$ is any $n \times (k+1)$ matrix with $\mathcal{C}(\tilde{\mathbf{X}}) = \mathcal{C}(\mathbf{X})$, then $\tilde{\mathbf{X}} = \mathbf{X}\mathbf{A}$ for an invertible $\mathbf{A}$.

Proof. **(i)** $\tilde{\mathbf{X}}\mathbf{v} = \mathbf{X}(\mathbf{A}\mathbf{v})$ shows $\mathcal{C}(\tilde{\mathbf{X}}) \subseteq \mathcal{C}(\mathbf{X})$, and $\mathbf{X}\mathbf{v} = \tilde{\mathbf{X}}(\mathbf{A}^{-1}\mathbf{v})$ the reverse inclusion. If $\tilde{\mathbf{X}}\mathbf{v} = \mathbf{0}$, then $\mathbf{X}(\mathbf{A}\mathbf{v}) = \mathbf{0}$, so $\mathbf{A}\mathbf{v} = \mathbf{0}$ by (A4) and $\mathbf{v} = \mathbf{0}$.

(ii) By Theorem 2.5, each fitted vector is the unique point of the common column space closest to $\mathbf{y}$, so the fitted values coincide, and with them the residuals, SSE and SSR $= \sum_i (\hat{y}_i - \bar{y})^2$; both designs have $k+1$ columns, so s^2 agrees as well, and R^2 and F are functions of these quantities.

(iii) Using $(\mathbf{A}'\mathbf{X}'\mathbf{X}\mathbf{A})^{-1} = \mathbf{A}^{-1}(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{A}')^{-1}$,

$$\hat{\tilde{\boldsymbol{\beta}}} = (\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}'\mathbf{y} = \mathbf{A}^{-1}(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{A}')^{-1}\mathbf{A}'\mathbf{X}'\mathbf{y} = \mathbf{A}^{-1}\hat{\boldsymbol{\beta}},$$

and Theorem 3.4 for the design $\tilde{\mathbf{X}}$ gives $\mathrm{var}(\hat{\tilde{\boldsymbol{\beta}}}) = \sigma^2(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1} = \sigma^2\mathbf{A}^{-1}(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{A}^{-1})'$.

(iv) Every column of $\tilde{\mathbf{X}}$ lies in $\mathcal{C}(\mathbf{X})$, so $\tilde{\mathbf{X}} = \mathbf{X}\mathbf{A}$ for some $(k+1) \times (k+1)$ matrix $\mathbf{A}$; likewise $\mathbf{X} = \tilde{\mathbf{X}}\mathbf{B}$ for some $\mathbf{B}$. Then $\mathbf{X}(\mathbf{A}\mathbf{B} - \mathbf{I}) = \mathbf{0}$, and (A4), applied to each column of $\mathbf{A}\mathbf{B} - \mathbf{I}$, gives $\mathbf{A}\mathbf{B} = \mathbf{I}$; so the square matrix $\mathbf{A}$ is invertible. □

Corollary 11.5 (Changes of Coding)**.** Each of the following changes of the design replaces $\mathbf{X}$ by $\mathbf{X}\mathbf{A}$ with $\mathbf{A}$ invertible, other columns being left unchanged, and therefore does not change the fit (Proposition 11.4):

(a) changing the baseline level of a factor entered by baseline coding;

(b) coding a binary regressor as ± 1, i.e. replacing its indicator D by $2D-\mathbf{1}$; with no other regressors the new intercept is $\frac{1}{2}(\bar{y}_1+\bar{y}_2)$ and the new slope is $\frac{1}{2}(\bar{y}_2-\bar{y}_1)$;

(c) centering a regressor, $\mathbf{X}_{\cdot j} \mapsto \mathbf{X}_{\cdot j} - \bar{x}_j\mathbf{1}$ $(j \geqslant 1)$;

(d) rescaling a regressor, $\mathbf{X}_{\cdot j} \mapsto a\mathbf{X}_{\cdot j}$ with $a \neq 0$, which replaces $\hat{\beta}_j$ by $\hat{\beta}_j/a$.

In (a) and (b), the F statistic of Corollary 7.6 for the hypothesis that all coefficients of the factor's indicators are zero has the same value under both codings. In (a) the intercept, the coefficients of the indicators and their t statistics in general change; in (b) the indicator's coefficient and its standard error are both halved, so its t statistic is unchanged (its square is that F statistic, Corollary 7.17), and only the intercept and its t statistic change in general. In both, the coefficients of the other columns, their standard errors and their t statistics are unchanged.

Proof. **(a)** Let $\mathbf{Z}$ denote the other columns. The design $[\mathbf{1} \quad D_2 \quad \cdots \quad D_g \quad \mathbf{Z}]$ and the design $[\mathbf{1} \quad D_1 \quad D_3 \quad \cdots \quad D_g \quad \mathbf{Z}]$ both have column space $\mathcal{C}([D_1 \quad \cdots \quad D_g \quad \mathbf{Z}])$, because $\mathbf{1} = \sum_l D_l$, $D_1 = \mathbf{1} - \sum_{l \geqslant 2} D_l$ and $D_2 = \mathbf{1} - \sum_{l \neq 2} D_l$. They have the same number of columns, so Proposition 11.4(iv) applies. **(b)** $[\mathbf{1} \quad 2D-\mathbf{1}] = [\mathbf{1} \quad D]\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$, an invertible matrix, extended by the identity on the other columns. Without other regressors, $\hat{\boldsymbol{\beta}} = (\bar{y}_1, \bar{y}_2 - \bar{y}_1)'$ by Proposition 11.2, and $\mathbf{A}^{-1} = \begin{pmatrix} 1 & 1/2 \\ 0 & 1/2 \end{pmatrix}$ gives $\mathbf{A}^{-1}\hat{\boldsymbol{\beta}} = (\frac{1}{2}(\bar{y}_1+\bar{y}_2), \frac{1}{2}(\bar{y}_2-\bar{y}_1))'$. **(c)** The new column is $\mathbf{X}(\boldsymbol{\iota}_j - \bar{x}_j\boldsymbol{\iota}_0)$, where $\boldsymbol{\iota}_0$ and $\boldsymbol{\iota}_j$ select the intercept column and column j; so $\mathbf{A} = \mathbf{I} - \bar{x}_j\boldsymbol{\iota}_0\boldsymbol{\iota}_j'$, whose inverse is $\mathbf{I} + \bar{x}_j\boldsymbol{\iota}_0\boldsymbol{\iota}_j'$ because $\boldsymbol{\iota}_j'\boldsymbol{\iota}_0 = 0$. **(d)** $\mathbf{A}$ is diagonal with a in position j and ones elsewhere, and the j-th entry of $\mathbf{A}^{-1}\hat{\boldsymbol{\beta}}$ is $\hat{\beta}_j/a$.

For the F statistic: in both (a) and (b), setting the coefficients of the factor's indicators to zero leaves the regression on $\mathbf{1}$ and $\mathbf{Z}$ (Corollary 7.16), the same under both codings, so SSE_R agrees; SSE_U agrees by Proposition 11.4(ii); and q and $n-k-1$ are the same. The coefficients $\mathbf{A}^{-1}\hat{\boldsymbol{\beta}}$, however, depend on $\mathbf{A}$. In (b), the row of $\mathbf{A}^{-1}$ belonging to the indicator is $\frac{1}{2}$ times the corresponding unit row, so by Proposition 11.4(iii) its new coefficient is $\frac{1}{2}$ times the old one, and since $(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1} = (\mathbf{A}'\mathbf{X}'\mathbf{X}\mathbf{A})^{-1} = \mathbf{A}^{-1}(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{A}^{-1})'$, the corresponding diagonal entry of $(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}$ is $\frac{1}{4}$ times the old one; s is unchanged by Proposition 11.4(ii), so the standard error is halved and the t statistic is unchanged. The intercept row of $\mathbf{A}^{-1}$ is $(1, \frac{1}{2}, 0, \ldots, 0)$, so the new intercept is $\hat{\beta}_0 + \frac{1}{2}\hat{\beta}_D$, where $\hat{\beta}_D$ is the old indicator coefficient. Finally, in both (a) and (b) the new columns of the factor are combinations of $\mathbf{1}$ and the indicators only, so $\mathbf{A}$ is block diagonal with the identity on the columns of $\mathbf{Z}$, and so is $\mathbf{A}^{-1}$; the rows of $\mathbf{A}^{-1}$ belonging to $\mathbf{Z}$ are therefore unit rows, and the same computation shows that the coefficients of $\mathbf{Z}$ and the corresponding diagonal entries of $(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}$, hence their standard errors and t statistics, are unchanged. □

11.2 Regressors Defined by Known Functions

Definition 11.6 (Linear in the Parameters)**.** The theory of these notes requires linearity in $\boldsymbol{\beta}$, not in the observed variables. A model $\mathsf{E}[y_i] = \beta_0 + \beta_1 g_1(\mathbf{z}_i) + \cdots + \beta_k g_k(\mathbf{z}_i)$, with known functions $g_1, \ldots, g_k$ of observed variables $\mathbf{z}_i$, is the model of Definition 1.1 with $x_{ij} = g_j(\mathbf{z}_i)$. Examples are *polynomial regression*, $g_j(z) = z^j$; logarithms, $g(z) = \ln z$; a threshold indicator $g(z) = 1\{z \geqslant \xi\}$; and the hinge $g(z) = 1\{z \geqslant \xi\}(z - \xi)$, with a known knot ξ. A model such as $\mathsf{E}[y_i] = \beta_0 + \exp(\beta_1 z_i)$ is not linear in the parameters.

Proposition 11.7 (Piecewise Linear Regression Functions)**.** Let ξ be known and $d(z) = 1\{z \geqslant \xi\}$.

(i) $f(z) = \beta_0 + \beta_1 z + \beta_2 d(z)(z - \xi)$ is continuous, with slope β_1 for $z < \xi$ and $\beta_1 + \beta_2$ for $z > \xi$.

(ii) $f(z) = \beta_0 + \beta_1 z + \beta_2 d(z) + \beta_3 d(z) z$ has slope β_1 for $z < \xi$ and $\beta_1 + \beta_3$ for $z > \xi$, and a jump at ξ of size $f(\xi) - \lim_{z \uparrow \xi} f(z) = \beta_2 + \beta_3 \xi$.

Proof. For $z < \xi$, $d(z) = 0$ and both functions equal $\beta_0 + \beta_1 z$, whose limit at ξ is $\beta_0 + \beta_1 \xi$. For $z \geqslant \xi$, $d(z) = 1$: the function in (i) is $(\beta_0 - \beta_2\xi) + (\beta_1 + \beta_2)z$, whose value at ξ is $\beta_0 + \beta_1\xi$, so f is continuous

at ξ (and, being linear on each side, everywhere); the function in (ii) is $(\beta_0+\beta_2)+(\beta_1+\beta_3)z$, whose value at ξ exceeds $\beta_0+\beta_1\xi$ by $\beta_2+\beta_3\xi$. □

11.3 Interactions

Definition 11.8 (Interaction)**.** For two regressors $\mathbf{X}_{\cdot 1}$ and $\mathbf{X}_{\cdot 2}$, the **interaction** is their elementwise product $\mathbf{X}_{\cdot 1} \circ \mathbf{X}_{\cdot 2}$, with entries $x_{i1}x_{i2}$, entered as a further regressor:

$$\mathsf{E}[y_i] = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1} x_{i2}.$$

$\mathbf{X}_{\cdot 1}$ and $\mathbf{X}_{\cdot 2}$ are the **main effects**. A model without interaction terms is called **additive**.

Proposition 11.9 (Effects in an Interaction Model)**.** Let $f(x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$. Then $\partial f/\partial x_1 = \beta_1 + \beta_3 x_2$, $\partial f/\partial x_2 = \beta_2 + \beta_3 x_1$, and $f(x_1+1, x_2) - f(x_1, x_2) = \beta_1 + \beta_3 x_2$: the effect of x_1 depends on the level of x_2. If $x_2 = z \in \{0, 1\}$ is an indicator, then $f(x_1, 0) = \beta_0 + \beta_1 x_1$ and $f(x_1, 1) = (\beta_0+\beta_2) + (\beta_1+\beta_3)x_1$: two lines whose intercepts differ by β_2 and whose slopes differ by β_3; they are parallel iff $\beta_3 = 0$.

Proof. Differentiate term by term, using $\partial(\beta_3 x_1 x_2)/\partial x_1 = \beta_3 x_2$ and symmetrically in x_2; the difference is $\beta_1 + \beta_3 x_2$ by direct expansion; and substituting $z = 0$ and $z = 1$ gives the two lines. □

Proposition 11.10 (The Hierarchical Principle)**.** Let $\mathbf{X} = [\mathbf{1} \;\; \mathbf{X}_{\cdot 1} \;\; \mathbf{X}_{\cdot 2} \;\; \mathbf{X}_{\cdot 1} \circ \mathbf{X}_{\cdot 2} \;\; \mathbf{Z}]$ satisfy (A4), where $\mathbf{Z}$ (possibly empty) collects further columns, and for constants a, b let $\tilde{\mathbf{X}}$ be obtained by replacing $\mathbf{X}_{\cdot 1}$, $\mathbf{X}_{\cdot 2}$ and $\mathbf{X}_{\cdot 1} \circ \mathbf{X}_{\cdot 2}$ by $\mathbf{X}_{\cdot 1} + a\mathbf{1}$, $\mathbf{X}_{\cdot 2} + b\mathbf{1}$ and $(\mathbf{X}_{\cdot 1} + a\mathbf{1}) \circ (\mathbf{X}_{\cdot 2} + b\mathbf{1})$. Then:

(i) $\tilde{\mathbf{X}} = \mathbf{X}\mathbf{A}$ with $\mathbf{A}$ invertible, so the two fits coincide (Proposition 11.4);

(ii) the coefficients become $\tilde\beta_3 = \hat\beta_3$, $\tilde\beta_1 = \hat\beta_1 - b\hat\beta_3$, $\tilde\beta_2 = \hat\beta_2 - a\hat\beta_3$, $\tilde\beta_0 = \hat\beta_0 - a\hat\beta_1 - b\hat\beta_2 + ab\hat\beta_3$, the coefficients of $\mathbf{Z}$ being unchanged;

(iii) the standard error and the t statistic (Theorem 7.1) of the interaction coefficient are unchanged;

(iv) if the main effects are omitted, i.e. for the design $[\mathbf{1} \;\; \mathbf{X}_{\cdot 1} \circ \mathbf{X}_{\cdot 2} \;\; \mathbf{Z}]$ and its shifted version, the column space changes unless $b\mathbf{X}_{\cdot 1} + a\mathbf{X}_{\cdot 2}$ lies in the column space of the unshifted design.

Proof. **(i)** Since $(\mathbf{X}_{\cdot 1} + a\mathbf{1}) \circ (\mathbf{X}_{\cdot 2} + b\mathbf{1}) = \mathbf{X}_{\cdot 1} \circ \mathbf{X}_{\cdot 2} + b\mathbf{X}_{\cdot 1} + a\mathbf{X}_{\cdot 2} + ab\mathbf{1}$, $\tilde{\mathbf{X}} = \mathbf{X}\mathbf{A}$ with $\mathbf{A}$ block diagonal, with the identity on the columns of $\mathbf{Z}$ and, on the first four columns,

$$\mathbf{A}_4 = \begin{pmatrix} 1 & a & b & ab \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & a \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

$\mathbf{A}_4$ is invertible: $\mathbf{A}_4\mathbf{v} = \mathbf{0}$ gives $v_4 = 0$ from the last row, then $v_3 = v_2 = 0$ from the third and second rows, and $v_1 = 0$ from the first. Hence $\mathbf{A}$ is invertible.

(ii) Writing the fitted function in the shifted variables, $\tilde\beta_0 + \tilde\beta_1(x_1+a) + \tilde\beta_2(x_2+b) + \tilde\beta_3(x_1+a)(x_2+b)$ expands to

$$(\tilde\beta_0 + a\tilde\beta_1 + b\tilde\beta_2 + ab\tilde\beta_3) + (\tilde\beta_1 + b\tilde\beta_3)x_1 + (\tilde\beta_2 + a\tilde\beta_3)x_2 + \tilde\beta_3 x_1 x_2.$$

By (i) the fitted values coincide, i.e. $\tilde{\mathbf{X}}\hat{\tilde{\boldsymbol\beta}} = \mathbf{X}\mathbf{A}\hat{\tilde{\boldsymbol\beta}} = \mathbf{X}\hat{\boldsymbol\beta}$, so $\mathbf{A}\hat{\tilde{\boldsymbol\beta}} = \hat{\boldsymbol\beta}$ by (A4); reading off the coordinates of $\mathbf{A}\hat{\tilde{\boldsymbol\beta}}$, which are the four brackets above and the coefficients of $\mathbf{Z}$, and solving from the interaction upward gives the stated values.

(iii) $\mathbf{A}_4^{-1}$ is again upper triangular with unit diagonal (its last row is $(0, 0, 0, 1)$ by the last row of $\mathbf{A}_4\mathbf{A}_4^{-1} = \mathbf{I}$), so for the coordinate vector $\mathbf{u}$ of the interaction, $\mathbf{u}'\mathbf{A}^{-1} = \mathbf{u}'$. By Proposition 11.4(iii), $(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1} = \mathbf{A}^{-1}(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{A}^{-1})'$, whence $\mathbf{u}'(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}\mathbf{u} = \mathbf{u}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{u}$. Since s is the same for both fits (Proposition 11.4(ii)) and $\tilde\beta_3 = \hat\beta_3$, the standard error and t statistic agree.

(iv) The shifted interaction column equals $\mathbf{X}_{\cdot 1} \circ \mathbf{X}_{\cdot 2} + (b\mathbf{X}_{\cdot 1} + a\mathbf{X}_{\cdot 2}) + ab\mathbf{1}$. If the two column spaces coincide, the shifted column lies in the unshifted column space, which contains $\mathbf{1}$ and $\mathbf{X}_{\cdot 1} \circ \mathbf{X}_{\cdot 2}$; hence so does $b\mathbf{X}_{\cdot 1} + a\mathbf{X}_{\cdot 2}$. □

Remark 11.11 (The meaning of the hierarchical principle)**.** With both main effects present, the fit and the inference on the interaction do not depend on the origins chosen for x_1 and x_2, while the main-effect coefficients are the slopes at $x_2 = 0$ and at $x_1 = 0$ (Proposition 11.9) and change with the origin; their individual t tests therefore answer origin-dependent questions. Dropping a main effect because its t statistic is small makes, by Proposition 11.10(iv), the column space, and in general the fit, depend on an arbitrary origin. (Part (iv) treats the omission of both main effects; if only $\mathbf{X}_{\cdot 2}$ is omitted, the same argument, with $\mathbf{1}$, $\mathbf{X}_{\cdot 1}$ and $\mathbf{X}_{\cdot 1} \circ \mathbf{X}_{\cdot 2}$ in the unshifted design, shows that the column space changes unless $a\mathbf{X}_{\cdot 2}$ lies in it, and symmetrically with $b\mathbf{X}_{\cdot 1}$ if only $\mathbf{X}_{\cdot 1}$ is omitted.) This is the **hierarchical principle**: if an interaction is included, its main effects are included as well, whatever their p-values.

12 Exercises

Tier A: Reproduce the Derivation — Estimation, Distribution Theory and the Overall F Test (Sections 2–7)

Exercise A1. Establish the matrix differentiation formulas of Lemma 2.2: for constant $\mathbf{a} \in \mathbb{R}^p$ and constant $\mathbf{A} \in \mathbb{R}^{p \times p}$, prove $D(\mathbf{a}'\boldsymbol{\beta}) = \mathbf{a}'$ and $D(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = \boldsymbol{\beta}'(\mathbf{A} + \mathbf{A}')$ by computing the partial derivatives $\partial/\partial\beta_j$ coordinate by coordinate. Then apply the formulas to $Q(\boldsymbol{\beta}) = \mathbf{y}'\mathbf{y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}$ to obtain $DQ(\boldsymbol{\beta}) = -2\mathbf{y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}$.

Exercise A2. Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be symmetric and idempotent ($\mathbf{A}^2 = \mathbf{A} = \mathbf{A}'$). (a) Show every eigenvalue of $\mathbf{A}$ lies in $\{0, 1\}$. (b) Conclude $\operatorname{tr}(\mathbf{A}) = \operatorname{rank}(\mathbf{A})$. (c) Verify the formula on $\mathbf{H}$ and $\mathbf{M}$: $\operatorname{tr}(\mathbf{H}) = k + 1$ and $\operatorname{tr}(\mathbf{M}) = n - k - 1$.

Exercise A3. Derive the OLS estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ by minimizing $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$. Verify the second-order condition.

Exercise A4. For $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and $\mathbf{M} = \mathbf{I} - \mathbf{H}$: (a) prove $\mathbf{H}$ and $\mathbf{M}$ are symmetric and idempotent; (b) prove $\mathbf{HX} = \mathbf{X}$, $\mathbf{MX} = \mathbf{0}$, and $\mathbf{HM} = \mathbf{0}$; (c) conclude $\operatorname{tr}(\mathbf{H}) = k + 1$ and $\operatorname{tr}(\mathbf{M}) = n - k - 1$.

Exercise A5. Prove that $\mathsf{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$ under (A2) and (A4), and derive $\operatorname{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ under (A3) and (A4).

Exercise A6. State and prove the Gauss–Markov theorem.

Exercise A7. Prove that $\mathrm{MSE} = \mathrm{SSE}/(n - k - 1)$ is an unbiased estimator of σ^2 under (A2)–(A4).

Exercise A8. Let $\mathbf{z} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. (a) For any constant matrix $\mathbf{A}$ and vector $\mathbf{b}$, prove $\mathbf{Az} + \mathbf{b} \sim N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b},\ \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$ via the moment generating function. (b) Specialize to $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$ and let $\mathbf{A}$ be symmetric idempotent of rank r; prove $\mathbf{z}'\mathbf{Az} \sim \chi^2_r$ via the spectral decomposition.

Exercise A9. (Independence of quadratic forms.) Let $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$ and let $\mathbf{A}_1, \mathbf{A}_2$ be symmetric idempotent with $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$. Prove that $\mathbf{z}'\mathbf{A}_1\mathbf{z}$ and $\mathbf{z}'\mathbf{A}_2\mathbf{z}$ are independent. *Hint:* use joint normality of $(\mathbf{A}_1\mathbf{z}, \mathbf{A}_2\mathbf{z})$ and the equivalence "uncorrelated ⇔ independent" under joint normality.

Exercise A10. Under (A1)–(A5), prove $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$ and that SSE is independent of $\hat{\boldsymbol{\beta}}$.

Exercise A11. Prove the variance decomposition $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$, and derive the overall F statistic including the proof that, under (A1)–(A5) and $k \geqslant 1$, $\mathrm{SSR}/\sigma^2 \sim \chi^2_k$ under $H_0\colon \beta_1 = \cdots = \beta_k = 0$.

Tier B: Reproduce and Specialize — Inference and Prediction (Sections 2, 3, 7, 9; Exercise B5 extends Section 3 to non-spherical errors)

Exercise B1. Let $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. (a) Show $0 \leqslant h_{ii} \leqslant 1$. (b) Show $\sum_{i=1}^{n} h_{ii} = k+1$. (c) Under (A3), derive $\operatorname{var}(e_i) = \sigma^2(1-h_{ii})$.

Exercise B2. Consider testing $H_0\colon \mathbf{c}'\boldsymbol{\beta} = r_0$ for a given nonzero vector $\mathbf{c}$, under (A1)–(A5). (a) Derive the distribution of $\mathbf{c}'\hat{\boldsymbol{\beta}}$. (b) Construct the t statistic and prove $t^2 \sim F_{1,n-k-1}$ under H_0.

Exercise B3. Consider two nested models: the full model with k regressors and a reduced model with $k_0 < k$ regressors. Let SSE_R and SSE_U denote the respective residual sums of squares. (a) Prove $\mathrm{SSE}_R \geqslant \mathrm{SSE}_U$. (b) Under (A1)–(A5), show that $\frac{(\mathrm{SSE}_R - \mathrm{SSE}_U)/(k-k_0)}{\mathrm{SSE}_U/(n-k-1)} \sim F_{k-k_0,n-k-1}$ under $H_0\colon \beta_{k_0+1} = \cdots = \beta_k = 0$, where the reduced model retains the intercept and the first k_0 regressors.

Exercise B4. (General linear hypothesis.) For $H_0\colon \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ with $\mathbf{R}$ a $q \times (k+1)$ matrix of rank q, derive the test statistic

$$F = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})/q}{\mathrm{SSE}/(n-k-1)}$$

and prove $F \sim F_{q,n-k-1}$ under H_0 and (A1)–(A5).

Exercise B5. Assume (A1), (A2) and (A4), but suppose that in place of (A3) we have $\operatorname{var}(\boldsymbol{\varepsilon}) = \sigma^2\boldsymbol{\Omega}$, where $\boldsymbol{\Omega}$ is known, symmetric, and positive definite (but $\neq \mathbf{I}$). (a) Show that OLS is unbiased but derive its actual (incorrect) variance. (b) Define $\hat{\boldsymbol{\beta}}_{\mathrm{GLS}} = (\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{y}$ and prove it is BLUE.

Exercise B6. Let $\hat{y}_0 = \mathbf{x}_0'\hat{\boldsymbol{\beta}}$ be the predicted value at a new point $\mathbf{x}_0$, where $y_0 = \mathbf{x}_0'\boldsymbol{\beta} + \varepsilon_0$ with $\mathsf{E}[\varepsilon_0] = 0$, $\operatorname{var}(\varepsilon_0) = \sigma^2$ and ε_0 independent of $\boldsymbol{\varepsilon}$. Assume (A1)–(A4). (a) Derive $\operatorname{var}(\hat{y}_0 - \mathsf{E}[y_0])$. (b) Derive $\operatorname{var}(\hat{y}_0 - y_0)$. (c) Explain why (b) always exceeds (a).

Tier C: Reproduce the Derivation — Partitioned Regression, Model Selection and Constraints (Sections 6, 8, 10)

Exercise C1. (Frisch–Waugh–Lovell.) Partition $\mathbf{X} = [\mathbf{X}_1 \;\; \mathbf{X}_2]$. Let $\mathbf{M}_2 = \mathbf{I} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'$. Prove: (a) $\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{M}_2\mathbf{y}$. (b) The full regression residuals equal those from regressing $\mathbf{M}_2\mathbf{y}$ on $\mathbf{M}_2\mathbf{X}_1$.

Exercise C2. (Omitted variable bias.) The true model is $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$, but only $\mathbf{y} = \mathbf{X}_1\boldsymbol{\gamma} + \mathbf{u}$ is estimated. Show $\mathsf{E}[\tilde{\boldsymbol{\gamma}}] = \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2$. Under what conditions is $\tilde{\boldsymbol{\gamma}}$ unbiased?

Exercise C3. (Monotonicity of R^2 and the adjusted R^2 criterion.) (a) Prove that adding a regressor cannot decrease R^2. (b) Show that $\bar{R}^2$ increases if and only if the added variable's $|t|$-statistic exceeds 1.

Exercise C4. (Constrained least squares.) Consider $\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ subject to $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$. (a) Derive the constrained estimator $\hat{\boldsymbol{\beta}}_R$ via Lagrange multipliers. (b) Show

$$\hat{\boldsymbol{\beta}}_R = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}).$$

(c) Prove $\mathrm{SSE}_R \geqslant \mathrm{SSE}_U$.

Tier D: Apply and Extend

The exercises in this tier apply the theory to objects that do not appear in the text: a data set to be worked by hand, situations in which one has to decide which assumption fails, and two small propositions not proved above (Exercises D3(a) and D4; Exercise D3(b) is the identity in the proof of Corollary 3.13 with $x_0 = x_i$, worked as a check on Exercise D1(d)).

Exercise D1. (A regression by hand.) Five observations of a regressor x and a response y are

x_i	1	2	3	4	5
y_i	1	2	2	4	6

Fit the model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ under (A1)–(A5). (a) Compute $\mathbf{X}'\mathbf{X}$, $\mathbf{X}'\mathbf{y}$, $(\mathbf{X}'\mathbf{X})^{-1}$ and $\hat{\boldsymbol{\beta}}$. (b) Compute the fitted values $\hat{\mathbf{y}}$ and residuals $\mathbf{e}$, and verify $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$ (Theorem 2.12). (c) Compute SSE and MSE $= s^2$. (d) Compute the leverages $h_{11}, \dots, h_{55}$ and verify $0 \leqslant h_{ii} \leqslant 1$ and $\sum_i h_{ii} = 2$ (Theorem 2.14). (e) Compute SST, SSR, verify SST $=$ SSR $+$ SSE (Theorem 6.4), and compute R^2 and $\bar{R}^2$. (f) Compute the t statistic for $H_0\colon \beta_1 = 0$ and the overall F statistic, and verify $F = t^2$ (Corollary 7.17). (g) Using $t_{3,0.975} = 3.182$, give a 95% confidence interval for β_1, and at $x_0 = 6$ a 95% confidence interval for the conditional mean and a 95% prediction interval for a new observation (Theorems 9.1 and 9.2).

Exercise D2. (Which assumption fails?) For each situation, state which of (A1)–(A5) is violated, and which of the following still hold: unbiasedness of $\hat{\boldsymbol{\beta}}$ (Theorem 3.3), the variance formula $\operatorname{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ (Theorem 3.4), the Gauss–Markov property (Theorem 3.5), and the exact t and F tests of Section 7.

(i) Household expenditure is regressed on income; the scatter of the observations around the fitted line widens visibly as income grows.

(ii) Wages are regressed on years of schooling, an intercept, and indicator variables for *all four* regions of a country.

(iii) Test scores are regressed on class size, but family income—which is correlated with class size and affects test scores—is not observed and is left out.

(iv) Monthly sales are regressed on monthly advertising expenditure; the errors of consecutive months are positively correlated.

(v) The response is the indicator $y_i \in \{0, 1\}$ of whether individual i is employed, regressed on age and schooling.

Exercise D3. (A lower bound for leverages.) (a) Show that in a model with an intercept, $h_{ii} \geqslant 1/n$ for every i, so that together with Theorem 2.14, $1/n \leqslant h_{ii} \leqslant 1$. *Hint:* apply the argument of Theorem 2.14(i) to the symmetric idempotent matrix $\mathbf{H} - \mathbf{J}$ of Lemma 7.2, Step 2. (b) For the simple regression model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, show that $h_{ii} = \dfrac{1}{n} + \dfrac{(x_i - \bar{x})^2}{\sum_{l=1}^{n}(x_l - \bar{x})^2}$, and check this formula against the leverages found in Exercise D1(d).

Exercise D4. (Adding an orthogonal regressor.) Let $\mathbf{X}_{k+1} = [\mathbf{X}_k \;\; \mathbf{x}]$ with $\mathbf{X}_k'\mathbf{x} = \mathbf{0}$ and $\mathbf{x} \neq \mathbf{0}$. Show: (a) $\hat{\beta}_{k+1} = \mathbf{x}'\mathbf{y}/(\mathbf{x}'\mathbf{x})$; (b) the coefficients on $\mathbf{X}_k$ are the same in the short regression (on $\mathbf{X}_k$) and in the enlarged regression (on $\mathbf{X}_{k+1}$); (c) $\mathrm{SSE}_k - \mathrm{SSE}_{k+1} = (\mathbf{x}'\mathbf{y})^2/(\mathbf{x}'\mathbf{x})$. (d) Relate (b) to Theorem 8.3.

13 Solutions to Exercises

Solutions to Tier A

Solution to Exercise A1. $\mathbf{a}'\boldsymbol{\beta} = \sum_i a_i\beta_i$ gives $\partial(\mathbf{a}'\boldsymbol{\beta})/\partial\beta_j = a_j$, i.e. $D(\mathbf{a}'\boldsymbol{\beta}) = \mathbf{a}'$. In $\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta} = \sum_{i,l}\beta_i A_{il}\beta_l$ the coordinate β_j occurs in the terms with $i = j$ and in those with $l = j$, so

$$\frac{\partial(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta})}{\partial\beta_j} = \sum_l A_{jl}\beta_l + \sum_i \beta_i A_{ij} = (\mathbf{A}\boldsymbol{\beta})_j + (\mathbf{A}'\boldsymbol{\beta})_j, \quad \text{i.e.} \quad D(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = \boldsymbol{\beta}'(\mathbf{A} + \mathbf{A}')$$

(this is the proof of Lemma 2.2). Applying (i) with $\mathbf{a} = \mathbf{X}'\mathbf{y}$ and (ii) with the symmetric $\mathbf{A} = \mathbf{X}'\mathbf{X}$ to $Q(\boldsymbol{\beta}) = \mathbf{y}'\mathbf{y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}$ gives $DQ(\boldsymbol{\beta}) = -2(\mathbf{X}'\mathbf{y})' + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X} = -2\mathbf{y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}$.

Solution to Exercise A2. (a) If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $\lambda\mathbf{v} = \mathbf{A}\mathbf{v} = \mathbf{A}^2\mathbf{v} = \lambda^2\mathbf{v}$, so $\lambda^2 = \lambda$ and $\lambda \in \{0, 1\}$. (b) $\operatorname{tr}(\mathbf{A}) = \operatorname{rank}(\mathbf{A})$ by Lemma 2.8, which holds for every idempotent matrix. For

symmetric $\mathbf{A}$ the trace can also be read off the spectral decomposition: by (L3), $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ with $\mathbf{P}$ orthogonal and $\mathbf{\Lambda}$ diagonal with entries in $\{0, 1\}$ by (a), so $\operatorname{tr}(\mathbf{A}) = \operatorname{tr}(\mathbf{\Lambda}\mathbf{P}'\mathbf{P}) = \operatorname{tr}(\mathbf{\Lambda})$ ((L2)) is the number of unit eigenvalues, and by the lemma this number is the rank. (c) $\mathbf{H}$ and $\mathbf{M}$ are symmetric idempotent (Theorem 2.9(i)) and $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$ has dimension $k+1$ (Theorem 2.9(iv), Step 1), so $\operatorname{tr}(\mathbf{H}) = \operatorname{rank}(\mathbf{H}) = k+1$ and $\operatorname{tr}(\mathbf{M}) = \operatorname{tr}(\mathbf{I}_n) - \operatorname{tr}(\mathbf{H}) = n-k-1$.

Solution to Exercise A3. Expand $Q(\boldsymbol{\beta}) = \mathbf{y}'\mathbf{y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}$ and differentiate with Lemma 2.2: $DQ(\boldsymbol{\beta}) = -2\mathbf{y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}$. Setting it to zero gives the normal equation $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$. Under (A4), $\mathbf{X}'\mathbf{X}\mathbf{v} = \mathbf{0}$ implies $\|\mathbf{X}\mathbf{v}\|^2 = \mathbf{v}'\mathbf{X}'\mathbf{X}\mathbf{v} = 0$, so $\mathbf{X}\mathbf{v} = \mathbf{0}$ and $\mathbf{v} = \mathbf{0}$; hence $\mathbf{X}'\mathbf{X}$ is invertible and $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. Second-order condition: the Hessian $2\mathbf{X}'\mathbf{X}$ satisfies $\mathbf{v}'(2\mathbf{X}'\mathbf{X})\mathbf{v} = 2\|\mathbf{X}\mathbf{v}\|^2 > 0$ for $\mathbf{v} \neq \mathbf{0}$; the minimum is global and unique by $Q(\boldsymbol{\beta}) = Q(\hat{\boldsymbol{\beta}}) + \|\mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})\|^2$, equation (7).

Solution to Exercise A4. (a) $\mathbf{H}' = \mathbf{X}[(\mathbf{X}'\mathbf{X})^{-1}]'\mathbf{X}' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{H}$ since $(\mathbf{X}'\mathbf{X})^{-1}$ is symmetric; $\mathbf{H}^2 = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{H}$; then $\mathbf{M}' = \mathbf{I} - \mathbf{H}' = \mathbf{M}$ and $\mathbf{M}^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H}^2 = \mathbf{M}$. (b) $\mathbf{H}\mathbf{X} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{X}$, $\mathbf{M}\mathbf{X} = \mathbf{X} - \mathbf{H}\mathbf{X} = \mathbf{0}$, $\mathbf{H}\mathbf{M} = \mathbf{H} - \mathbf{H}^2 = \mathbf{0}$. (c) By Lemma 2.8, $\operatorname{tr}(\mathbf{H}) = \operatorname{rank}(\mathbf{H})$; every column of $\mathbf{H}$ is $\mathbf{X}$ times a vector, so $\mathcal{C}(\mathbf{H}) \subseteq \mathcal{C}(\mathbf{X})$, and $\mathbf{X} = \mathbf{H}\mathbf{X}$ gives the reverse inclusion; hence $\operatorname{rank}(\mathbf{H}) = \dim \mathcal{C}(\mathbf{X}) = k+1$ by (A4), and $\operatorname{tr}(\mathbf{M}) = n - (k+1)$.

Solution to Exercise A5. Substituting $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$: $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$, where (A4) is what makes $(\mathbf{X}'\mathbf{X})^{-1}$ exist. Taking expectations and using $\mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ (A2) gives $\mathsf{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$. For the variance, $\hat{\boldsymbol{\beta}}$ is the constant $\boldsymbol{\beta}$ plus the linear transformation $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$, so Lemma 3.2(i) and (A3) give $\operatorname{var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.

Solution to Exercise A6. *Statement:* under (A1)–(A4), for every linear estimator $\tilde{\boldsymbol{\beta}} = \mathbf{C}\mathbf{y}$ that is unbiased for all $\boldsymbol{\beta}$, $\operatorname{var}(\tilde{\boldsymbol{\beta}}) - \operatorname{var}(\hat{\boldsymbol{\beta}})$ is positive semi-definite, with equality iff $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$. *Proof:* $\mathsf{E}[\mathbf{C}\mathbf{y}] = \mathbf{C}\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}$ for all $\boldsymbol{\beta}$ forces $\mathbf{C}\mathbf{X} = \mathbf{I}_{k+1}$. Write $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}$; then $\mathbf{D}\mathbf{X} = \mathbf{C}\mathbf{X} - \mathbf{I} = \mathbf{0}$. By Lemma 3.2 and (A3), $\operatorname{var}(\tilde{\boldsymbol{\beta}}) = \sigma^2\mathbf{C}\mathbf{C}' = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} + \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}' + \sigma^2\mathbf{D}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \sigma^2\mathbf{D}\mathbf{D}'$, and the two middle terms vanish because $\mathbf{D}\mathbf{X} = \mathbf{0}$ and $\mathbf{X}'\mathbf{D}' = (\mathbf{D}\mathbf{X})'$. Hence $\operatorname{var}(\tilde{\boldsymbol{\beta}}) - \operatorname{var}(\hat{\boldsymbol{\beta}}) = \sigma^2\mathbf{D}\mathbf{D}' \succeq \mathbf{0}$, with equality iff $\mathbf{D} = \mathbf{0}$, i.e. $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$.

Solution to Exercise A7. $\mathbf{e} = \mathbf{M}\mathbf{y} = \mathbf{M}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \mathbf{M}\boldsymbol{\varepsilon}$ since $\mathbf{M}\mathbf{X} = \mathbf{0}$, so $\mathrm{SSE} = \boldsymbol{\varepsilon}'\mathbf{M}'\mathbf{M}\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}$ ($\mathbf{M}$ symmetric idempotent). By Lemma 3.6 with $\mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ (A2) and $\operatorname{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$ (A3), $\mathsf{E}[\mathrm{SSE}] = \operatorname{tr}(\mathbf{M} \cdot \sigma^2\mathbf{I}) = \sigma^2 \operatorname{tr}(\mathbf{M}) = \sigma^2(n-k-1)$ (Theorem 2.9(iv)). Dividing by $n-k-1$ gives $\mathsf{E}[\mathrm{MSE}] = \sigma^2$.

Solution to Exercise A8. (a) $M_{\mathbf{A}\mathbf{z}+\mathbf{b}}(\mathbf{t}) = \mathsf{E}[e^{\mathbf{t}'(\mathbf{A}\mathbf{z}+\mathbf{b})}] = e^{\mathbf{t}'\mathbf{b}}M_{\mathbf{z}}(\mathbf{A}'\mathbf{t}) = \exp\big(\mathbf{t}'(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}) + \frac{1}{2}\mathbf{t}'\mathbf{A}\mathbf{\Sigma}\mathbf{A}'\mathbf{t}\big)$, the MGF of $N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\mathbf{\Sigma}\mathbf{A}')$; conclude by uniqueness (N1). (b) Write $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ ((L3)) with $\mathbf{\Lambda} = \operatorname{diag}(1, \ldots, 1, 0, \ldots, 0)$ (r ones: the eigenvalues lie in $\{0, 1\}$ by idempotency and $\operatorname{tr}\mathbf{A} = \operatorname{rank}\mathbf{A} = r$). Then $\mathbf{w} = \mathbf{P}'\mathbf{z} \sim N(\mathbf{0}, \mathbf{P}'\mathbf{P}) = N(\mathbf{0}, \mathbf{I}_n)$ by (a), its coordinates are i.i.d. $N(0,1)$ (the MGF $e^{\mathbf{t}'\mathbf{t}/2}$ factorizes, (N2)), and $\mathbf{z}'\mathbf{A}\mathbf{z} = \mathbf{w}'\mathbf{\Lambda}\mathbf{w} = \sum_{i=1}^{r} w_i^2 \sim \chi^2_r$ by definition (N5).

Solution to Exercise A9. $(\mathbf{u}', \mathbf{v}')' = ((\mathbf{A}_1\mathbf{z})', (\mathbf{A}_2\mathbf{z})')'$ is a linear image of $\mathbf{z}$, hence jointly normal (Theorem 4.2), with $\operatorname{cov}(\mathbf{u}, \mathbf{v}) = \mathbf{A}_1\mathbf{I}\mathbf{A}_2' = \mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$ (Lemma 3.2(ii)). By Theorem 4.3, $\mathbf{u} \perp\!\!\!\perp \mathbf{v}$. Since $\mathbf{A}_i$ is symmetric idempotent, $\mathbf{z}'\mathbf{A}_i\mathbf{z} = \|\mathbf{A}_i\mathbf{z}\|^2$ is a function of $\mathbf{u}$ (resp. $\mathbf{v}$) alone, so the two quadratic forms are independent ((N3)).

Solution to Exercise A10. Let $\mathbf{z} = \boldsymbol{\varepsilon}/\sigma \sim N(\mathbf{0}, \mathbf{I}_n)$ (Theorem 4.2). Then $\mathrm{SSE}/\sigma^2 = \mathbf{z}'\mathbf{M}\mathbf{z}$ with $\mathbf{M}$ symmetric idempotent of trace $n-k-1$, so $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$ by Theorem 4.5. For independence, stack $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$ and $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$: the pair is jointly normal by Theorem 4.2, with cross-covariance $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{M} = \mathbf{0}$ because $\mathbf{X}'\mathbf{M} = (\mathbf{M}\mathbf{X})' = \mathbf{0}$. Theorem 4.3 gives $\hat{\boldsymbol{\beta}} \perp\!\!\!\perp \mathbf{e}$, hence $\hat{\boldsymbol{\beta}} \perp\!\!\!\perp \mathrm{SSE} = \mathbf{e}'\mathbf{e}$ ((N3)).

Solution to Exercise A11. *Decomposition:* $\mathbf{M}_0\mathbf{y} = \mathbf{M}_0\hat{\mathbf{y}} + \mathbf{M}_0\mathbf{e} = \mathbf{M}_0\hat{\mathbf{y}} + \mathbf{e}$ because $\mathbf{1}'\mathbf{e} = 0$ (Theorem 2.12(ii)); the cross term $\mathbf{e}'\mathbf{M}_0\hat{\mathbf{y}} = \mathbf{e}'\hat{\mathbf{y}} - \bar{y}\,\mathbf{e}'\mathbf{1} = 0$ (Theorem 2.12(ii), (iii)), so $\mathrm{SST} = \|\mathbf{M}_0\mathbf{y}\|^2 = \|\mathbf{M}_0\hat{\mathbf{y}}\|^2 + \|\mathbf{e}\|^2 = \mathrm{SSR} + \mathrm{SSE}$ (using $\bar{\hat{y}} = \bar{y}$). *F statistic:* $\hat{\mathbf{y}} - \bar{y}\mathbf{1} = (\mathbf{H} - \mathbf{J})\mathbf{y}$, and $\mathbf{H} - \mathbf{J}$ is symmetric idempotent of rank k because $\mathbf{H}\mathbf{J} = \mathbf{J} = \mathbf{J}\mathbf{H}$ (from $\mathbf{H}\mathbf{1} = \mathbf{1}$). Under H_0, $\mathbf{X}\boldsymbol{\beta} = \beta_0\mathbf{1}$ and $(\mathbf{H} - \mathbf{J})\mathbf{1} = \mathbf{0}$, so $\mathrm{SSR} = \boldsymbol{\varepsilon}'(\mathbf{H} - \mathbf{J})\boldsymbol{\varepsilon}$ and $\mathrm{SSR}/\sigma^2 \sim \chi^2_k$ (Theorem 4.5); $(\mathbf{H} - \mathbf{J})\mathbf{M} = \mathbf{0}$ gives $\mathrm{SSR} \perp\!\!\!\perp \mathrm{SSE}$

(Theorem 4.6); with $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$ (Theorem 5.2), $F = (\mathrm{SSR}/k)/(\mathrm{SSE}/(n-k-1)) \sim F_{k,n-k-1}$ by definition (N5).

Solutions to Tier B

Solution to Exercise B1. (a) $\mathbf{H} = \mathbf{H}'\mathbf{H}$ gives $h_{ii} = \sum_j h_{ij}^2 = h_{ii}^2 + \sum_{j\neq i} h_{ij}^2$, so $h_{ii} \geqslant 0$ and $h_{ii} \geqslant h_{ii}^2$, i.e. $0 \leqslant h_{ii} \leqslant 1$. (b) $\sum_i h_{ii} = \mathrm{tr}(\mathbf{H}) = \mathrm{rank}(\mathbf{H}) = k+1$ (Lemma 2.8, Theorem 2.9(iv)). (c) $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$, so $\mathrm{var}(\mathbf{e}) = \mathbf{M}(\sigma^2\mathbf{I})\mathbf{M}' = \sigma^2\mathbf{M}$ (Lemma 3.2, A3), whose (i,i) entry is $\sigma^2(1-h_{ii})$.

Solution to Exercise B2. (a) By Theorem 4.2, $\mathbf{c}'\hat{\boldsymbol{\beta}} \sim N(\mathbf{c}'\boldsymbol{\beta},\ \sigma^2\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c})$. (b) Under H_0, $Z = (\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0)/(\sigma\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}) \sim N(0,1)$ by (a); $V = \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$ is independent of $\hat{\boldsymbol{\beta}}$ (Theorem 5.2), hence of Z ((N3)); so $t = Z/\sqrt{V/(n-k-1)} = (\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0)/(s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}) \sim t_{n-k-1}$ ((N5)). Squaring, $t^2 = (Z^2/1)/(V/(n-k-1))$ with $Z^2 \sim \chi^2_1$ (definition (N5) with $r=1$) independent of V, so $t^2 \sim F_{1,n-k-1}$; it coincides with the F statistic of Theorem 7.4 for $\mathbf{R} = \mathbf{c}'$, $\mathbf{r} = r_0$ (Corollary 7.8).

Solution to Exercise B3. Let $\mathbf{X}_U$ be the full design matrix and $\mathbf{X}_R$ the submatrix of its first k_0+1 columns (intercept and the k_0 retained regressors), with hat matrices $\mathbf{H}_U = \mathbf{H}$, $\mathbf{H}_R = \mathbf{X}_R(\mathbf{X}_R'\mathbf{X}_R)^{-1}\mathbf{X}_R'$ and residual makers $\mathbf{M}_U = \mathbf{M}$, $\mathbf{M}_R = \mathbf{I} - \mathbf{H}_R$; the reduced model is the full model under $H_0\colon \beta_{k_0+1} = \cdots = \beta_k = 0$.

(a) The reduced model minimizes $\|\mathbf{y} - \mathbf{X}_U\boldsymbol{\beta}\|^2$ over the subspace $\{\boldsymbol{\beta} : \beta_{k_0+1} = \cdots = \beta_k = 0\}$ of $\mathbb{R}^{k+1}$, whereas the unconstrained minimum SSE_U is taken over all of $\mathbb{R}^{k+1}$; hence $\mathrm{SSE}_R \geqslant \mathrm{SSE}_U$. (Alternatively, by (b) below, $\mathrm{SSE}_R - \mathrm{SSE}_U = \|(\mathbf{H}_U - \mathbf{H}_R)\mathbf{y}\|^2 \geqslant 0$.)

(b) *Nested projections.* Each column of $\mathbf{X}_R$ is a column of $\mathbf{X}_U$, so $\mathbf{H}_U\mathbf{X}_R = \mathbf{X}_R$ (Theorem 2.9(ii)); hence $\mathbf{H}_U\mathbf{H}_R = \mathbf{H}_U\mathbf{X}_R(\mathbf{X}_R'\mathbf{X}_R)^{-1}\mathbf{X}_R' = \mathbf{H}_R$ and, transposing, $\mathbf{H}_R\mathbf{H}_U = \mathbf{H}_R$. Therefore $\mathbf{M}_R - \mathbf{M}_U = \mathbf{H}_U - \mathbf{H}_R$ is symmetric, idempotent since $(\mathbf{H}_U - \mathbf{H}_R)^2 = \mathbf{H}_U - \mathbf{H}_R - \mathbf{H}_R + \mathbf{H}_R = \mathbf{H}_U - \mathbf{H}_R$, of rank $\mathrm{tr}(\mathbf{H}_U) - \mathrm{tr}(\mathbf{H}_R) = (k+1) - (k_0+1) = k - k_0$ (Lemma 2.8), and $(\mathbf{M}_R - \mathbf{M}_U)\mathbf{M}_U = (\mathbf{H}_U - \mathbf{H}_R)(\mathbf{I} - \mathbf{H}_U) = \mathbf{H}_U - \mathbf{H}_U - \mathbf{H}_R + \mathbf{H}_R = \mathbf{0}$. *Distribution.* $\mathrm{SSE}_R - \mathrm{SSE}_U = \mathbf{y}'\mathbf{M}_R\mathbf{y} - \mathbf{y}'\mathbf{M}_U\mathbf{y} = \mathbf{y}'(\mathbf{M}_R - \mathbf{M}_U)\mathbf{y}$. Under H_0, $\mathbf{y} = \mathbf{X}_R\boldsymbol{\beta}_R + \boldsymbol{\varepsilon}$ (with $\boldsymbol{\beta}_R$ the first k_0+1 coefficients) and $\mathbf{M}_R\mathbf{X}_R = \mathbf{0} = \mathbf{M}_U\mathbf{X}_R$, so $\mathrm{SSE}_R - \mathrm{SSE}_U = \boldsymbol{\varepsilon}'(\mathbf{M}_R - \mathbf{M}_U)\boldsymbol{\varepsilon}$ and likewise $\mathrm{SSE}_U = \boldsymbol{\varepsilon}'\mathbf{M}_U\boldsymbol{\varepsilon}$. With $\mathbf{z} = \boldsymbol{\varepsilon}/\sigma \sim N(\mathbf{0}, \mathbf{I}_n)$, Theorem 4.5 gives $(\mathrm{SSE}_R - \mathrm{SSE}_U)/\sigma^2 \sim \chi^2_{k-k_0}$, and Theorem 4.6 (with $\mathbf{A}_1 = \mathbf{M}_R - \mathbf{M}_U$, $\mathbf{A}_2 = \mathbf{M}_U$, $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$) gives independence from $\mathrm{SSE}_U/\sigma^2 \sim \chi^2_{n-k-1}$. Hence $F = \frac{(\mathrm{SSE}_R - \mathrm{SSE}_U)/(k-k_0)}{\mathrm{SSE}_U/(n-k-1)} \sim F_{k-k_0,n-k-1}$ by definition (N5). *Via Theorem 7.4.* The same F is the general F statistic for $\mathbf{R} = [\mathbf{0} \mid \mathbf{I}_{k-k_0}]$ ($q = k - k_0$) and $\mathbf{r} = \mathbf{0}$: by Corollary 7.6, its numerator $(\mathbf{R}\hat{\boldsymbol{\beta}})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}})/q$ equals $(\mathrm{SSE}_R - \mathrm{SSE}_U)/q$.

Solution to Exercise B4. By Theorems 5.1 and 4.2, $\mathbf{R}\hat{\boldsymbol{\beta}} \sim N(\mathbf{R}\boldsymbol{\beta}, \sigma^2\boldsymbol{\Sigma}_R)$ with $\boldsymbol{\Sigma}_R = \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$, which is positive definite because $\mathrm{rank}\,\mathbf{R} = q$ (for $\mathbf{v} \neq \mathbf{0}$, $\mathbf{R}'\mathbf{v} \neq \mathbf{0}$ and $(\mathbf{X}'\mathbf{X})^{-1}$ is positive definite by Theorem 2.4, Step 4). Under H_0, $\mathbf{w} = \sigma^{-1}\boldsymbol{\Sigma}_R^{-1/2}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}) \sim N(\mathbf{0}, \mathbf{I}_q)$ ((L4), Theorem 4.2), so $W = \mathbf{w}'\mathbf{w} = (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'\boldsymbol{\Sigma}_R^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})/\sigma^2 \sim \chi^2_q$ (Theorem 4.5 with $\mathbf{A} = \mathbf{I}_q$). W is a function of $\hat{\boldsymbol{\beta}}$, hence independent of $V = \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$ (Theorem 5.2, (N3)), and $F = (W/q)/(V/(n-k-1))$ is the displayed statistic (the factors σ^{-2} cancel), distributed as $F_{q,n-k-1}$ by (N5).

Solution to Exercise B5. (a) Unbiasedness: $\mathsf{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathsf{E}[\boldsymbol{\varepsilon}] = \boldsymbol{\beta}$ (unchanged; unbiasedness does not use (A3)). Actual variance:

$$\mathrm{var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\boldsymbol{\Omega})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1},$$

which in general differs from $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$, so the usual standard errors are wrong. (Not for *every* $\boldsymbol{\Omega} \neq \mathbf{I}$, however: if $\boldsymbol{\Omega} = \mathbf{I} + \mathbf{v}\mathbf{v}'$ with $\mathbf{X}'\mathbf{v} = \mathbf{0}$, then $\mathbf{X}'\boldsymbol{\Omega}\mathbf{X} = \mathbf{X}'\mathbf{X}$ and the two expressions coincide.)

(b) Let $\boldsymbol{\Omega} = \mathbf{L}\mathbf{L}'$ with $\mathbf{L}$ invertible ((L4); for instance the Cholesky factor, or the symmetric square root $\boldsymbol{\Omega}^{1/2}$). Transform: $\mathbf{L}^{-1}\mathbf{y} = \mathbf{L}^{-1}\mathbf{X}\boldsymbol{\beta} + \mathbf{L}^{-1}\boldsymbol{\varepsilon}$, where $\mathrm{var}(\mathbf{L}^{-1}\boldsymbol{\varepsilon}) = \sigma^2\mathbf{L}^{-1}\boldsymbol{\Omega}(\mathbf{L}^{-1})' = \sigma^2\mathbf{I}$. The OLS estimator applied to the transformed model is

$$\hat{\boldsymbol{\beta}}_{\mathrm{GLS}} = [(\mathbf{L}^{-1}\mathbf{X})'(\mathbf{L}^{-1}\mathbf{X})]^{-1}(\mathbf{L}^{-1}\mathbf{X})'(\mathbf{L}^{-1}\mathbf{y}) = (\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{y}.$$

Since the transformed model satisfies (A1)–(A4) (its regressor matrix $\mathbf{L}^{-1}\mathbf{X}$ has full column rank because $\mathbf{L}^{-1}$ is invertible), Gauss–Markov (Theorem 3.5) guarantees that $\hat{\boldsymbol{\beta}}_{\mathrm{GLS}}$ is BLUE in the transformed model. It is also BLUE in the original model, because the two models have the *same* class of linear unbiased estimators with the *same* variances: an estimator $\mathbf{C}\mathbf{y}$ that is linear in $\mathbf{y}$ equals $(\mathbf{C}\mathbf{L})(\mathbf{L}^{-1}\mathbf{y})$, which is linear in the transformed response, and conversely every $\tilde{\mathbf{C}}(\mathbf{L}^{-1}\mathbf{y})$ equals $(\tilde{\mathbf{C}}\mathbf{L}^{-1})\mathbf{y}$; unbiasedness in the original model, $\mathbf{C}\mathbf{X} = \mathbf{I}$, is the same condition as $(\mathbf{C}\mathbf{L})(\mathbf{L}^{-1}\mathbf{X}) = \mathbf{I}$ in the transformed model; and $\operatorname{var}(\mathbf{C}\mathbf{y})$ is the variance of the same random vector under both descriptions. Hence the estimator that minimizes the variance (in the positive semi-definite order) over the transformed class, namely $\hat{\boldsymbol{\beta}}_{\mathrm{GLS}}$, minimizes it over the original class.

Solution to Exercise B6. (a) $\hat{y}_0 - \mathsf{E}[y_0] = \mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$, so by Lemma 3.2 and Theorem 3.4, $\operatorname{var}(\hat{y}_0 - \mathsf{E}[y_0]) = \mathbf{x}_0' \operatorname{var}(\hat{\boldsymbol{\beta}})\mathbf{x}_0 = \sigma^2 \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$. (b) $\hat{y}_0 - y_0 = \mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - \varepsilon_0$, where ε_0 is independent of $\boldsymbol{\varepsilon}$ and hence of $\mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$; the variance of a sum of independent terms is the sum of the variances (fact (N7)), so $\operatorname{var}(\hat{y}_0 - y_0) = \sigma^2 \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 + \sigma^2 = \sigma^2(1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)$ (Theorem 9.2, Step 2). (c) The prediction variance exceeds the estimation variance by exactly σ^2, the irreducible noise variance of ε_0.

Solutions to Tier C

Solution to Exercise C1. (a) The second block of the normal equations $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$ reads $\mathbf{X}_2'\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}_2'\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = \mathbf{X}_2'\mathbf{y}$, so $\hat{\boldsymbol{\beta}}_2 = (\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'(\mathbf{y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1)$. Substituting into the first block, $\mathbf{X}_1'\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}_1'\mathbf{H}_2(\mathbf{y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1) = \mathbf{X}_1'\mathbf{y}$ with $\mathbf{H}_2 = \mathbf{I} - \mathbf{M}_2$, which simplifies to $\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 = \mathbf{X}_1'\mathbf{M}_2\mathbf{y}$; here $\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1 = (\mathbf{M}_2\mathbf{X}_1)'(\mathbf{M}_2\mathbf{X}_1)$ is positive definite because $\mathbf{M}_2\mathbf{X}_1\mathbf{v} = \mathbf{0}$ would put $\mathbf{X}_1\mathbf{v}$ in $\mathcal{C}(\mathbf{X}_2)$, contradicting (A4) unless $\mathbf{v} = \mathbf{0}$. (b) The residual from regressing $\mathbf{M}_2\mathbf{y}$ on $\mathbf{M}_2\mathbf{X}_1$ is $\mathbf{M}_2\mathbf{y} - \mathbf{M}_2\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 = \mathbf{M}_2(\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 + \mathbf{e}) = \mathbf{M}_2\mathbf{e} = \mathbf{e}$, using $\mathbf{M}_2\mathbf{X}_2 = \mathbf{0}$ and $\mathbf{X}_2'\mathbf{e} = \mathbf{0}$ (Theorem 2.12(i)).

Solution to Exercise C2. Substituting the true model into the short-regression estimator, $\tilde{\boldsymbol{\gamma}} = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'(\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}) = \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\boldsymbol{\varepsilon}$; taking expectations with $\mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ (A2) gives the formula. Unbiasedness holds iff $\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{0}$, which occurs when $\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$ (every column of $\mathbf{X}_1$ is orthogonal to every column of $\mathbf{X}_2$) or $\boldsymbol{\beta}_2 = \mathbf{0}$.

Solution to Exercise C3. (a) With $\mathcal{C}_k \subseteq \mathcal{C}_{k+1}$ the two column spaces, $\hat{\mathbf{y}}_{k+1}$ is the closest point of $\mathcal{C}_{k+1}$ to $\mathbf{y}$ (Theorem 2.5) and $\hat{\mathbf{y}}_k \in \mathcal{C}_k \subseteq \mathcal{C}_{k+1}$, so $\mathrm{SSE}_{k+1} = \|\mathbf{y} - \hat{\mathbf{y}}_{k+1}\|^2 \leqslant \|\mathbf{y} - \hat{\mathbf{y}}_k\|^2 = \mathrm{SSE}_k$; since SST is the same in both models, $R^2 = 1 - \mathrm{SSE}/\mathrm{SST}$ cannot decrease. (b) $\bar{R}^2_{k+1} > \bar{R}^2_k \iff \mathrm{MSE}_{k+1} < \mathrm{MSE}_k \iff \mathrm{SSE}_{k+1}/(n-k-2) < (\mathrm{SSE}_{k+1} + \hat{\beta}^2_{k+1}\,\mathbf{x}'\mathbf{M}_k\mathbf{x})/(n-k-1)$ (Lemma 6.9) $\iff \mathrm{SSE}_{k+1} < (n-k-2)\hat{\beta}^2_{k+1}\,\mathbf{x}'\mathbf{M}_k\mathbf{x} \iff 1 < \hat{\beta}^2_{k+1}\,\mathbf{x}'\mathbf{M}_k\mathbf{x}/\mathrm{MSE}_{k+1}$; the right-hand side is t^2_{k+1} because $\mathrm{SE}(\hat{\beta}_{k+1})^2 = \mathrm{MSE}_{k+1}[(\mathbf{X}_{k+1}'\mathbf{X}_{k+1})^{-1}]_{k+1,k+1} = \mathrm{MSE}_{k+1}/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$ by (31).

Solution to Exercise C4. (a) With $\mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\lambda}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + 2\boldsymbol{\lambda}'(\mathbf{R}\boldsymbol{\beta} - \mathbf{r})$, Lemma 2.2 gives $\partial\mathcal{L}/\partial\boldsymbol{\beta}' = -2\mathbf{y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X} + 2\boldsymbol{\lambda}'\mathbf{R} = \mathbf{0}'$, i.e. $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}}_R = \mathbf{X}'\mathbf{y} - \mathbf{R}'\hat{\boldsymbol{\lambda}}$, so $\hat{\boldsymbol{\beta}}_R = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}}$; imposing $\mathbf{R}\hat{\boldsymbol{\beta}}_R = \mathbf{r}$ gives $\hat{\boldsymbol{\lambda}} = [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})$ ($\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$ is positive definite since $\operatorname{rank}\mathbf{R} = q$). (b) Substitute $\hat{\boldsymbol{\lambda}}$ into the expression for $\hat{\boldsymbol{\beta}}_R$. (c) For any $\boldsymbol{\beta}$ with $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$, $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R\|^2 + \|\mathbf{X}(\hat{\boldsymbol{\beta}}_R - \boldsymbol{\beta})\|^2$, because the cross term is $(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R)'\mathbf{X}(\hat{\boldsymbol{\beta}}_R - \boldsymbol{\beta}) = \hat{\boldsymbol{\lambda}}'\mathbf{R}(\hat{\boldsymbol{\beta}}_R - \boldsymbol{\beta}) = 0$ (Theorem 10.3, Step 3); so $\hat{\boldsymbol{\beta}}_R$ is the constrained minimizer, and $\mathrm{SSE}_R \geqslant \mathrm{SSE}_U$ because the unconstrained minimum SSE_U is taken over a larger set.

Solutions to Tier D

Solution to Exercise D1. (a) $\mathbf{X}'\mathbf{X} = \begin{pmatrix} 5 & 15 \\ 15 & 55 \end{pmatrix}$, $\mathbf{X}'\mathbf{y} = (15, 57)'$, $\det(\mathbf{X}'\mathbf{X}) = 275 - 225 = 50$, $(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{50}\begin{pmatrix} 55 & -15 \\ -15 & 5 \end{pmatrix} = \begin{pmatrix} 1.1 & -0.3 \\ -0.3 & 0.1 \end{pmatrix}$, and $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (16.5 - 17.1,\ -4.5 + 5.7)' = (-0.6,\ 1.2)'$. (b) $\hat{y}_i = -0.6 + 1.2x_i$: $\hat{\mathbf{y}} = (0.6, 1.8, 3.0, 4.2, 5.4)'$ and $\mathbf{e} = (0.4, 0.2, -1.0, -0.2, 0.6)'$; $\sum e_i = 0$ and $\sum x_i e_i = 0.4 + 0.4 - 3.0 - 0.8 + 3.0 = 0$. (c) $\mathrm{SSE} = 0.16 + 0.04 + 1 + 0.04 + 0.36 = 1.6$, $\mathrm{MSE} = 1.6/3 = 0.5333$, $s = 0.7303$. (d) $h_{ii} = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i = 1.1 - 0.6x_i + 0.1x_i^2$, giving $(0.6, 0.3, 0.2, 0.3, 0.6)$: all in $[0, 1]$ and summing to $2 = k + 1$. (e) $\bar{y} = 3$, $\mathrm{SST} = 4 + 1 + 1 + 1 + 9 = 16$, $\mathrm{SSR} = \sum(\hat{y}_i - 3)^2 = 5.76 + 1.44 + 0 + 1.44 + 5.76 = 14.4$, and indeed $14.4 + 1.6 = 16$; $R^2 = 14.4/16 = 0.9$

and $\bar{R}^2 = 1 - (1.6/3)/(16/4) = 0.8667$. (f) $\mathrm{SE}(\hat{\beta}_1) = s\sqrt{0.1} = \sqrt{0.05333} = 0.2309$, $t = 1.2/0.2309 = 5.196$; $F = \mathrm{MSR}/\mathrm{MSE} = 14.4/0.5333 = 27.0 = t^2$. (g) For β_1: $1.2 \pm 3.182 \times 0.2309 = 1.2 \pm 0.735$, i.e. $(0.465, 1.935)$. At $\mathbf{x}_0 = (1, 6)'$: $\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = 1.1 - 3.6 + 3.6 = 1.1$ and $\hat{y}_0 = -0.6 + 7.2 = 6.6$; the confidence interval for the conditional mean is $6.6 \pm 3.182 \times 0.7303\sqrt{1.1} = 6.6 \pm 2.437$, i.e. $(4.163, 9.037)$, and the prediction interval is $6.6 \pm 3.182 \times 0.7303\sqrt{2.1} = 6.6 \pm 3.368$, i.e. $(3.232, 9.968)$.

Solution to Exercise D2. (i) Heteroscedasticity: (A3) fails ($\mathrm{var}(\varepsilon_i)$ depends on income) while (A1)/(A2) may well hold. Unbiasedness survives (Theorem 3.3 uses only (A2) and (A4)); the variance formula, the Gauss–Markov property, $\mathsf{E}[\mathrm{MSE}] = \sigma^2$ and the exact t and F tests all rely on (A3) and fail in general (compare Exercise B5, where an exception is exhibited). (ii) The dummy variable trap: (A4) fails, because the four indicators sum to the intercept column, $\mathbf{1} = D_1 + D_2 + D_3 + D_4$; $\mathbf{X}'\mathbf{X}$ is singular and $\hat{\boldsymbol{\beta}}$ is not defined. Drop one indicator (or the intercept). (iii) Omitted variable: the unobserved income is absorbed into the error, which is then correlated with class size, so (A2)—equivalently (A1)—fails for the regression that is estimated. Taking the four properties in turn: unbiasedness fails, the estimated coefficient vector being the short-regression estimator $\tilde{\boldsymbol{\gamma}}$ of Theorem 8.3, with $\mathsf{E}[\tilde{\boldsymbol{\gamma}}] = \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 \neq \boldsymbol{\beta}_1$ in general; the variance formula survives as a statement about dispersion, since if the true error $\boldsymbol{\varepsilon}$ satisfies (A3) then Lemma 3.2 gives $\mathrm{var}(\tilde{\boldsymbol{\gamma}}) = \sigma^2(\mathbf{X}_1'\mathbf{X}_1)^{-1}$, but it is dispersion around the wrong center; the Gauss–Markov property does not apply, because $\tilde{\boldsymbol{\gamma}}$ is not in the class of unbiased estimators that Theorem 3.5 compares; and the exact t and F tests fail, because under (A5) the statistics are still built from a normal vector, but one centered at $\boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2$ rather than at $\boldsymbol{\beta}_1$, so the null distributions of Section 7 no longer hold. (iv) Serial correlation: (A3) fails through $\mathrm{cov}(\varepsilon_t, \varepsilon_{t+1}) \neq 0$; the consequences are as in (i)—unbiased $\hat{\boldsymbol{\beta}}$, wrong variance formula, invalid tests. (v) A binary response: (A5) fails, since given $\mathbf{x}_i$ the error $\varepsilon_i = y_i - \mathbf{x}_i'\boldsymbol{\beta}$ takes only the two values $1 - \mathbf{x}_i'\boldsymbol{\beta}$ and $-\mathbf{x}_i'\boldsymbol{\beta}$; (A3) also fails, since $\mathrm{var}(\varepsilon_i) = p_i(1 - p_i)$ with $p_i = \mathbf{x}_i'\boldsymbol{\beta}$ varies with i (and (A1) itself, which requires $p_i = \mathbf{x}_i'\boldsymbol{\beta} \in [0, 1]$ for all i, is questionable). Taking the four properties in turn: if (A1) holds, $\hat{\boldsymbol{\beta}}$ remains unbiased (Theorem 3.3 needs only (A2) and (A4)); the variance formula fails in general, since Lemma 3.2 gives $\mathrm{var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\mathrm{var}(\boldsymbol{\varepsilon})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$ with $\mathrm{var}(\boldsymbol{\varepsilon})$ having diagonal entries $p_i(1 - p_i)$ that vary with i, so $\mathrm{var}(\boldsymbol{\varepsilon}) \neq \sigma^2\mathbf{I}$; the Gauss–Markov property fails together with (A3), exactly as in (i); and the exact t and F tests fail, since both (A3) and (A5) are violated and the normal-theory distributions of Sections 5–7 are unavailable.

Solution to Exercise D3. (a) By Lemma 7.2, Step 2, $\mathbf{H} - \mathbf{J}$ is symmetric idempotent whenever $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, so $\mathbf{H} - \mathbf{J} = (\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J})$ and, comparing (i, i) entries exactly as in the proof of Theorem 2.14(i), $h_{ii} - \frac{1}{n} = \sum_{j=1}^{n}(h_{ij} - \frac{1}{n})^2 \geqslant 0$. (b) (The inverse below is Corollary 2.18(ii), and the result is the identity in the proof of Corollary 3.13 with $x_0 = x_i$.) For $\mathbf{X} = [\mathbf{1} \quad \mathbf{x}]$, $\mathbf{X}'\mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum_l x_l^2 \end{pmatrix}$ has determinant $n\sum_l x_l^2 - n^2\bar{x}^2 = nS_{xx}$, where $S_{xx} := \sum_l (x_l - \bar{x})^2 = \sum_l x_l^2 - n\bar{x}^2$; so $(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{nS_{xx}}\begin{pmatrix} \sum_l x_l^2 & -n\bar{x} \\ -n\bar{x} & n \end{pmatrix}$ and

$$h_{ii} = (1, x_i)\,(\mathbf{X}'\mathbf{X})^{-1}(1, x_i)' = \frac{\sum_l x_l^2 - 2n\bar{x}x_i + nx_i^2}{nS_{xx}} = \frac{S_{xx} + n(x_i - \bar{x})^2}{nS_{xx}} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}.$$

In Exercise D1, $\bar{x} = 3$ and $S_{xx} = 10$, giving $h_{ii} = 0.2 + (x_i - 3)^2/10 = (0.6, 0.3, 0.2, 0.3, 0.6)$, as found there.

Solution to Exercise D4. $\mathbf{X}_k'\mathbf{x} = \mathbf{0}$ gives $\mathbf{H}_k\mathbf{x} = \mathbf{X}_k(\mathbf{X}_k'\mathbf{X}_k)^{-1}\mathbf{X}_k'\mathbf{x} = \mathbf{0}$, so $\mathbf{M}_k\mathbf{x} = \mathbf{x}$, $\mathbf{x}'\mathbf{M}_k\mathbf{x} = \mathbf{x}'\mathbf{x} > 0$ (hence Lemma 6.9 applies) and $\mathbf{x}'\mathbf{M}_k\mathbf{y} = \mathbf{x}'\mathbf{y}$. (a) Equation (27): $\hat{\beta}_{k+1} = \mathbf{x}'\mathbf{y}/(\mathbf{x}'\mathbf{x})$. (b) In the proof of Lemma 6.9, $\hat{\boldsymbol{\beta}}_k^* = \hat{\boldsymbol{\beta}}_k - (\mathbf{X}_k'\mathbf{X}_k)^{-1}\mathbf{X}_k'\mathbf{x}\,\hat{\beta}_{k+1} = \hat{\boldsymbol{\beta}}_k$ because $\mathbf{X}_k'\mathbf{x} = \mathbf{0}$. (c) Equation (28): $\mathrm{SSE}_k - \mathrm{SSE}_{k+1} = (\mathbf{x}'\mathbf{M}_k\mathbf{y})^2/(\mathbf{x}'\mathbf{M}_k\mathbf{x}) = (\mathbf{x}'\mathbf{y})^2/(\mathbf{x}'\mathbf{x})$. (d) Read the enlarged model as the true model and the short regression as the one omitting $\mathbf{x}$: Theorem 8.3 with $\mathbf{X}_1 = \mathbf{X}_k$ and $\mathbf{X}_2 = \mathbf{x}$ says that the omitted-variable bias is $(\mathbf{X}_k'\mathbf{X}_k)^{-1}\mathbf{X}_k'\mathbf{x}\,\beta_{k+1} = \mathbf{0}$; part (b) is the exact, sample-level counterpart of this statement about expectations.